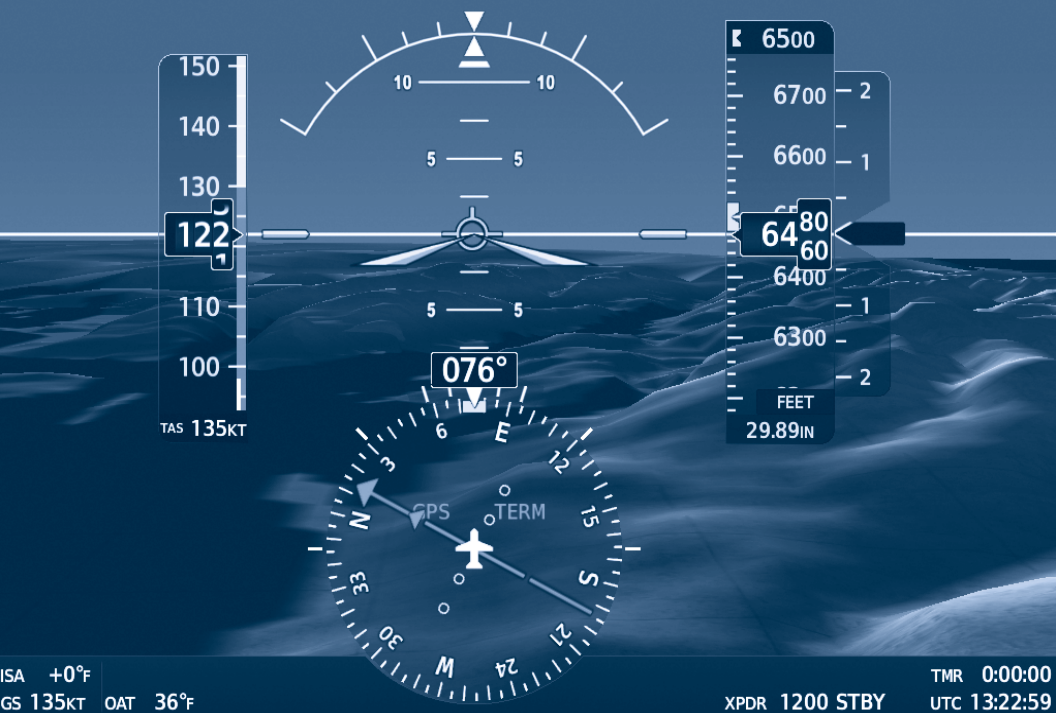


GARMIN[®]

G1000[®] NXi

Cockpit Reference Guide



Diamond DA40NG

System Software Version 1929.11 or later

FLIGHT INSTRUMENTS

ENGINE INDICATION SYSTEM (EIS)

NAV/COM/TRANSPONDER/AUDIO PANEL

FLIGHT MANAGEMENT SYSTEM

HAZARD AVOIDANCE

AUTOMATIC FLIGHT CONTROL SYSTEM

ADDITIONAL FEATURES

ABNORMAL OPERATION

ANNUNCIATIONS & ALERTS

APPENDIX

INDEX

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This manual reflects the operation of System Software version 1929.11 or later for the Diamond DA40NG. Some differences in operation may be observed when comparing the information in this manual to earlier or later software versions.

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WARNING: Do not use terrain avoidance displays as the sole source of information for maintaining separation from terrain and obstacles. Garmin obtains terrain and obstacle data from third party sources and cannot independently verify the accuracy of the information.



WARNING: Always refer to current aeronautical charts and NOTAMs for verification of displayed aeronautical information. Displayed aeronautical data may not incorporate the latest NOTAM information.



WARNING: Do not use geometric altitude for compliance with air traffic control altitude requirements. The primary barometric altimeter must be used for compliance with all air traffic control altitude regulations, requirements, instructions, and clearances.



WARNING: Do not use basemap information (land and water data) as the sole means of navigation. Basemap data is intended only to supplement other approved navigation data sources and should be considered only an aid to enhance situational awareness.



WARNING: Do not rely solely upon the display of traffic information to accurately depict all of the traffic within range of the aircraft. Due to lack of equipment, poor signal reception, and/or inaccurate information from aircraft or ground stations, traffic may be present that is not represented on the display.



WARNING: Do not use data link weather information for maneuvering in, near, or around areas of hazardous weather. Information contained within data link weather products may not accurately depict current weather conditions.



WARNING: Do not use the indicated data link weather product age to determine the age of the weather information shown by the data link weather product. Due to time delays inherent in gathering and processing weather data for data link transmission, the weather information shown by the data link weather product may be older than the indicated weather product age.



WARNING: Do not rely on the displayed minimum safe altitude (MSAs) as the sole source of obstacle and terrain avoidance information. Always refer to current aeronautical charts for appropriate minimum clearance altitudes.



WARNING: Do not operate this equipment without first obtaining qualified instruction.



WARNING: Do not use a QFE altimeter setting with this system. System functions will not operate properly with a QFE altimeter setting. Use only a QNH altimeter setting for height above mean sea level, or the standard pressure setting, as applicable.



WARNING: Do not use GPS to navigate to any active waypoint identified as a 'NON WGS84 WPT' by a system message. 'NON WGS84 WPT' waypoints are derived from an unknown map reference datum that may be incompatible with the map reference datum used by GPS (known as WGS84) and may be positioned in error as displayed.



WARNING: Do not rely on the autopilot to level the aircraft at the MDA/DH when flying an approach with vertical guidance. The autopilot will not level the aircraft at the MDA/DH even if the MDA/DH is set in the altitude preselect.



WARNING: Do not rely solely upon the display of traffic information for collision avoidance maneuvering. The traffic display does not provide collision avoidance resolution advisories and does not under any circumstances or conditions relieve the pilot's responsibility to see and avoid other aircraft.



WARNING: Do not rely on the accuracy of attitude and heading indications in the following geographic areas (due to variations in the earth's magnetic field): North of 72° North latitude at all longitudes; South of 70° South latitude at all longitudes; North of 65° North latitude between longitude 75° W and 120° W. (Northern Canada); North of 70° North latitude between longitude 70° W and 128° W. (Northern Canada); North of 70° North latitude between longitude 85° E and 114° E. (Northern Russia); South of 55° South latitude between longitude 120° E and 165° E. (Region south of Australia and New Zealand).



WARNING: Do not rely on information from a lightning detection system display as the sole basis for hazardous weather avoidance. Range limitations and interference may cause the system to display inaccurate or incomplete information. Refer to documentation from the lightning detection system manufacturer for detailed information about the system.



WARNING: Use appropriate primary systems for navigation, and for terrain, obstacle, and traffic avoidance. Garmin SVT is intended as an aid to situational awareness only and may not provide either the accuracy or reliability upon which to solely base decisions and/or plan maneuvers to avoid terrain, obstacles, or traffic.



WARNING: Do not use the Garmin SVT runway depiction as the sole means for determining the proximity of the aircraft to the runway or for maintaining the proper approach path angle during landing.



WARNING: Do not use TAWS information for primary terrain or obstacle avoidance. TAWS is intended only to enhance situational awareness.



CAUTION: Do not clean display surfaces with abrasive cloths or cleaners containing ammonia. They will harm the anti-reflective coating.



CAUTION: Do not allow repairs to be made by anyone other than an authorized Garmin service center. Unauthorized repairs or modifications could void both the warranty and affect the airworthiness of the aircraft.



NOTE: Do not rely solely upon data link services to provide Temporary Flight Restriction (TFR) information. Always confirm TFR information through official sources such as Flight Service Stations or Air Traffic Control.



NOTE: All visual depictions contained within this document, including screen images of the system panel and displays, are subject to change and may not reflect the most current system and aviation databases. Depictions of equipment may differ slightly from the actual equipment.



NOTE: The United States government operates the Global Positioning System and is solely responsible for its accuracy and maintenance. The GPS system is subject to changes which could affect the accuracy and performance of all GPS equipment. Portions of the system utilize GPS as a precision electronic NAVigation AID (NAVAID). Therefore, as with all NAVAIDs, information presented by the system can be misused or misinterpreted and, therefore, become unsafe.



NOTE: This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.



NOTE: Interference from GPS repeaters operating inside nearby hangars can cause an intermittent loss of attitude and heading displays while the aircraft is on the ground. Moving the aircraft more than 100 yards away from the source of the interference should alleviate the condition.



NOTE: Use of polarized eyewear may cause the flight displays to appear dim or blank.



NOTE: This product, its packaging, and its components contain chemicals known to the State of California to cause cancer, birth defects, or reproductive harm. This notice is being provided in accordance with California's Proposition 65. If you have any questions or would like additional information, please refer to our web site at www.garmin.com/prop65.



NOTE: Operating the system in the vicinity of metal buildings, metal structures, or electromagnetic fields can cause sensor differences that may result in nuisance miscompare annunciations during start up, shut down, or while taxiing. If one or more of the sensed values are unavailable, the annunciation indicates no comparison is possible.



NOTE: The system responds to a terminal procedure based on data coded within that procedure in the Navigation Database. Differences in system operation may be observed among similar types of procedures due to differences in the Navigation Database coding specific to each procedure.



NOTE: The FAA has asked Garmin to remind pilots who fly with Garmin database-dependent avionics of the following:

- It is the pilot's responsibility to remain familiar with all FAA regulatory and advisory guidance and information related to the use of databases in the National Airspace System.
- Garmin equipment will only recognize and use databases that are obtained from Garmin or Jeppesen. Databases obtained from Garmin or Jeppesen are assured compliance with all data quality requirements (DQRs) by virtue of a Type 2 Letter of Authorization (LOA) from the FAA. A copy of the Type 2 LOA is available for each database and can be viewed at <http://fly.garmin.com> by selecting 'Type 2 LOA Status.'
- Use of a current Garmin or Jeppesen database in your Garmin equipment is required for compliance with established FAA regulatory guidance, but does not constitute authorization to fly any and all terminal procedures that may be presented by the system. It is the pilot's responsibility to operate in accordance with established AFM(S) and regulatory guidance or limitations as applicable to the pilot, the aircraft, and installed equipment.



NOTE: The pilot/operator must review and be familiar with Garmin's database exclusion list as discussed in SAIB CE-14-04 to determine what data may be incomplete. The database exclusion list can be viewed at www.flygarmin.com by selecting 'Database Exclusions List.'



NOTE: The pilot/operator must have access to Garmin and Jeppesen database alerts and consider their impact on the intended aircraft operation. The database alerts can be viewed at www.flygarmin.com by selecting 'Aviation Database Alerts.'



NOTE: If the pilot/operator wants or needs to adjust the database, contact Garmin Product Support.



NOTE: Garmin requests the flight crew report any observed discrepancies related to database information. These discrepancies could come in the form of an incorrect procedure; incorrectly identified terrain, obstacles and fixes; or any other displayed item used for navigation or communication in the air or on the ground. Go to FlyGarmin.com and select 'Aviation Data Error Report'.



NOTE: Operate G1000NXi system power through at least one cycle in a period of four days of continuous operation to avoid an autonomous system reboot.



NOTE: Electronic aeronautical charts displayed on this system have been shown to meet the guidance in AC 120 76D as a Type B Electronic Flight Bag (EFB) for FlightCharts and ChartView. The accuracy of the charts is subject to the chart data provider. Own-ship position on airport surface charts cannot be guaranteed to meet the accuracy specified in AC 120-76D. Possible additional requirements may make a secondary source of aeronautical charts, such as traditional paper charts or an additional electronic display, necessary on the aircraft and available to the pilot. If the secondary source of aeronautical charts is a Portable Electronic Device (PED), its use must be consistent with the guidance in AC 120-76D.



NOTE: When using Stormscope, there are several atmospheric phenomena in addition to nearby thunderstorms that can cause isolated discharge points in the strike display mode. However, clusters of two or more discharge points in the strike display mode do indicate thunderstorm activity if these points reappear after the screen has been cleared.



NOTE: Intruder aircraft at or below 500 ft. AGL may not appear on the Garmin SVT display or may appear as a partial symbol..



NOTE: The purpose of this Cockpit Reference Guide is to provide the pilot a resource with which to find operating instructions on the major features of the system more easily. It is not intended to be a comprehensive operating guide. Complete operating procedures for the system are found in the Pilot's Guide for this aircraft.

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Part Number	Change Summary
190-02453-00	Initial Release

Revision	Date of Revision	Affected Pages	Description
A	June 2018	All	Production release at GDU 20.80
B	October 2021	All	Clerical Updates

Blank Page

FLIGHT INSTRUMENTS	1
Configuring Vspeed Bugs Individually	1
Modifying Vsports As A Group	1
Starting The PFD Generic Timer	1
Setting the Selected Altitude	2
Displaying The Metric altitude Overlay	2
Selecting The Altimeter Barometric Pressure Setting	2
Selecting Standard Barometric Pressure (29.92 in Hg)	2
Changing Altimeter Barometric Pressure Setting Units	2
Setting the Baro Transition Alert	2
Enabling/disabling the HSI Map on the PFD	3
Adjusting The Selected Heading	3
Adjusting The Selected Course	3
Changing The Navigation Angle True/Magnetic Setting	4
Selecting Bearing Display And Changing Sources	4
Displaying The DME Information Window	4
Change Navigation Sources	4
Changing The Selected GPS CDI Setting	5
Enabling/Disabling OBS Mode While Navigating A GPS Flight Plan	5
Changing temperature display units	5
Displaying Wind Data	5
Setting the Minimum Descent Altitude/Decision Height	6
Activating And Deactivating SVT	6
Activating And Deactivating Pathways	7
Activating And Deactivating Horizon Headings	7
Activating And Deactivating Airport Signs	7
Enabling/Disabling SVT Field Of View On The Navigation Map	7
ENGINE INDICATION SYSTEM	9
Engine Display	9
Engine Page	11
NAV/COM/TRANSPONDER/AUDIO PANEL	13
ADF Tuning (Optional)	13
DME Tuning (Optional)	13
Selecting Transponder Mode	13
Entering a Transponder Code	13
Selecting a COM Radio	14
Selecting a NAV Radio	14
NAV/COM Tuning	14
Auto-Tuning Radios On The PFD	14
Auto-Tuning Radios On The MFD	14
Digital Clearance Recorder and Player	15

Intercom System (ICS) Isolation	15
3D Audio	16
Blue-Select Mode (Telephone/Entertainment Distribution)	17
Bluetooth®	17
FLIGHT MANAGEMENT	19
Changing A Field In The MFD Navigation Status Box	19
USING MAP DISPLAYS	19
Changing The Navigation Map Orientation	19
Enabling/disabling Auto North Up And Selecting The Minimum Switching Range	19
Configuring Automatic Zoom	20
Panning The Map	20
Reviewing Information For An Airport, NAVAID, Or User Waypoint	20
Viewing Airspace Information For A Special-Use Or Controlled Airspace	21
Displaying/Removing Topographic Data On The PFD Inset Map Or The HSI Map	21
Displaying/Removing Topo Data And Selecting A Topo Data Range Using The Navigation Map Page Menu	21
Setting Up The 'Land', 'Aviation' Or 'Airspace' Group Items	21
Decluttering The Map	22
Decluttering The PFD Map	22
Displaying/Removing Airways	22
Selecting An Airway Range (Low ALT Airways or High ALT Airways)	22
Setting Up Additional 'Map' Group Items	23
IFR/VFR Charts	23
WAYPOINTS	24
Selecting An Airport For Review By Identifier, Facility Name, Or Location	24
Selecting A Runway	24
Viewing A Destination Airport	24
Viewing Information For A Nearest Airport On The PFD	24
Viewing Information For A Nearest Airport On The MFD	25
Viewing Runway Information For A Specific Airport	25
Selecting An Intersection	25
Selecting An NDB	25
Selecting A VOR	26
Selecting A VRP	26
Selecting A User Waypoint	26
Creating User Waypoints From The User Waypoint Information Page	26
Creating User Waypoints From Map Pages	27

AIRSPACES	28
Changing The Altitude Buffer Distance Setting	28
Turning An Airspace Alert On Or Off	28
Selecting And Viewing An Airspace Alert With Its Associated Information	28
Turning Smart Airspace On Or Off	28
DIRECT-TO NAVIGATION	29
Entering A Waypoint Identifier, Facility Name, Or City As A Direct-To Destination	29
Selecting An Active Flight Plan Waypoint As A Direct-To Destination	29
Selecting A Nearest, Recent Or User Waypoint As A Direct-To Destination	29
Cancelling A Direct-To	30
Selecting A Manual Direct-To Course	30
Entering A VNV Altitude And Along-Track Offset For The Waypoint	30
Removing A VNV Altitude Constraint	31
FLIGHT PLANNING	31
Creating An Active Flight Plan	31
Adding A Waypoint To The Active Flight Plan	32
Creating And Adding User Waypoints To The Active Flight Plan	32
Adding An Airway To A Flight Plan	33
Collapse/Expand Airways in a Flight Plan	33
Storing An Active Flight Plan From The Active Flight Plan	
Page or the Active Flight Plan Window	33
Activating A Stored Flight Plan On The MFD	34
Inverting And Activating A Stored Flight Plan On The MFD	34
Import a Flight Plan from an SD Card	34
Export a Flight Plan to an SD Card	35
Previewing A Pending Flight Plan Transfer From A Wireless Connection	35
Ignoring A Pending Flight Plan Transfer From A Wireless Connection	36
Storing A Pending Flight Plan	36
Activating A Pending Flight Plan	36
Deleting A Pending Flight Plan	37
Deleting all pending flight plans	37
Deleting The Active Flight Plan	37
Deleting An Individual Waypoint From The Active Flight Plan	37
Deleting An Entire Airway From The Active Flight Plan	38
Deleting An Entire Procedure From The Active Flight Plan	38
Entering An Along Track Offset Distance	38
Activating Parallel Track	39
Cancelling Parallel Track	39
Activating A Flight Plan Leg	39
Inverting The Active Flight Plan	40
Inverting And Activating A Stored Flight Plan	40

Switching Between Leg-To-Leg Waypoint Distance And Cumulative Waypoint Distance.....	40
Switching between wide and narrow view.....	40
Determining The Closest Point Along The Active Flight Plan To A Selected Waypoint.....	41
Creating A User-Defined Hold At An Active Flight Plan Waypoint.....	41
Creating A User-Defined Hold At The Aircraft Present Position.....	42
Creating A User-Defined Hold At A Direct-To Waypoint.....	42
Exiting A User-Defined Hold Inserted Into The Active Flight Plan.....	43
Removing A User-Defined Hold Inserted Into The Active Flight Plan.....	43
Removing A User-Defined Hold At An Off-Route Direct-To.....	43
VERTICAL NAVIGATION	43
Enabling VNV Guidance.....	43
Disabling VNV Guidance.....	44
Activating A Vertical Navigation Direct-To.....	44
Modifying The VS TGT And FPA.....	44
Altitude Constraints.....	44
Designating A Waypoint Altitude To Be Used For Vertical Guidance.....	46
Entering/Modifying An Altitude Constraint.....	46
Deleting An Altitude Constraint Provided By The Navigation Database.....	47
Deleting An Altitude Constraint That Has Been Manually Entered.....	47
Reverting A Manually Entered Altitude Constraint Back To The Navigation Database Value.....	47
Modifying A System Calculated Altitude Constraint.....	48
Enabling the Vertical Situation Display.....	48
Disabling the Vertical Situation Display.....	48
PROCEDURES.....	48
Viewing Available Procedures At An Airport.....	48
Loading A Procedure Into The Active Flight Plan From The <Procedure> Information Page.....	49
Loading A Departure Into The Active Flight Plan Using The PROC Key.....	50
Removing A Departure Procedure From The Active Flight Plan.....	50
Loading An Arrival Into The Active Flight Plan Using The PROC Key.....	51
Removing An Arrival From The Active Flight Plan.....	51
Loading An Approach Into The Active Flight Plan Using The PROC Key.....	52
Activating A Previously Loaded Approach.....	52
Activating A Previously Loaded Approach With Vectors To Final.....	53
Removing An Approach From The Active Flight Plan.....	53
Activating A Missed Approach In The Active Flight Plan.....	53
Activating Temperature Compensated Altitude.....	53
Cancelling Temperature Compensated Altitude.....	53

TRIP PLANNING	54
Selecting Automatic Or Manual Page Mode	54
Selecting Flight Plan Or Waypoint Mode	54
Selecting A Flight Plan And Leg For Trip Statistics	54
Selecting Waypoints For Waypoint Mode	54
Entering Manual Data For Trip Statistics Calculations	54
RAIM PREDICTION	55
Predicting RAIM Availability At A Selected Waypoint	55
Predicting RAIM Availability At The Aircraft Present Position	56
Disabling SBAS	56
HAZARD AVOIDANCE	57
Data Link Weather	57
FIS-B Weather Status	72
Stormscope Lightning Detection System	72
Terrain Displays	74
Traffic Information Service (TIS)	76
Avidyne TAS600 Traffic	78
ADS-B Traffic	81
AUTOMATIC FLIGHT CONTROL SYSTEM	85
Activating The Flight Director	85
Vertical Modes	86
Lateral Modes	86
Combination Modes (VNV, APR, NAV, BC, GA)	87
ADDITIONAL FEATURES	89
Terminal Procedure Charts	89
Airport Directory	91
SiriusXM™ Radio Entertainment	92
Satellite Telephone	93
Scheduler	103
Crew Profiles	104
Connex® Bluetooth® Management	107
Electronic Stability and Protection (ESP™)	109
Electronic Checklists	110
ABNORMAL OPERATION	113
Reversionary Mode	113
Abnormal COM Operation	113
Hazard Displays with Loss of GPS Position	113
Unusual Attitudes	114
Dead Reckoning	115
Garmin Connex Weather Request Status	116

ANNUNCIATIONS & ALERTS 119

 Crew Alerting System (CAS) Messages 119

 System Annunciations 120

 AFCS Alerts 121

 Relative Obstacle Symbols and Colors 122

 Relative Wind Turbine Obstacles and Colors 123

 Terrain-SVT and TAWS-B Potential Impact Area With Annunciations 123

 Terrain-SVT Alerts 123

 Terrain-SVT System Status Annunciations 124

 TIS System Status 125

 TIS Traffic Status Annunciations 126

 TAS Traffic Modes 127

 TAS Failure Annunciations 127

 TAS Traffic Status Annunciations 127

 ADS-B Traffic Modes 128

 ADS-B Traffic Failure Annunciations 128

 ADS-B Traffic Status Annunciations 128

 Aux-ADS-B Status Page Messages 129

 ADS-B Ground Station Status 130

 GDL 69A SXM Data Link Receiver Messages 130

 Voice Alerts 131

 System Message Advisories 132

 Flight Plan Import/Export Messages 149

 Crew Profile Import/Export Messages 151

APPENDIX 153

 Softkey Function 153

 PFD Softkey Maps 153

 MFD Softkey Maps 156

 Loading Updated Databases 159

 Database Deletion Feature 168

 Magnetic Field Variation Database Update 169

 Map Symbols 170

 Miscellaneous Symbols 174

INDEX Index-1

FLIGHT INSTRUMENTS

CONFIGURING VSPEED BUGS INDIVIDUALLY

- 1) Press the **TMR/REF** Softkey to show the 'References' Window.
- 2) Turn the large **FMS** Knob to highlight the desired Vspeed.
- 3) Use the small **FMS** Knob to change the highlighted Vspeed in 1-kt increments. When a speed has been changed from its default value, an asterisk appears next to the changed speed.
- 4) Press the **ENT** Key or turn the large **FMS** Knob to highlight the On/Off field
- 5) Turn the small **FMS** Knob clockwise to 'On' or counterclockwise to 'Off'.
- 6) To remove the window, press the **CLR** Key or the **TMR/REF** Softkey.

MODIFYING VSPEEDS AS A GROUP

- 1) Press the **TMR/REF** Softkey.
- 2) Press the **MENU** Key.
- 3) Turn the **FMS** Knob to highlight the desired selection (All On, All Off, Restore Defaults).
- 4) Press the **ENT** Key.
- 5) To remove the window, press the **CLR** Key or the **TMR/REF** Softkey.

STARTING THE PFD GENERIC TIMER

- 1) Press the **TMR/REF** Softkey, then turn the large **FMS** Knob to select the time field (hh/mm/ss). Turn the **FMS** Knobs to set the desired time, then press the **ENT** Key. The Up/Dn field is now highlighted.
- 2) Turn the small **FMS** Knob to display the Up/Down window. Turn the **FMS** Knob to select 'Up' or 'Dn', then press the **ENT** Key. 'Start?' is now highlighted.
- 3) Press the **ENT** Key to start the timer (if the timer is counting down, it will start counting up after reaching zero). Subsequent presses of the **ENT** Key will stop, then reset the timer. Press the **CLR** Key or the **TMR/REF** Softkey to remove the window.

SETTING THE SELECTED ALTITUDE

Turn the **ALT** Knob to set the Selected Altitude. Turn the large knob for 1000-ft increments, the small knob for 100-ft increments. If set to Metric mode, the large knob adjusts the Selected Altitude in 500-meter increments; the small knob adjusts the Selected Altitude in 50-meter increments.

If a Minimum Descent Altitude/Decision Height (MDA/DH) value has been set, this altitude is also available for the Selected Altitude.

DISPLAYING THE METRIC ALTITUDE OVERLAY

- 1) Press the **PFD Opt** Softkey to display the second-level softkeys.
- 2) Press the **ALT Units** Softkey.
- 3) Press the **Meters** Softkey to turn on metric altitude readouts.
- 4) Press the **Back** Softkey twice to return to the top-level softkeys.

SELECTING THE ALTIMETER BAROMETRIC PRESSURE SETTING

Turn the **BARO** Knob to select the desired setting.

SELECTING STANDARD BAROMETRIC PRESSURE (29.92 IN HG)

- 1) Press the **PFD Opt** Softkey to display the second-level softkeys.
- 2) Press the **STD Baro** Softkey.

CHANGING ALTIMETER BAROMETRIC PRESSURE SETTING UNITS

- 1) Press the **PFD Opt** Softkey to display the second-level softkeys.
- 2) Press the **ALT Units** Softkey.
- 3) Press the **IN** Softkey to display the barometric pressure setting in inches of mercury (in Hg).

Or:

Press the **HPA** Softkey to display the barometric pressure setting in hectopascals (hPa).

- 4) Press the **Back** Softkey twice to return to the top-level softkeys.

SETTING THE BARO TRANSITION ALERT

- 1) Use the **FMS** Knob to select the 'Aux - System Setup 1' Page on the MFD.
- 2) Press the **FMS** Knob to activate the cursor.

- 3) To enable/disable the Baro Transition Alert based on altitude, turn the large **FMS** Knob to highlight the 'On' or 'Off' field for the BARO Transition Alert Altitude in the BARO Transition Alert Box.
- 4) If desired, turn the small **FMS** Knob to set the BARO Transition Alert Altitude On or Off.
- 5) Turn the large **FMS** Knob to highlight the Altitude field.
- 6) Use the **FMS** Knobs to change the altitude and press the **ENT** Key to accept or press the **CLR** Key to return to the previous altitude selection.
- 7) Turn the large **FMS** Knob to highlight the 'On' or 'Off' field for the BARO Transition Alert Level.
- 8) If desired, turn the small **FMS** Knob to set the BARO Transition Alert Flight Level On or Off.
- 9) Turn the large **FMS** Knob to highlight the Flight Level field.
- 10) Use the **FMS** Knobs to change the Flight Level for the alert and press the **ENT** Key to accept or press the **CLR** Key to return to the previous altitude selection.
- 11) Push the **FMS** Knob to deactivate the cursor.

ENABLING/DISABLING THE HSI MAP ON THE PFD

- 1) Press the **Map/HSI** Softkey.
- 2) Press the **Layout** Softkey.
- 3) Press the **HSI Map** Softkey to enable the HSI Map.

Or:

Press the **Map Off** Softkey to disable the HSI Map.

ADJUSTING THE SELECTED HEADING

Turn the **HDG** Knob to set the Selected Heading.

Press the **HDG** Knob to synchronize the bug to the current heading.

ADJUSTING THE SELECTED COURSE

Turn the **CRS** Knob to set the Selected Course.

Press the **CRS** Knob to re-center the CDI and return the course pointer to the bearing of the active waypoint or navigation station.

CHANGING THE NAVIGATION ANGLE TRUE/MAGNETIC SETTING

- 1) Use the **FMS** Knob to select the 'Aux - System Setup 1' Page on the MFD.
- 2) Press the **FMS** Knob to activate the cursor.
- 3) Turn the large **FMS** Knob to highlight 'Nav Angle' in the 'Display Units' Box.
- 4) Turn the small **FMS** Knob to highlight the desired setting and press the **ENT** Key.
 - True - References angles to true north (T)
 - Magnetic - Angles corrected to the computed magnetic variation (Mag Var)

SELECTING BEARING DISPLAY AND CHANGING SOURCES

- 1) Press the **PFD Opt** Softkey.
- 2) Press either the **Bearing 1** or **Bearing 2** Softkey to display the desired bearing pointer and information window with a NAV source.
- 3) Press either the **Bearing 1** or **Bearing 2** Softkey again to change the bearing source to GPS.
- 4) Press either the **Bearing 1** or **Bearing 2** Softkey a third time to change the bearing source to ADF (note: ADF radio installation is optional).
- 5) To remove the bearing pointer and information window, press either the **Bearing 1** or **Bearing 2** Softkey again.

DISPLAYING THE DME INFORMATION WINDOW

- 1) Press the **PFD Opt** Softkey.
- 2) Press the **DME** Softkey to display the 'DME' Information Window above the Bearing1 Information Window.
- 3) To remove the 'DME' Information Window, press the **DME** Softkey again.

CHANGE NAVIGATION SOURCES

- 1) Press the **CDI** Softkey to change from GPS to VOR1 or LOC1. This places the cyan tuning box over the NAV1 standby frequency in the upper left corner of the PFD.
- 2) Press the **CDI** Softkey again to change to VOR2 or LOC2. This places the cyan tuning box over the NAV2 standby frequency.
- 3) Press the **CDI** Softkey a third time to return to GPS.

CHANGING THE SELECTED GPS CDI SETTING

- 1) Select the 'Aux - System Setup 1' Page on the MFD.
- 2) If necessary, press the **Setup 1** Softkey to display the System Setup 1 Page.
- 3) Press the **FMS** Knob to activate the cursor.
- 4) Turn the large **FMS** Knob to highlight 'Format Allowed' in the 'GPS CDI' Box.
- 5) Turn the small **FMS** Knob to highlight the desired setting and press the **ENT** Key.
- 6) To cancel the selection, press the **FMS** Knob or the **CLR** Key.

ENABLING/DISABLING OBS MODE WHILE NAVIGATING A GPS FLIGHT PLAN

- 1) Press the **OBS** Softkey to select OBS Mode.
- 2) Turn a **CRS** Knob to select the desired course to/from the waypoint. Push the **CRS** Knob to synchronize the Selected Course with the bearing to the next waypoint.
- 3) Press the **OBS** Softkey again to disable OBS Mode.

CHANGING TEMPERATURE DISPLAY UNITS

- 1) Select the 'Aux - System Setup 1' Page on the MFD.
- 2) Press the **FMS** Knob to activate the cursor.
- 3) Turn the large **FMS** Knob to highlight the 'Temperature' field in the 'Display Units' Box.
- 4) Turn the small **FMS** Knob to highlight either 'Celsius' or 'Fahrenheit' and press the **ENT** Key to confirm the selection.
- 5) To cancel the selection, press the **FMS** Knob or the **CLR** Key.

DISPLAYING WIND DATA

- 1) Press the **PFD Opt** Softkey.
- 2) Press the **Wind** Softkey.
- 3) Press one of the Option softkeys to change how wind data is displayed:
 - Option 1: Headwind/tailwind and crosswind arrows with numeric speed components
 - Option 2: Wind direction arrow and speed
 - Option 3: Wind direction arrow with headwind/tailwind and crosswind components
- 4) To remove the window, press the **Off** Softkey.

SETTING THE MINIMUM DESCENT ALTITUDE/DECISION HEIGHT

- 1) Press the **TMR/REF** Softkey.
- 2) Turn the large **FMS** Knob to highlight the Off/BARO/TEMP COMP/RAD ALT field to the right of 'MINS'.
- 3) Turn the small **FMS** Knob clockwise to select BARO, TEMP COMP, or RAD ALT. Off is selected by default.
- 4) Press the **ENT** Key or turn the large **FMS** Knob to highlight the next field.
- 5) Use the small **FMS** Knob to enter the desired altitude (BARO or TEMP COMP from zero to 16,000 feet). If TEMP COMP was selected, a field for entering the airport temperature will appear. Press the **ENT** Key or turn the large **FMS** Knob to highlight this field, and use the small **FMS** Knob to enter the temperature (-59°C to 59°C).
- 6) Press the **ENT** Key.
- 7) To remove the window, press the **CLR** Key or the **TMR/REF** Softkey.



WARNING: Use appropriate primary systems for navigation, and for terrain, obstacle, and traffic avoidance. SVT is intended as an aid to situational awareness only and may not provide the accuracy and/or fidelity upon which to solely base decisions and/or plan maneuvers to avoid terrain, obstacles, or traffic.



WARNING: Do not use SVT runway depiction as the sole means for determining the proximity of the aircraft to the runway or for maintaining the proper approach path angle during landing.



NOTE: Do not use the flight path marker as a flight director.

ACTIVATING AND DEACTIVATING SVT

- 1) Press the **PFD Opt** Softkey.
- 2) Press the **SVT** Softkey.
- 3) Press the **Terrain** Softkey. The SVT display will cycle on or off with the **Terrain** Softkey.

ACTIVATING AND DEACTIVATING PATHWAYS

- 1) Press the **PFD Opt** Softkey.
- 2) Press the **SVT** Softkey.
- 3) Press the **Terrain** Softkey.
- 4) Press the **Pathways** Softkey. The Pathway feature will cycle on or off with the **Pathways** Softkey.

ACTIVATING AND DEACTIVATING HORIZON HEADINGS

- 1) Press the **PFD Opt** Softkey.
- 2) Press the **SVT** Softkey.
- 3) Press the **Terrain** Softkey.
- 4) Press the **HDG LBL** Softkey. The horizon heading display will cycle on or off with the **HDG LBL** Softkey.

ACTIVATING AND DEACTIVATING AIRPORT SIGNS

- 1) Press the **PFD Opt** Softkey.
- 2) Press the **SVT** Softkey.
- 3) Press the **Terrain** Softkey.
- 4) Press the **APT Sign** Softkey. Display of airport signs will cycle on or off with the **APT Sign** Softkey.

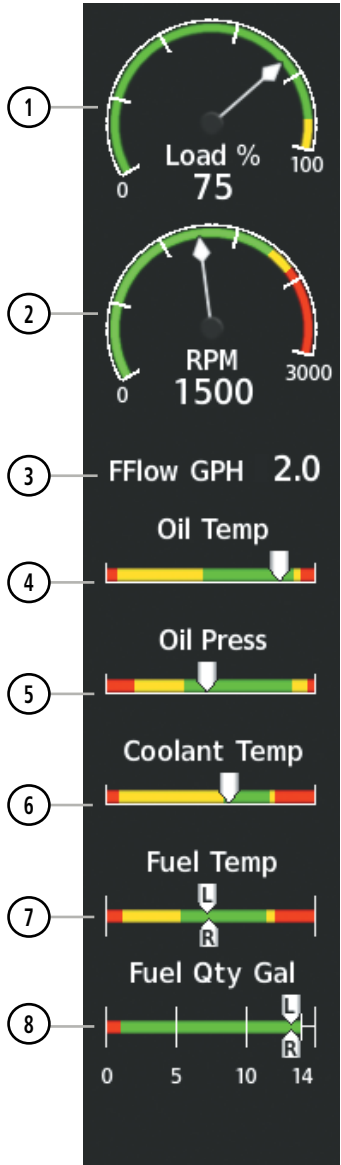
ENABLING/DISABLING SVT FIELD OF VIEW ON THE NAVIGATION MAP

- 1) While viewing the 'Map - Navigation Map' Page, press the **MENU** Key to display the 'Page Menu' Window.
- 2) Turn the large **FMS** Knob to highlight 'Map Settings' and press the **ENT** Key.
- 3) Turn the small **FMS** Knob to select the Map Group and press the **ENT** Key.
- 4) Turn the large **FMS** Knob to scroll through the Map Group options to 'Field of View'.
- 5) Turn the small **FMS** Knob to select 'On' or 'Off'.
- 6) Press the **FMS** Knob to return to the 'Map - Navigation Map' Page.

Blank Page

ENGINE INDICATION SYSTEM

ENGINE DISPLAY



Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCs
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

- | | | |
|---|---|---|
| ① | Engine Load Indicator (LOAD %) | Displays current engine loads as percentages |
| ② | Tachometer (RPM) | Displays propeller speed in revolutions per minute (rpm) |
| ③ | Fuel Flow Indicator (Fuel Flow GPH) | Displays digital value of current fuel flow in gallons per hour (gph) |
| ④ | Oil Temperature Indicator (Oil Temp) | Displays engine oil temperature ranges |
| ⑤ | Oil Pressure Indicator (Oil Press) | Displays engine oil pressure ranges |
| ⑥ | Coolant Temperature Indicator (Coolant Temp) | Displays coolant temperature ranges |
| ⑦ | Fuel Temperature Indicator (Fuel Temp) | Displays fuel temperature range; pointers labeled L and R indicate the fuel temperature in each tank |
| ⑧ | Fuel Quantity Indicator (Fuel Qty Gal) | Displays the fuel quantity in gallons (gal) of fuel in the main tanks; pointers labeled L and R indicate the fuel quantity in each tank |

ENGINE PAGE



- 1 **Oil Temperature Indicator (Oil °C)** Displays engine oil temperature ranges in degrees Celsius (°C)
- 2 **Oil Pressure Indicator (Oil bar)** Displays engine oil pressure ranges in bar
- 3 **Tachometer (RPM)** Displays propeller speed in revolutions per minute (rpm)
- 4 **Engine Load Indicator (Load %)** Displays current engine load as a percentage
- 5 **Fluids** Displays coolant temperature ranges
Displays the temperature of the gearbox in degrees Celsius (°C)
- 6 **Electrical** Displays the primary bus voltage
Displays each alternator load in amperes (amps)
- 7 **Total Service** Displays the aircraft's total flight hours (hrs); activated when the aircraft becomes airborne

8 Fuel System

Displays fuel quantity in the main tanks in gallons (gal)

Displays fuel temperature for the main tanks in degrees Celsius (°C)

Displays current fuel flow in gallons per hour (GPH)

9 Fuel Calculator

Gallons Remain - Displays current fuel remaining (gal) as set by the pilot and adjusted for fuel burn since last set

Gallons Used - Displays quantity of fuel used (gal) based on fuel flow since last reset

Endurance - Displays flight time remaining in hours and minutes (HH:MM) based on the calculated fuel remaining

Range NM - Displays aircraft range in nautical miles (nm) based on the calculated fuel remaining

Fuel Calculations



NOTE: Fuel calculations do not use the aircraft fuel quantity indicators and are calculated from the last time the fuel was reset.

Adjusting the fuel totalizer quantity

On the Engine Page, use the **DEC Fuel** and **INC Fuel** Softkeys to obtain the desired fuel remaining (Gallons Remain).

Resetting the fuel totalizer

On the Engine Page, press the **RST Fuel** Softkey; this resets displayed fuel remaining (Gallons Remain) to the maximum fuel capacity for the aircraft and fuel used to zero.

NAV/COM/TRANSPONDER/AUDIO PANEL

ADF TUNING (OPTIONAL)

- 1) Press the **ADF/DME** Softkey.
- 2) Turn the small **FMS** Knob to enter the first digit of the desired ADF frequency.
- 3) Turn the large **FMS** Knob to select the next desired field.
- 4) Turn the small **FMS** Knob to enter the desired number.
- 5) Repeat steps 3 and 4 until the desired ADF frequency is entered.
- 6) Press the **ENT** Key to accept the new frequency.
- 7) Press the **ENT** Key again to transfer the frequency to the active field.
- 8) Turn the large **FMS** Knob to select the Mode field.
- 9) Turn the small **FMS** Knob to select ANT, ADF, ADF/BFO, or ANT/BFO.
- 10) Press the **ENT** Key to complete the selection.

DME TUNING (OPTIONAL)

- 1) Press the **ADF/DME** or **DME** Softkey.
- 2) Turn the large **FMS** to select the DME mode field.
- 3) Turn the small **FMS** Knob to select the desired NAV radio.
- 4) Press the **ENT** Key to complete the selection.

SELECTING TRANSPONDER MODE

- 1) Press the **XPDR** Softkey to display the Transponder Mode Selection Softkeys.
- 2) Press the desired softkey to activate the transponder mode.

ENTERING A TRANSPONDER CODE

- 1) Press the **XPDR** Softkey to display the transponder mode selection softkeys.
- 2) Press the **Code** Softkey to display the transponder code selection softkeys, for digit entry.
- 3) Press the digit softkeys to enter the code in the code field. When entering the code, the next key in sequence must be pressed within 10 seconds, or the entry is cancelled and restored to the previous code. Five seconds after the fourth digit has been entered, the transponder code becomes active.

SELECTING A COM RADIO

Transmit/Receive

Press the **COM1 MIC** or **COM2 MIC** Key on the audio panel.

Receive Only

Press the **COM1** or **COM2** Key on the audio panel.

SELECTING A NAV RADIO

- 1) To begin navigating using a navigation radio, press the **CDI** Softkey on the PFD to select VOR1/LOC1 (NAV1) or VOR2/LOC2 (NAV2).
- 2) Press the **NAV1**, **NAV2**, **DME**, or **ADF** Key on the audio panel to select or deselect the navigation radio audio source. All radio keys can be selected individually or together.

NAV/COM TUNING

- 1) Press the small tuning knob to select the desired radio for tuning. A cyan box highlights the radio frequency to be tuned.
- 2) Turn the respective tuning knobs to enter the desired frequency into the standby frequency field. The large knob enters MHz and the small knob enters kHz.
- 3) Press the Frequency Transfer Key to place the frequency into the active frequency field.

AUTO-TUNING RADIOS ON THE PFD

- 1) Press the **Nearest** Softkey to display the Nearest Airports Window.
- 2) Turn either **FMS** Knob to highlight the desired frequency.
- 3) Press the **ENT** Key to place the frequency in the standby field of the active radio.
- 4) Press the Frequency Transfer Key to place the frequency in the active field.

NAV frequencies are entered automatically in the NAV frequency active or standby field (depending in CDI selection) upon approach loading or approach activation.

AUTO-TUNING RADIOS ON THE MFD

Auto-tuning on the MFD is done in much the same way as on the PFD. Use the **FMS** Knobs to select the desired frequency on any of the information pages. Pressing the **ENT** Key then loads the selected frequency in the tuning box as a standby frequency.

DIGITAL CLEARANCE RECORDER AND PLAYER



NOTE: Only the audio for the selected **COM MIC** Key is recorded.

- Pressing **PLAY** once plays the latest recorded memory block, then returns to normal operation. Pressing the **PLAY** Key again during play of a memory block stops play.
- Pressing **MKR/MUTE** while playing a memory block stops play.
- Pressing the **PLAY** Key twice within one-half second while audio is playing plays the previous block of recorded audio. Each subsequent two presses of the **PLAY** Key within one-half second backtracks through the recorded memory blocks to reach and play any recorded block.

INTERCOM SYSTEM (ICS) ISOLATION

Press the **PILOT ICS**, **COPLT ICS**, and/or the **PASS ICS** Key to enable intercom audio for the selected position. If the annunciators are lit, those positions will share intercom audio. If an annunciator is NOT lit that position is isolated from the others.

Audio Panel	Pilot Hears	Copilot Hears	Passenger Hears
 or  or 	Selected Radios Aural Alerts	Selected Radios Aural Alerts	Passenger MIC
	Selected Radios Aural Alerts	Selected Radios Aural Alerts	Selected Radios Aural Alerts Passenger MIC
	Selected Radios Aural Alerts Copilot MIC	Selected Radios Aural Alerts Pilot MIC	Passenger MIC

Audio Panel	Pilot Hears	Copilot Hears	Passenger Hears
	Selected Radios Aural Alerts	Selectd Radios Aural Alerts Passenger MIC	Selected Radios Aural Alerts Copilot MIC Passenger Mic
	Selected Radios Aural Alerts Passenger MIC	Selected Radios Aural Alerts	Selected Radios Aural Alerts Pilot MIC Passenger MIC
	Selected Radios Aural Alerts Copilot MIC Passenger MIC	Selected Radios Aural Alerts Pilot MIC Passenger MIC	Pilot MIC Copilot MIC Passenger MIC

ICS Isolation Modes

3D AUDIO

3D Audio is useful when multiple audio sources are present. By using different responses in each ear, 3D audio processing creates the illusion that each audio source is coming from a unique location or seat position.

Enabling/Disabling 3D Audio

- 1) Use the **FMS** Knob and softkeys on the MFD to select the 'Aux - System Setup 2' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight the 'Pilot 3D Audio' or 'Copilot 3D Audio' field.
- 4) Use the **FMS** Knob to select the desired setting.

BLUE-SELECT MODE (TELEPHONE/ENTERTAINMENT DISTRIBUTION)

The music and telephone (**TEL**, **MUS1**, and **MUS2**) audio are distributed using the Blue-Select Mode.

The Blue-Select Mode is entered by pressing the small knob when the volume control cursor (flashing white annunciator) is not active. If the volume control cursor is active, press the small knob twice. The first press will cancel the volume control cursor, the second will activate Blue-Select Mode.

The annunciator over the **TEL** Button will be flashing blue. Any combination of the annunciators over the **PILOT ICS**, **COPLT ICS**, and **PASS ICS** buttons may be blue. Select the desired button to turn the blue annunciator on or off to distribute the telephone audio to selected crew/passenger positions. Turn the large knob to select **MUS1** or **MUS2**, and select the crew/passenger positions to receive the music audio.

Selecting any button other than **PILOT ICS**, **COPLT ICS**, **PASS ICS**, **TEL**, **MUS1**, or **MUS2** will cancel Blue-Select Mode. Pressing the small knob will also cancel Blue-Select Mode. After approximately ten seconds with no input, the Blue-Select Mode will automatically cancel.

BLUETOOTH®



NOTE: Pairing is only necessary during the first attempt to connect a Bluetooth device to the audio panel. Once paired, the audio panel and the device will connect automatically.

Pairing a Bluetooth Device With The Audio Panel

Press and hold the inner knob for two seconds. The Bluetooth Annunciator flashes to indicate the unit is discoverable and the aural message “**Bluetooth discoverable**” is heard. The audio panel will remain discoverable for 90 seconds or until a successful pair is established. Once paired, the Bluetooth Annunciator turns steady blue and the aural message “**Bluetooth paired**” is heard.

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCs
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

Assigning An Audio Source To The Bluetooth Device

Press the **TEL**, **MUS1**, or **MUS2** key until the annunciator turns blue (the audio from the Bluetooth source will not be heard until this step is complete). The key annunciator cycles OFF-WHITE-BLUE. WHITE selects the wired audio source. BLUE selects the Bluetooth audio source. The BLUE source assignment will persist through Bluetooth audio connection disruptions.



NOTE: *The Bluetooth audio can only be assigned to one source at a time. Once the Bluetooth audio is assigned to an audio source, the remaining entertainment audio sources will only cycle between OFF and WHITE.*

Bluetooth audio will maintain a separate volume level and Blue Select distribution from the wired audio source. If the Bluetooth connection is supporting a phone call, all intercom positions listening to that source can also speak on the call through the headset MICs.

Additional Bluetooth Control Functions

When the Audio Panel detects a recording device as the Bluetooth connected device, the Pilot ICS Annunciator will turn blue. All audio heard by the Pilot will be recorded. Press and hold the PILOT ICS Key to enable/disable Bluetooth recording mode.

FLIGHT MANAGEMENT

Changing A Field In The MFD Navigation Status Box

- 1) Use the **FMS** Knob to select the 'Aux - System Setup 1' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight the desired field number in the MFD Data Bar Fields Box.
- 4) Turn the small **FMS** Knob to display and scroll through the data options list to select the desired data.
- 5) Press the **ENT** Key. Pressing the **Defaults** Softkey returns all fields to the default setting.

USING MAP DISPLAYS

CHANGING THE NAVIGATION MAP ORIENTATION

- 1) With the 'Map - Navigation Map' Page displayed, press the **MENU** Key. The cursor flashes on the 'Map Settings' option.
- 2) Press the **ENT** Key to display the Map Settings Window.
- 3) Turn the large **FMS** Knob, or press the **ENT** Key once, to select the 'Orientation' field.
- 4) Turn the small **FMS** Knob to select the desired orientation.
- 5) Press the **ENT** Key to select the new orientation.
- 6) Press the **FMS** Knob to return to the base page.

ENABLING/DISABLING AUTO NORTH UP AND SELECTING THE MINIMUM SWITCHING RANGE

- 1) Press the **MENU** Key with the 'Map - Navigation Map Page' displayed. The cursor flashes on the 'Map Settings' option.
- 2) Press the **ENT** Key. The 'Map Settings' Menu is displayed.
- 3) Select the 'Map' group.
- 4) Press the **ENT** Key.
- 5) Highlight the 'North Up Above' Field.
- 6) Select 'On' or 'Off' using the small **FMS** Knob.
- 7) Press the **ENT** Key to accept the selected option. The flashing cursor highlights the range Field.

- 8) Use the small **FMS** Knob to select the desired range.
- 9) Press the **ENT** Key to accept the selected option.
- 10) Press the **FMS** Knob to return to the 'Map - Navigation Map' Page.

CONFIGURING AUTOMATIC ZOOM

- 1) Press the **MENU** Key with the 'Map - Navigation Map' Page displayed. The cursor flashes on the 'Map Settings' option.
- 2) Press the **ENT** Key. The Map Settings Menu is displayed.
- 3) Select the 'Map' group.
- 4) Press the **ENT** Key.
- 5) Highlight the 'Auto Zoom' on/off field, and select 'Off' or 'On' using the small **FMS** Knob.
- 6) Press the **ENT** Key to accept the selected option. The flashing cursor highlights the 'Auto Zoom' display selection field.
- 7) Select 'MFD', 'PFD', or 'All' using the small **FMS** Knob.
- 8) Press the **ENT** Key to accept the selected option. The flashing cursor highlights the 'Max Look FWD' field. Times are from zero to 999 minutes.
- 9) Use the **FMS** Knobs to set the time. Press the **ENT** Key.
- 10) Repeat step 9 for 'Min Look FWD' (zero to 99 minutes) and 'Time Out' (zero to 99 minutes).
- 11) Press the **FMS** Knob to return to the Navigation Map Page.

PANNING THE MAP

- 1) Press the **Joystick** to display the Map Pointer.
- 2) Move the **Joystick** to move the Map Pointer around the map.
- 3) Press the **Joystick** to remove the Map Pointer and recenter the map on the aircraft's current position.

REVIEWING INFORMATION FOR AN AIRPORT, NAVAID, OR USER WAYPOINT

- 1) Place the Map Pointer on a waypoint.
- 2) Press the **ENT** Key to display the Waypoint Information Page for the selected waypoint.
- 3) Press the **Go Back** Softkey, the **CLR** Key, or the **ENT** Key to exit the Waypoint Information Page and return to the Navigation Map showing the selected waypoint.

VIEWING AIRSPACE INFORMATION FOR A SPECIAL-USE OR CONTROLLED AIRSPACE

- 1) Place the Map Pointer on the boundary of an airspace.
- 2) Push the **Joystick** to remove the Map Pointer and center the map on the aircraft.

DISPLAYING/REMOVING TOPOGRAPHIC DATA ON THE PFD INSET MAP OR THE HSI MAP

- 1) Press the **Map/HSI** Softkey.
- 2) Press the **Topo** Softkey.
- 3) Press the **Topo** Softkey again to remove topographic data from the PFD Inset Map or the HSI Map. When topographic data is removed from the PFD Map or HSI Map, all navigation data is presented on a black background.

DISPLAYING/REMOVING TOPO DATA AND SELECTING A TOPO DATA RANGE USING THE NAVIGATION MAP PAGE MENU

- 1) Press the **MENU** Key with the 'Map - Navigation Map' Page displayed. The cursor flashes on the "Map Setting" option.
- 2) Press the **ENT** Key. The 'Map Setting' Menu is displayed.
- 3) Select the 'Map' group.
- 4) Press the **ENT** Key.
- 5) Highlight the 'Terrain Display' Field.
- 6) Select 'Topo' or 'Off'.
- 7) Press the **ENT** Key. The cursor will move to the 'Terrain Display' range field. Ranges are from 1 nm to 1000 nm.
- 8) Turn the small **FMS** Knob to display the range list.
- 9) Select the desired range using the small **FMS** Knob.
- 10) Press the **ENT** Key.
- 11) Press the **FMS** Knob to return to the 'Map - Navigation Map' Page.

SETTING UP THE 'LAND', 'AVIATION' OR 'AIRSPACE' GROUP ITEMS

- 1) Press the **MENU** Key with the Navigation Map Page displayed. The cursor flashes on the 'Map Settings' option.
- 2) Press the **ENT** Key. The Map Settings Menu is displayed.

- 3) Turn the small **FMS** Knob to select the desired group.
- 4) Press the **ENT** Key. The cursor flashes on the first field.
- 5) Turn the large **FMS** Knob to select the desired option.
- 6) Turn the small **FMS** Knob to select the desired setting (e.g. On/Off or maximum range).
- 7) Press the **ENT** Key to accept the selected option and move the cursor to the next item.
- 8) Repeat steps 5-7 as necessary.
- 9) Press the **FMS** Knob to return to the Navigation Map Page.

DECLUTTERING THE MAP

Press the **Detail** Softkey with the 'Map - Navigation Map' Page displayed. The current declutter level is shown. With each softkey selection, another level of map information is removed.

DECLUTTERING THE PFD MAP

- 1) Press the **Map/HSI** Softkey.
- 2) Press the **Detail** Softkey. The current declutter level is shown. With each selection, another level of map information is removed.

DISPLAYING/REMOVING AIRWAYS

- 1) Press the **Map Opt** Softkey.
- 2) Press the **AWY** Softkey. Both High and Low Altitude Airways are displayed (AWY On).
- 3) Press the softkey again to display Low Altitude Airways only (AWY LO).
- 4) Press the softkey again to display High Altitude Airways only (AWY HI).
- 5) Press the softkey again to remove High Altitude Airways. No airways are displayed (AWY Off).

SELECTING AN AIRWAY RANGE (LOW ALT AIRWAYS OR HIGH ALT AIRWAYS)

- 1) Press the **MENU** Key with the 'Map - Navigation Map' Page displayed. The cursor flashes on the 'Map Settings' option.
- 2) Press the **ENT** Key. The Map Settings Menu is displayed.
- 3) Turn the small **FMS** Knob to select the 'Airways' group, and press the **ENT** Key.
- 4) Highlight the 'Low ALT Airways' or 'High ALT Airways' range field.

- 5) To change the range setting, turn the small **FMS** Knob to display the range list.
- 6) Select the desired range using the small **FMS** Knob.
- 7) Press the **ENT** Key.
- 8) Press the **FMS** Knob to return to the Navigation Map Page.

Navigation maps can display some additional items. These items (e.g. track vector, wind vector, fuel range ring, SVT field of view, and selected altitude intercept arc) can be displayed/removed individually.

SETTING UP ADDITIONAL 'MAP' GROUP ITEMS

- 1) Press the **MENU** Key with the Navigation Map Page displayed. The cursor flashes on the 'Map Settings' option.
- 2) Press the **ENT** Key. The Map Settings Menu is displayed.
- 3) Turn the small **FMS** Knob to select the 'Map' group.
- 4) Press the **ENT** Key. The cursor flashes on the first field.
- 5) Turn the large **FMS** Knob to select the desired option.
- 6) Turn the small **FMS** Knob to select 'On' or 'Off'.

Or:

If it is a data field, use the FMS Knob to select the range or time value.

- 7) Press the **ENT** Key to accept the selected option and move the cursor to the next item.
- 8) Repeat steps 5-7 as necessary.
- 9) Press the **FMS** Knob to return to the Navigation Map Page.

IFR/VFR CHARTS

Modifying the chart settings

- 1) Press the **MENU** Key with the IFR Low / IFR High Chart displayed. The cursor flashes on the 'Settings' option.
- 2) Press the **ENT** Key. The cursor flashes on the 'Color Scheme' setting.
- 3) Turn the small **FMS** Knob to select 'Day', 'Auto', or 'Night'. The IFR/VFR Chart display changes to reflect the setting.
- 4) If 'Auto' was selected, turn the large **FMS** Knob to select the backlight threshold field (sets value for automatic day/night switching).

Selecting the chart to display

- 1) Press the **MENU** Key with the 'Map - IFR/VFR Charts' Page displayed.
- 2) Turn the **FMS** Knob to highlight the 'Display VFR' Field, the 'Display IFR Low' Field, or the 'Display IFR High' Field.
- 3) Press the **ENT** Key. The selected chart is displayed.

WAYPOINTS

SELECTING AN AIRPORT FOR REVIEW BY IDENTIFIER, FACILITY NAME, OR LOCATION

- 1) From the 'WPT - Airport Information' Page, press the **FMS** Knob.
- 2) Use the **FMS** Knobs and enter an identifier, facility name, or location.
- 3) Press the **ENT** Key.
- 4) Press the **FMS** Knob to remove the cursor.

SELECTING A RUNWAY

- 1) With the 'WPT - Airport Information' Page displayed, press the **FMS** Knob to activate the cursor.
- 2) Turn the large **FMS** Knob to place the cursor in the 'Runways' Box, on the runway designator.
- 3) Turn the small **FMS** Knob to display the desired runway (if more than one) for the selected airport.
- 4) To remove the flashing cursor, press the **FMS** Knob.

VIEWING A DESTINATION AIRPORT

From the 'WPT - Airport Information' Page press the **MENU** Key. Select 'View Destination Airport'. The Destination Airport is displayed.

VIEWING INFORMATION FOR A NEAREST AIRPORT ON THE PFD

- 1) Press the **Nearest** Softkey to display the 'Nearest Airports' Window.
- 2) Highlight the airport identifier with the **FMS** Knob and press the **ENT** Key to display the 'Airport Information' Window.
- 3) To return to the 'Nearest Airports' Window press the **ENT** Key (with the cursor on 'BACK') or press the **CLR** Key. The cursor is now on the next airport in the nearest airports list. (Repeatedly pressing the **ENT** Key moves through the airport list, alternating between the 'Nearest Airports' Window and the 'Airport Information' Window.)

- 4) Press the **CLR** Key or the **Nearest** Softkey to close the PFD Nearest Airports Window.

VIEWING INFORMATION FOR A NEAREST AIRPORT ON THE MFD

- 1) Turn the large **FMS** Knob to select the NRST page group.
- 2) Turn the small **FMS** Knob to select the 'NRST - Nearest Airports' Page (it is the first page of the group, so it may already be selected). If there are no Nearest Airports available, "None Within 200nm" is displayed.
- 3) Press the **APT** Softkey; or press the **FMS** Knob; or press the **MENU** Key, highlight 'Select Airport Window' and press the **ENT** Key. The cursor is placed in the 'Nearest Airports' Box. The first airport in the nearest airports list is highlighted.
- 4) Turn the **FMS** Knob to highlight the desired airport. (Pressing the **ENT** Key also moves to the next airport.)
- 5) Press the **FMS** Knob to remove the flashing cursor.

VIEWING RUNWAY INFORMATION FOR A SPECIFIC AIRPORT

- 1) With the 'NRST - Nearest Airports' displayed, press the **RNWWY** Softkey; or press the **MENU** Key, highlight 'Select Runway Window'; and press the **ENT** Key. The cursor is placed in the 'Runways' Box.
- 2) Turn the small **FMS** Knob to select the desired runway.
- 3) Press the **FMS** Knob to remove the flashing cursor.

SELECTING AN INTERSECTION

- 1) With the 'WPT - Intersection Information' Page displayed, enter an identifier in the Intersection Box.
- 2) Press the **ENT** Key.
- 3) Press the **FMS** Knob to remove the flashing cursor.

SELECTING AN NDB

- 1) With the 'WPT - NDB Information' Page displayed, enter an identifier, the name of the NDB, or the city in which it's located in the NDB Box.
- 2) Press the **ENT** Key.
- 3) Press the **FMS** Knob to remove the flashing cursor.

SELECTING A VOR

- 1) With the 'WPT - VOR Information' Page displayed, enter an identifier, the name of the VOR, or the city in which it's located in the VOR Box.
- 2) Press the **ENT** Key.
- 3) Press the **FMS** Knob to remove the flashing cursor.

SELECTING A VRP

- 1) With the 'WPT - VRP Information' Page displayed, enter the identifier or the name of the VRP in the VRP Box.
- 2) Press the **ENT** Key.
- 3) Press the **FMS** Knob to remove the flashing cursor.

SELECTING A USER WAYPOINT

- 1) With the 'WPT - User WPT Information' Page displayed, enter the name of the User Waypoint, or scroll to the desired waypoint in the User Waypoint List using the large **FMS** Knob.
- 2) Press the **ENT** Key.
- 3) Press the **FMS** Knob to remove the flashing cursor.

CREATING USER WAYPOINTS FROM THE USER WAYPOINT INFORMATION PAGE

- 1) With the 'WPT - User WPT Information' Page displayed, press the **New** Softkey, **or** press the **MENU** Key and select 'Create New User Waypoint'.
- 2) Enter a user waypoint name (up to six characters).
- 3) Press the **ENT** Key. The current aircraft position is the default location of the new waypoint.
- 4) If desired, define the type and location of the waypoint in one of the following ways:

Select "RAD/RAD" using the small **FMS** Knob, press the **ENT** Key, and enter the two reference waypoint identifiers and radials into the Reference Waypoints window using the **FMS** Knobs.

Or:

Select "RAD/DIS" using the small **FMS** Knob, press the **ENT** Key, and enter the reference waypoint identifier, the radial, and the distance into the Reference Waypoints window using the **FMS** Knobs.

Or:

Select "LAT/LON" using the small **FMS** Knob, press the **ENT** Key, and enter the latitude and longitude into the Information window using the **FMS** Knobs.

- 5) Press the **ENT** Key to accept the new waypoint.
- 6) If desired, change the storage method of the waypoint to "Temporary" or "Normal" by moving the cursor to "Temporary" and pressing the **ENT** Key to check or uncheck the box.
- 7) Press the **FMS** Knob to remove the flashing cursor.

CREATING USER WAYPOINTS FROM MAP PAGES

- 1) Press the **Joystick** to activate the panning function and pan to the map location of the desired user waypoint.
- 2) Press the **ENT** Key. The 'WPT - User WPT Information' Page is displayed with the captured position.
- 3) Enter a user waypoint name (up to six characters).
- 4) Press the **ENT** Key to accept the selected name. The first reference waypoint box is highlighted.
- 5) If desired, define the type and location of the waypoint in one of the following ways:

Select "RAD/RAD" using the small **FMS** Knob, press the **ENT** Key, and enter the two reference waypoint identifiers and radials into the Reference Waypoints window using the **FMS** Knobs.

Or:

Select "RAD/DIS" using the small **FMS** Knob, press the **ENT** Key, and enter the reference waypoint identifier, the radial, and the distance into the Reference Waypoints window using the **FMS** Knobs.

Or:

Select "LAT/LON" using the small **FMS** Knob, press the **ENT** Key, and enter the latitude and longitude into the Information window using the **FMS** Knobs.

- 6) Press the **ENT** Key to accept the new waypoint.
- 7) If desired, change the storage method of the waypoint to "Temporary" or "Normal" by moving the cursor to "Temporary" and pressing the **ENT** Key to check or uncheck the box.
- 8) Press the **FMS** Knob to remove the flashing cursor.
- 9) Press the **Go Back** Softkey to return to the map page.

AIRSPACES

CHANGING THE ALTITUDE BUFFER DISTANCE SETTING

- 1) Use the **FMS** Knob to select the 'AUX - System Setup 1' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight the altitude buffer field in the Airspace Alerts Box.
- 4) Use the **FMS** Knob to enter an altitude buffer value and press the **ENT** Key.
- 5) Press the **FMS** Knob to remove the flashing cursor.

TURNING AN AIRSPACE ALERT ON OR OFF

- 1) Use the **FMS** Knob to select the 'Aux - System Setup 1' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight the desired field in the Airspace Alerts Box.
- 4) Turn the small **FMS** Knob clockwise to turn the airspace alert 'On' or counterclockwise to turn the alert 'Off'.
- 5) Press the **FMS** Knob to remove the flashing cursor.

SELECTING AND VIEWING AN AIRSPACE ALERT WITH ITS ASSOCIATED INFORMATION

- 1) Select the 'NRST - Nearest Airspaces' Page.
- 2) Press the **Alerts** Softkey; or press the **FMS** Knob; or press the **MENU** Key, highlight 'Select Alerts Window', and press the **ENT** Key. The cursor is placed in the 'Airspace Alerts' Box.
- 3) Select the desired airspace.
- 4) Press the **FMS** Knob to remove the flashing cursor.

TURNING SMART AIRSPACE ON OR OFF

- 1) Use the **FMS** Knob to select the 'Map - Navigation Map' Page.
- 2) Press the **MENU** Key, and press the **ENT** Key. The cursor is placed in the 'Group' Box.
- 3) Turn the small **FMS** Knob to highlight the 'Airspace' in the 'Group' Box and press the **ENT** Key.
- 4) Turn the large **FMS** Knob to highlight the 'Smart Airspace' Field in the 'Map Settings' Window.

- 5) Turn the small **FMS** Knob clockwise to turn smart airspace 'On' or counterclockwise to turn smart airspace 'Off'.
- 6) Press the **FMS** Knob to remove the flashing cursor.

DIRECT-TO NAVIGATION

ENTERING A WAYPOINT IDENTIFIER, FACILITY NAME, OR CITY AS A DIRECT-TO DESTINATION

- 1) Press the **Direct-to** Key. The 'Direct To' Window is displayed (with the active flight plan waypoint as the default selection or a blank waypoint field if no flight plan is active).
- 2) Turn the small **FMS** Knob clockwise to begin entering a waypoint identifier (turning it counter-clockwise brings up the waypoint selection submenu - press the **CLR** Key to remove it), or turn the large **FMS** Knob to select the facility name, or city Field and turn the small **FMS** Knob to begin entering a facility name or city. If duplicate entries exist for the entered facility or city name, additional entries can be viewed by turning the small **FMS** Knob during the selection process.
- 3) Press the **ENT** Key. The 'Activate?' Field is highlighted.
- 4) Press the **ENT** Key to activate the direct-to.

SELECTING AN ACTIVE FLIGHT PLAN WAYPOINT AS A DIRECT-TO DESTINATION

- 1) While navigating an active flight plan, press the **Direct-to** Key. The 'Direct To' Window is displayed with the active flight plan waypoint as the default selection.
- 2) Turn the small **FMS** Knob counter-clockwise to display a list of flight plan waypoints (the FPL list is populated only when navigating a flight plan).
- 3) Select the desired waypoint.
- 4) Press the **ENT** Key. The cursor is now displayed on 'Activate?'.
- 5) Press the **ENT** Key again to activate the direct-to.

SELECTING A NEAREST, RECENT OR USER WAYPOINT AS A DIRECT-TO DESTINATION

- 1) Press the **Direct-to** Key. The 'Direct To' Window is displayed (with the active flight plan destination as the default selection or a blank destination if no flight plan is active).

- 2) Turn the small **FMS** Knob counter-clockwise to display a list of flight plan waypoints (the FPL list is populated only when navigating a flight plan).
- 3) Turn the small **FMS** Knob clockwise to display the Nearest, Recent or User waypoints.
- 4) Turn the large **FMS** Knob clockwise to select the desired waypoint.
- 5) Press the **ENT** Key. The cursor is now displayed on 'Activate?'.
- 6) Press the **ENT** Key again to activate the direct-to.

CANCELLING A DIRECT-TO

- 1) Press the **Direct-to** Key to display the 'Direct To' Window.
- 2) Press the **MENU** Key.
- 3) With 'Cancel Direct-To NAV' highlighted, press the **ENT** Key. If a flight plan is still active, the system resumes navigating the flight plan along the closest leg.

SELECTING A MANUAL DIRECT-TO COURSE

- 1) Press the **Direct-to** Key. The 'Direct To' Window is displayed with the destination Field highlighted.
- 2) Highlight the 'Course' Field.
- 3) Enter the desired course.
- 4) Press the **ENT** Key. The cursor is now displayed on 'Activate?'.
- 5) Press the **ENT** Key again to activate the direct-to.

ENTERING A VNV ALTITUDE AND ALONG-TRACK OFFSET FOR THE WAYPOINT

- 1) Press the **Direct-to** Key to display the 'Direct To' Window.
- 2) Turn the large **FMS** Knob to place the cursor over the 'VNV' altitude Field.
- 3) Enter the desired altitude.
- 4) Press the **ENT** Key to accept the altitude constraint; if the selected waypoint is an airport, an additional choice is displayed. Turn the small **FMS** Knob to choose 'MSL' or 'AGL', and press the **ENT** Key to accept the altitude.
- 5) Turn the small **FMS** Knob to select 'MSL' or 'AGL'.
- 6) Press the **ENT** Key. The cursor is now flashing in the VNV offset distance Field.
- 7) Enter the desired along-track distance before the waypoint.

- 8) Press the **ENT** Key. The 'Activate?' Field is highlighted.
- 9) Press the **ENT** Key to activate.

REMOVING A VNV ALTITUDE CONSTRAINT

- 1) Press the **Direct-to** Key to display the 'Direct To' Window.
- 2) Press the **MENU** Key.
- 3) With 'Clear Vertical Constraints' highlighted, press the **ENT** Key.

FLIGHT PLANNING

CREATING AN ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key.
- 2) Press the small **FMS** Knob to activate the cursor (only on MFD).
- 3) Select the origin airport and runway.
 - a) Turn the small **FMS** Knob to display the 'Waypoint Information' Window. (Turning it clockwise displays a blank 'Waypoint Information' Window, turning it counter-clockwise displays the 'Waypoint Information' Window with a waypoint selection submenu allowing selection of active flight plan, nearest, recent, or user waypoints).
 - b) Enter the identifier, facility, or city name of the origin airport or select an airport from the submenu of waypoints and press the **ENT** Key. The 'Set Runway' Window is displayed with the Runway field highlighted.
 - c) Turn the small **FMS** Knob to select the runway and press the **ENT** Key.
 - d) Press the **ENT** Key again to add the runway to the flight plan.
- 4) Turn the large **FMS** Knob to highlight the destination airport identifier field.
- 5) Select the destination airport and runway.
 - a) Turn the small **FMS** Knob to display the 'Waypoint Information' Window. (Turning it clockwise displays a blank 'Waypoint Information' Window, turning it counter-clockwise displays the 'Waypoint Information' Window with a waypoint selection submenu allowing selection of active flight plan, nearest, recent, or user waypoints).
 - b) Enter the identifier, facility, or city name of the destination airport or select an airport from the submenu of waypoints and press the **ENT** Key. The 'Set Runway' Window is displayed with the Runway field highlighted.

- c) Turn the small **FMS** Knob to select the runway and press the **ENT** Key.
- d) Press the **ENT** Key again to add the runway to the flight plan.
- 6) Turn the large **FMS** Knob to highlight the Enroute Header, an enroute waypoint, or the dashes below the header. (If the header is selected, the new waypoint is placed following the header. If an enroute waypoint or the dashes are selected, the new waypoint will be placed ahead of the selected item.)
 - a) Turn the small **FMS** Knob to display the 'Waypoint Information' Window. (Turning it clockwise displays a blank 'Waypoint Information' Window, turning it counter-clockwise displays the 'Waypoint Information' Window with a waypoint selection submenu allowing selection of active flight plan, nearest, recent, or user waypoints).
 - b) Enter the identifier, facility, or city name of the departure waypoint or select a waypoint from the submenu of waypoints and press the **ENT** Key. The active flight plan is modified as each waypoint is entered.
- 7) Repeat step number 6 to enter each additional enroute waypoint.
- 8) When all waypoints have been entered, press the **FMS** Knob to remove the cursor.

ADDING A WAYPOINT TO THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key.
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD).
- 3) Select the point in the flight plan to add the new waypoint. The new waypoint is placed prior to the highlighted waypoint.
- 4) Turn the small **FMS** Knob to display the 'Waypoint Information' Window. (Turning it clockwise displays a blank 'Waypoint Information' Window, turning it counter-clockwise displays the 'Waypoint Information' Window with a waypoint selection submenu allowing selection of active flight plan, nearest, recent, user, or airway waypoints).
- 5) Enter the identifier, facility, or city name of the waypoint or select a waypoint from the submenu of waypoints and press the **ENT** Key. The active flight plan is modified as each waypoint is entered.

CREATING AND ADDING USER WAYPOINTS TO THE ACTIVE FLIGHT PLAN

- 1) Push the **Joystick** to activate the panning function on the 'FPL - Active Flight Plan' Page and pan to the map location of the desired user waypoint.

- 2) Select the **LD WPT** Softkey; or press the **MENU** Key, select 'Load Waypoint', and press the **ENT** Key. The user waypoint is created with a name of USRxxx (using the next available in sequence) and is added to the active flight plan.

ADDING AN AIRWAY TO A FLIGHT PLAN

- 1) Press the **FPL** Key.
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD).
- 3) Turn the large **FMS** Knob to highlight the waypoint after the desired airway entry point. If this waypoint is not a valid airway entry point, a valid entry point should be entered at this time.
- 4) Turn the small **FMS** Knob one click clockwise and select the **LD AIRWY** Softkey, or press the **MENU** Key and select 'Load Airway'. The 'FPL - Select Airway' Page is displayed. The **LD AIRWY** Softkey or the 'Load Airway' menu item is available only when a valid airway entry waypoint has been chosen (the waypoint ahead of the cursor position).
- 5) Turn the **FMS** Knob to select the desired airway from the list, and press the **ENT** Key. Low altitude airways are shown first in the list, followed by 'all' altitude airways, and then high altitude airways.
- 6) Turn the **FMS** Knob to select the desired airway exit point from the list, and press the **ENT** Key. 'Load?' is highlighted.
- 7) Press the **ENT** Key. The system returns to editing the flight plan with the new airway inserted.

COLLAPSE/EXPAND AIRWAYS IN A FLIGHT PLAN

- 1) Press the **FPL** Key.
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD).
- 3) Press the **MENU** Key, highlight 'Collapse Airways' or 'Expand Airways' and press the **ENT** Key. All airways are collapsed/expanded.

STORING AN ACTIVE FLIGHT PLAN FROM THE ACTIVE FLIGHT PLAN PAGE OR THE ACTIVE FLIGHT PLAN WINDOW

- 1) Press the **MENU** Key.
- 2) Highlight 'Store Flight Plan'.
- 3) Press the **ENT** Key.
- 4) With 'OK' highlighted, press the **ENT** Key. The flight plan is stored in the next available position in the flight plan list on the 'FPL - Flight Plan Catalog' Page.

ACTIVATING A STORED FLIGHT PLAN ON THE MFD

- 1) Press the **FPL** Key and turn the small **FMS** Knob to display the 'FPL - Flight Plan Catalog' Page.
- 2) Press the **FMS** Knob to activate the cursor, and turn the **FMS** Knob to highlight the desired flight plan.
- 3) Select the **Activate** Softkey; or press the **ENT** Key twice; or press the **MENU** Key, highlight 'Activate Flight Plan', and press the **ENT** Key. The 'activate stored flight plan?' Window is displayed.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.

INVERTING AND ACTIVATING A STORED FLIGHT PLAN ON THE MFD

- 1) Press the **FPL** Key and turn the small **FMS** Knob to display the 'FPL - Flight Plan Catalog' Page.
- 2) Press the **FMS** Knob to activate the cursor, and turn the **FMS** Knob to highlight the desired flight plan.
- 3) Select the **Invert** Softkey; or press the **MENU** Key, highlight 'Invert & Activate FPL?', and press the **ENT** Key. The 'Invert and activate stored flight plan?' Window is displayed.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.

IMPORT A FLIGHT PLAN FROM AN SD CARD

- 1) Insert the SD card containing the flight plan in the top card slot on the MFD.
- 2) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 3) Turn the small **FMS** Knob to select the 'FPL - Flight Plan Catalog' Page.
- 4) Press the **FMS** Knob to activate the cursor.
- 5) Turn either **FMS** Knob to highlight an empty or existing flight plan.
- 6) Select the **Import** Softkey; or press the **MENU** Key, select 'Import Flight Plan', and press the **ENT** Key.

If an empty slot is selected, a list of the available flight plans on the SD card will be displayed.

Or:

If an existing flight plan is selected, an 'Overwrite existing flight plan? OK or CANCEL' prompt is displayed. Press the **ENT** Key to choose to overwrite the selected flight plan and see the list of available flight plans on the SD card. If overwriting the existing flight plan is not desired, select 'CANCEL' using the **FMS** Knob, press the **ENT** Key, select another flight plan slot, and select the **Import** Softkey again.

- 7) Turn the small **FMS** Knob to highlight the desired flight plan for importing.
- 8) Press the **ENT** Key to initiate the import.
- 9) Press the **ENT** Key again to confirm the import.

EXPORT A FLIGHT PLAN TO AN SD CARD

- 1) Insert the SD card into the top card slot on the MFD.
- 2) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 3) Turn the small **FMS** Knob to select the 'FPL - Flight Plan Catalog' Page.
- 4) Press the **FMS** Knob to activate the cursor.
- 5) Turn the large **FMS** Knob to highlight the flight plan to be exported.
- 6) Select the **Export** Softkey; or press the **MENU** Key, select 'Export Flight Plan'.
- 7) If desired, change the name for the exported file by turning the large **FMS** Knob to the left to highlight the name, then use the small and large **FMS** knobs to enter the new name, and press the **ENT** Key.
- 8) Press the **ENT** Key to initiate the export.
- 9) Press the **ENT** Key to confirm the export.

PREVIEWING A PENDING FLIGHT PLAN TRANSFER FROM A WIRELESS CONNECTION

- 1) When a flight plan transfer has been initiated from a mobile device, a Pending Flight Plan pop-up alert appears in the lower right corner of the MFD, and a Connex announcement appears to the right of the MFD page name.
- 2) Press the **ENT** Key to display the 'FPL - Preview Flight Plan' Page on the MFD.

IGNORING A PENDING FLIGHT PLAN TRANSFER FROM A WIRELESS CONNECTION

- 1) When a flight plan transfer has been initiated from a mobile device, a Pending Flight Plan pop-up alert appears in the lower right corner of the MFD, and a Connex announcement appears to the right of the MFD page name.
- 2) Press the **CLR** Key to remove the pop-up alert and ignore the pending flight plan. The pending flight plan will still be available on the 'FPL - Flight Plan Catalog' Page.

Or:

Press the **Ignore** Softkey to remove the pop-up alert and ignore the pending flight plan. The pending flight plan will still be available on the 'FPL - Flight Plan Catalog' Page.

STORING A PENDING FLIGHT PLAN

- 1) Press the **FPL** Key.
- 2) Turn the small **FMS** Knob to display the 'FPL - Flight Plan Catalog' Page.
- 3) Press the **FMS** Knob to activate the cursor.
- 4) Turn the **FMS** Knob to highlight the pending flight plan.
- 5) Press the **ENT** Key to display the 'FPL - Preview Flight Plan' Page on the MFD.
- 6) Press the **Store** Softkey to store the flight plan. The pending flight plan is stored and the pending announcement is removed.

ACTIVATING A PENDING FLIGHT PLAN

- 1) Press the **FPL** Key and turn the small **FMS** Knob to display the 'FPL - Flight Plan Catalog' Page.
- 2) Press the **FMS** Knob to activate the cursor, and turn the **FMS** Knob to highlight the pending flight plan.
- 3) Press the **ENT** Key to display the 'FPL - Preview Flight Plan' Page on the MFD.
- 4) Press the **Activate** Softkey. The 'Activate Flight Plan?' Window is displayed.
- 5) With 'OK' highlighted, press the **ENT** Key to activate the pending flight plan. The pending flight plan becomes the active flight plan and is removed from the 'FPL - Flight Plan Catalog' Page. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.

DELETING A PENDING FLIGHT PLAN

- 1) Press the **FPL** Key.
- 2) Turn the small **FMS** Knob to display the 'FPL - Flight Plan Catalog' Page.
- 3) Press the **FMS** Knob to activate the cursor.
- 4) Turn the **FMS** Knob to highlight the desired pending flight plan.
- 5) Press the **Delete** Softkey. The 'Delete Flight Plan XX?' Window is displayed.
- 6) With 'OK' highlighted, press the **ENT** Key to delete the pending flight plan. The pending flight plan is removed from the 'FPL - Flight Plan Catalog' Page. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.

DELETING ALL PENDING FLIGHT PLANS

- 1) Press the **FPL** Key.
- 2) Turn the small **FMS** Knob to display the 'FPL - Flight Plan Catalog' Page.
- 3) Press the **MENU** Key.
- 4) Turn the **FMS** Knob to highlight 'Delete All Pending'.
- 5) Press the **ENT** Key. A 'Delete all pending flight plans?' confirmation Window is displayed.
- 6) With 'OK' highlighted, press the **ENT** Key to delete all pending flight plans. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.

DELETING THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, highlight 'Delete Flight Plan', and press the **ENT** Key. The 'Delete all waypoints in flight plan?' window is displayed.
- 3) With 'OK' highlighted, press the **ENT** Key to delete the active flight plan. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.

DELETING AN INDIVIDUAL WAYPOINT FROM THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD) and turn the large **FMS** Knob to highlight the waypoint to be deleted.

- 3) Press the **CLR** Key. The 'Remove XXXXX?' window is displayed.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.
- 5) Press the **FMS** Knob to remove the flashing cursor.

DELETING AN ENTIRE AIRWAY FROM THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD) and turn the large **FMS** Knob to highlight the header of the airway to be deleted.
- 3) Press the **CLR** Key. The 'Remove <airway name> from flight plan?' window is displayed.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.
- 5) Press the **FMS** Knob to remove the flashing cursor.

DELETING AN ENTIRE PROCEDURE FROM THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD) and turn the large **FMS** Knob to highlight the header of the procedure to be deleted.
- 3) Press the **CLR** Key. The 'Remove <procedure name> from flight plan?' window is displayed.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.
- 5) Press the **FMS** Knob to remove the flashing cursor.

ENTERING AN ALONG TRACK OFFSET DISTANCE

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD) and turn the large **FMS** Knob to highlight the waypoint for the along track offset.
- 3) Press the **ATK OFS** Softkey (MFD only); or press the **MENU** Key, highlight 'Create ATK Offset Waypoint', and press the **ENT** Key.

- 4) Enter a positive or negative offset distance in the range of +/- 1 to 999 nm (limited by leg distances).
- 5) Press the **ENT** Key to create the offset waypoint.
- 6) Press the **FMS** Knob to remove the flashing cursor.

ACTIVATING PARALLEL TRACK

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, highlight 'Parallel Track', and press the **ENT** Key. The 'Parallel Track' Window is displayed with the 'Direction' field highlighted.
- 3) Turn the small **FMS** Knob to select 'LEFT' or 'RIGHT' and press the **ENT** Key. The 'Distance' field is highlighted.
- 4) Turn the small **FMS** Knob to enter a distance from 1-99 nm and press the **ENT** Key. 'Activate Parallel Track' is highlighted.
- 5) Press the **ENT** Key to activate parallel track. Press the **FMS** Knob or the **CLR** Key to cancel the parallel track activation.

CANCELLING PARALLEL TRACK

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, highlight 'Parallel Track', and press the **ENT** Key. The 'Parallel Track' Window is displayed with 'Cancel Parallel Track?' highlighted.
- 3) Press the **ENT** Key.

ACTIVATING A FLIGHT PLAN LEG

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD) and turn the large **FMS** Knob to highlight the destination waypoint for the desired leg.
- 3) Select the **ACT Leg** Softkey (MFD only); or press the **MENU** Key, highlight 'Activate Leg', and press the **ENT** Key. A confirmation Window is displayed with 'Activate' highlighted.
- 4) Press the **ENT** Key to activate the flight plan leg. To cancel, press the **CLR** Key, or highlight 'Cancel' and press the **ENT** Key.
- 5) Press the **FMS** Knob to remove the flashing cursor.

INVERTING THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, highlight 'Invert Flight Plan', and press the **ENT** Key. An 'Invert active flight plan?' confirmation Window is displayed.
- 3) Select 'OK'.
- 4) Press the **ENT** Key to invert and activate the active flight plan. To cancel, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.

INVERTING AND ACTIVATING A STORED FLIGHT PLAN

- 1) Press the **FPL** Key and turn the small **FMS** Knob to display the 'FPL - Flight Plan Catalog' Page.
- 2) Press the **FMS** Knob to activate the cursor, and turn the **FMS** Knob to highlight the desired flight plan.
- 3) Press the **Invert** Softkey; or press the **MENU** Key, highlight 'Invert & Activate FPL?', and press the **ENT** Key. The 'Invert and activate stored flight plan?' window is displayed.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the request, press the **CLR** Key, or highlight 'CANCEL' and press the **ENT** Key.

SWITCHING BETWEEN LEG-TO-LEG WAYPOINT DISTANCE AND CUMULATIVE WAYPOINT DISTANCE

- 1) Press the **FPL** Key on the MFD to display the 'FPL - Active Flight Plan' Page.
- 2) Press the **View** Softkey to display the **CUM** and **Leg-Leg** Softkeys.
- 3) Press the **CUM** Softkey to view cumulative waypoint distance, or press the **Leg-Leg** Softkey to view leg-to-leg waypoint distance.
- 4) Press the **Back** Softkey to return to the top level active flight plan softkeys.

SWITCHING BETWEEN WIDE AND NARROW VIEW

- 1) Press the **FPL** Key on the MFD to display the 'FPL - Active Flight Plan' Page.
- 2) Press the **View** Softkey to display the **Wide** and **Narrow** Softkeys.
- 3) Press the **Wide** Softkey to display the wide view, or press the **Narrow** Softkey to display the narrow view.
- 4) Press the **Back** Softkey to return to the top level active flight plan softkeys.

DETERMINING THE CLOSEST POINT ALONG THE ACTIVE FLIGHT PLAN TO A SELECTED WAYPOINT

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, highlight 'Closest Point Of FPL', and press the **ENT** Key. A window appears with a highlighted field within the 'From' Box.
- 3) Enter the identifier of the reference waypoint. The system displays the bearing (BRG) and distance (DIS) to the closest point along the flight plan to the selected reference waypoint and creates a user waypoint at this location. Press the **ENT** Key. The name for the new user waypoint is derived from the identifier of the reference waypoint.

CREATING A USER-DEFINED HOLD AT AN ACTIVE FLIGHT PLAN WAYPOINT

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD) and turn the large **FMS** Knob to highlight the waypoint for the hold.
- 3) Press the **MENU** Key, highlight 'Hold At Waypoint', and press the **ENT** Key. The 'Hold at' Window appears with the 'Course' Field highlighted.
- 4) Use the **FMS** Knobs to edit the entry course, and press the **ENT** Key.
- 5) Use the small **FMS** Knob to select 'Inbound' or 'Outbound' course direction, and press the **ENT** Key.
- 6) Use the small **FMS** Knob to select 'Time' or 'Distance' length mode, and press the **ENT** Key.
- 7) Use the **FMS** Knobs to edit the length, and press the **ENT** Key.
- 8) Use the small **FMS** Knob to select 'Right' or 'Left' turn direction, and press the **ENT** Key.
- 9) Use the **FMS** Knobs to edit the Expect Further Clearance Time (EFC Time), and press the **ENT** Key.
- 10) Press the **ENT** Key while 'Load?' is highlighted to insert the hold into the flight plan.

CREATING A USER-DEFINED HOLD AT THE AIRCRAFT PRESENT POSITION

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, highlight 'Hold At Present Position', and press the **ENT** Key. The 'Hold at' Window appears with the 'Course' Field highlighted.
- 3) If desired, use the **FMS** Knobs to edit the entry course, and press the **ENT** Key.
- 4) Use the small **FMS** Knob to select 'Inbound' or 'Outbound' course direction, and press the **ENT** Key.
- 5) Use the small **FMS** Knob to select 'Time' or 'Distance' length mode, and press the **ENT** Key.
- 6) Use the **FMS** Knobs to edit the length, and press the **ENT** Key.
- 7) Use the small **FMS** Knob to select 'Right' or 'Left' turn direction, and press the **ENT** Key.
- 8) Use the **FMS** Knobs to edit the Expect Further Clearance Time (EFC Time), and press the **ENT** Key.
- 9) Press the **ENT** Key while 'Activate?' is highlighted to create an Offroute Direct-to hold waypoint at the aircraft present position and activate the hold.

CREATING A USER-DEFINED HOLD AT A DIRECT-TO WAYPOINT

- 1) Press a **Direct-to** Key and set up the direct-to waypoint as desired, but select 'Hold?' instead of 'Activate?' when finished (MFD or PFD).
- 2) Use the **FMS** Knobs to edit the entry course, and press the **ENT** Key.
- 3) Use the small **FMS** Knob to select 'Inbound' or 'Outbound' course direction, and press the **ENT** Key.
- 4) Use the small **FMS** Knob to select 'Time' or 'Distance' length mode, and press the **ENT** Key.
- 5) Use the **FMS** Knobs to edit the length, and press the **ENT** Key.
- 6) Use the small **FMS** Knob to select 'Right' or 'Left' turn direction, and press the **ENT** Key.
- 7) Use the **FMS** Knobs to edit the Expect Further Clearance Time (EFC TIME), and press the **ENT** Key.

- 8) Press the **ENT** Key while 'Activate?' is highlighted to activate the direct-to with the user-defined hold defined at the direct-to waypoint. (If the direct-to waypoint is part of the active flight plan, the HOLD is inserted into the active flight plan. If the direct-to waypoint is not part of the active flight plan, an off-route direct-to hold is created.)

EXITING A USER-DEFINED HOLD INSERTED INTO THE ACTIVE FLIGHT PLAN

Press the **SUSP** Softkey. The system will provide guidance to follow the holding pattern to the inbound course and resume automatic waypoint sequencing.

REMOVING A USER-DEFINED HOLD INSERTED INTO THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **FMS** Knob to activate the cursor (not required on the PFD), and turn the large **FMS** Knob to highlight the HOLD waypoint.
- 3) Press the **CLR** Key. A 'Remove Holding Pattern?' confirmation Window is displayed.
- 4) Select 'OK' and press the **ENT** Key. The holding pattern is removed from the active flight plan. Select 'CANCEL' and press the **ENT** Key to cancel the removal of the holding pattern.

REMOVING A USER-DEFINED HOLD AT AN OFF-ROUTE DIRECT-TO

- 1) Press a **Direct To** Key to display the Direct To Window (PFD or MFD).
- 2) Press the **MENU** Key to display the Page Menu with the cursor on the 'Cancel Direct-To NAV' selection.
- 3) Press the **ENT** Key. The holding pattern is removed.

VERTICAL NAVIGATION

ENABLING VNV GUIDANCE

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **ENBL VNV** Softkey; or press the **MENU** Key, highlight 'Enable VNV', and press the **ENT** Key. Vertical navigation is enabled, and vertical guidance begins with the waypoint shown in the 'Active VNV Profile' Box (defaults first waypoint in the active flight plan with an altitude enabled for vertical navigation).

DISABLING VNV GUIDANCE

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **Cncl VNV** Softkey; or press the **MENU** Key, highlight 'Cancel VNV', and press the **ENT** Key. Vertical navigation is disabled.

ACTIVATING A VERTICAL NAVIGATION DIRECT-TO

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **FMS** Knob to activate the cursor and turn the **FMS** Knob to highlight the desired waypoint.
- 3) Press the **VNV Direct-To** Softkey; or press the **MENU** Key, highlight 'VNV Direct-To', and press the **ENT** Key. An 'Activate vertical Direct-to to: NNNNNFT at XXXXXX?' confirmation Window is displayed.
- 4) Press the **ENT** Key. Vertical guidance begins to the altitude constraint for the selected waypoint.
- 5) Press the **FMS** Knob to remove the flashing cursor.



NOTE: The selected waypoint must have a designated altitude constraint (cyan number) to be used. If not, the first waypoint in the flight plan with a designated altitude constraint is selected.

MODIFYING THE VS TGT AND FPA

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **VNV Prof** Softkey; or press the **MENU** Key, highlight 'Select VNV Profile Window', and press the **ENT** Key. The cursor is now located in the 'Active VNV Profile' Box.
- 3) Turn the **FMS** Knobs as needed to edit the values.
- 4) Press the **FMS** Knob to remove the flashing cursor.

ALTITUDE CONSTRAINTS

The system can use altitude constraints associated with lateral waypoints to give guidance for vertical navigation. These altitudes are, depending on the specific instance, manually entered or retrieved from the published altitudes in the navigation database.

Active Flight Plan
KMKC / KCOS

	DTK	DIS	ALT
BRK	353°	21.5NM	12000FT
KCOS-RNAV _{GPS} Y 35R LPV			
HABUK iaf	164°	8.0NM	10280FT
FALUR	290°	5.3NM	8600FT
CEGIX faf	352°	6.4NM	8100FT
RW35R map	352°	6.1NM	
6600FT	352°	0.9NM	6600FT
ADANE mahp	056°	15.6NM	9000FT
HOLD	297°	7.0NM	

Displayed Text Examples

White Text

Cyan Text with Pencil Icon

Cyan Text

White Text with Altitude Restriction Bar

Altitude Constraint Examples

2300FT

Cross AT 2,300 ft

5000FT

Cross AT or ABOVE 5,000 ft

3000FT

Cross AT or BELOW 3,000 ft

6646FT

Temperature Compensated

White Text	Cyan Text
5000FT Altitude calculated by the system estimating the altitude of the aircraft as it passes over the navigation point. No white line above or below to indicate a potential constraint. 5000FT Altitude retrieved from the navigation database. White line above or below indicates the type of constraint, as shown in the preceding figure. These altitudes are provided as a reference, and are not designated to be used in determining vertical guidance.	5000FT Altitude is designated for use in determining vertical guidance. A pencil icon indicates manual designation or manual data entry. 5000FT The system cannot use this altitude in determining vertical guidance because of an invalid constraint condition.

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCs
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

Altitudes associated with arrival and approach procedures are “auto-designated”. This means the system automatically uses the altitudes loaded with the arrival or approach for giving vertical speed and deviation guidance. Note that these altitudes are displayed as cyan text up to, but not including, the FAF. The FAF is always a “reference only” altitude and cannot be designated, unless the selected approach does not provide vertical guidance. In this case, the FAF altitude can be designated.

Altitudes that have been designated for use in vertical guidance can be “un-designated” using the **CLR** Key. The altitude is now displayed only as a reference. It is not used to give vertical guidance. Other displayed altitudes may change due to re-calculations or be rendered invalid as a result of manually changing an altitude to a non-designated altitude.

DESIGNATING A WAYPOINT ALTITUDE TO BE USED FOR VERTICAL GUIDANCE

- 1) Press the **FPL** Key to display the ‘FPL - Active Flight Plan’ Page on the MFD.
- 2) Press the **FMS** Knob, and turn to highlight the desired waypoint altitude.
- 3) Turn the small **FMS** Knob to enter editing mode.
- 4) Press the **ENT** Key. The altitude is now shown in cyan, indicating it is usable for vertical guidance.

ENTERING/MODIFYING AN ALTITUDE CONSTRAINT

- 1) Press the **FPL** Key to display the ‘FPL - Active Flight Plan’ Page on the MFD.
- 2) Press the **FMS** Knob, and turn to highlight the desired waypoint altitude constraint.
- 3) Enter an altitude constraint value using the **FMS** Knobs. To enter altitudes as a flight level, turn the small **FMS** Knob counter-clockwise past zero or clockwise past 9 on the first character, and the system automatically changes to show units of Flight Level. Turn the large **FMS** Knob clockwise to highlight the first zero and enter the three digit Flight Level.
- 4) Press the **ENT** Key to accept the altitude constraint; if the selected waypoint is an airport, an additional choice is displayed. Turn the small **FMS** Knob to choose ‘MSL’ or ‘AGL’, and press the **ENT** Key to accept the altitude.

An altitude constraint is invalid if:

- Meeting the constraint requires the aircraft to climb
- Meeting the constraint requires the maximum flight path angle or maximum vertical speed to be exceeded

- The altitude constraint results in a TOD behind the aircraft present position
- The constraint is within a leg type for which altitude constraints are not supported
- The altitude constraint is added to the FAF of an approach that provides vertical guidance (i.e., ILS or GPS SBAS approach)
- The altitude constraint is added to a waypoint past the FAF

DELETING AN ALTITUDE CONSTRAINT PROVIDED BY THE NAVIGATION DATABASE

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **FMS** Knob, and turn to highlight the desired waypoint altitude constraint.
- 3) Press the **CLR** Key. A 'Remove VNV altitude?' confirmation Window is displayed.
- 4) Select 'OK' and press the **ENT** Key.

DELETING AN ALTITUDE CONSTRAINT THAT HAS BEEN MANUALLY ENTERED

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **FMS** Knob, and turn to highlight the desired waypoint altitude constraint.
- 3) Press the **CLR** Key. A 'Remove or Revert to published VNV altitude of nnnnnFT?' confirmation Window is displayed.
- 4) Select 'REMOVE' and press the **ENT** Key. The manually entered altitude is deleted (it is replaced by a system calculated altitude, if available).

REVERTING A MANUALLY ENTERED ALTITUDE CONSTRAINT BACK TO THE NAVIGATION DATABASE VALUE

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **FMS** Knob, and turn to highlight the desired waypoint altitude constraint.
- 3) Press the **CLR** Key. A 'Remove or Revert to published VNV altitude of nnnnnFT?' confirmation Window is displayed.
- 4) Select 'REVERT' and press the **ENT** Key. The altitude is changed to the navigation database value.
- 5) Press the **FMS** Knob to remove the flashing cursor.

MODIFYING A SYSTEM CALCULATED ALTITUDE CONSTRAINT

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **FMS** Knob, and turn to highlight the desired waypoint altitude constraint.
- 3) Press the **CLR** Key. An 'Edit or Revert to published VNV altitude of nnnnnFT?' confirmation Window is displayed.
- 4) Select 'EDIT' and press the **ENT** Key.
- 5) Edit the value using the **FMS** Knobs, and press the **ENT** Key.
- 6) Press the **FMS** Knob to remove the flashing cursor.

ENABLING THE VERTICAL SITUATION DISPLAY

- 1) Select the 'Map - Navigation Map' Page.
- 2) Select the **Map Opt** Softkey.
- 3) Press the **Inset** Softkey.
- 4) Press the **VSD** Softkey to enable the Vertical Situation Display.
- 5) Press the **VSD** Mode Softkey to choose between **Auto**, **FPL**, or **TRK**.

DISABLING THE VERTICAL SITUATION DISPLAY

- 1) Select the 'Map - Navigation Map' Page.
- 2) Press the **Map Opt** Softkey.
- 3) Press the **Off** Softkey.

PROCEDURES

VIEWING AVAILABLE PROCEDURES AT AN AIRPORT

- 1) From the 'WPT - Airport Information' Page (first page in the WPT group):
Press the **DP** Softkey. The 'WPT - Departure Information' Page is displayed, defaulting to the airport displayed on the 'WPT - Airport Information' Page.
Or.
Press the **STAR** Softkey. The 'WPT - Arrival Information' Page is displayed, defaulting to the airport displayed on the 'WPT - Airport Information' Page.
Or.
Press the **APR** Softkey. The 'WPT - Approach Information' Page is displayed, defaulting to the airport displayed on the 'WPT - Airport Information' Page.

- 2) To select another airport, press the **FMS** Knob to activate the cursor, enter an identifier/facility name/city, and press the **ENT** Key.
- 3) Turn the large **FMS** Knob to highlight the procedure. The procedure is previewed on the map.
- 4) Turn the small **FMS** Knob to view the available procedures. Press the **ENT** Key to select the procedure. The cursor moves to the next box (runway or transition). The procedure is previewed on the map.
- 5) Turn the small **FMS** Knob to view the available runway or transition. Press the **ENT** Key to select the runway or transition. The cursor moves to the next box (if available). The procedure is previewed on the map.
- 6) Turn the small **FMS** Knob to view the available runway or transition. Press the **ENT** Key to select the runway or transition. The cursor moves to the 'Sequence' Box or the 'Minimums' Box. The procedure is previewed on the map.
- 7) Press the **Info 1** Softkey or the **Info 2** Softkey to return to the 'WPT - Airport Information' Page.

LOADING A PROCEDURE INTO THE ACTIVE FLIGHT PLAN FROM THE <PROCEDURE> INFORMATION PAGE

- 1) From the 'WPT - Airport Information' Page (first page in the WPT group):
Press the **DP** Softkey. The 'WPT - Departure Information' Page is displayed, defaulting to the airport displayed on the 'WPT - Airport Information' Page.
Or.
Press the **STAR** Softkey. The 'WPT - Arrival Information' Page is displayed, defaulting to the airport displayed on the 'WPT - Airport Information' Page.
Or.
Press the **APR** Softkey. The 'WPT - Approach Information' Page is displayed, defaulting to the airport displayed on the 'WPT - Airport Information' Page.
- 2) To select another airport, press the **FMS** Knob to activate the cursor, enter an identifier/facility name/city, and press the **ENT** Key.
- 3) Select a different procedure, if desired.
 - a) Turn the large **FMS** Knob to highlight the procedure. The procedure is previewed on the map.

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

- b) Turn the small **FMS** Knob to view the available procedures. Press the **ENT** Key to select the procedure. The cursor moves to the next box (runway or transition). The procedure is previewed on the map.
 - c) Turn the small **FMS** Knob to view the available runway or transition. Press the **ENT** Key to select the runway or transition. The cursor moves to the next box (if available). The procedure is previewed on the map.
 - d) Turn the small **FMS** Knob to view the available transitions. Press the **ENT** Key to select the transition. The cursor moves to the 'Sequence' Box or the 'Minimums' Box. The procedure is previewed on the map.
- 4) Press the **MENU** Key to display the 'WPT - <Procedure> Information' 'Page Menu'.
 - 5) Turn the **FMS** Knob to highlight 'Load <procedure>'.
 - 6) Press the **ENT** Key to load the procedure into the active flight plan.

LOADING A DEPARTURE INTO THE ACTIVE FLIGHT PLAN USING THE PROC KEY

- 1) Press the **PROC** Key. The 'Procedures' Window is displayed.
- 2) Highlight 'Select Departure'.
- 3) Press the **ENT** Key. The 'PROC - Departure Loading' Page is displayed.
- 4) Use the **FMS** Knob to select an airport, if necessary, and press the **ENT** Key.
- 5) Select a departure from the list and press the **ENT** Key.
- 6) Select a runway (if required) and press the **ENT** Key.
- 7) Select a transition (if required) and press the **ENT** Key. 'Load?' is highlighted.
- 8) Press the **ENT** Key to load the departure procedure.

REMOVING A DEPARTURE PROCEDURE FROM THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the active 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, and highlight 'Remove Departure'.
- 3) Press the **ENT** Key. A confirmation window is displayed listing the departure procedure.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the removal request, highlight 'CANCEL' and press the **ENT** Key.

LOADING AN ARRIVAL INTO THE ACTIVE FLIGHT PLAN USING THE PROC KEY

- 1) Press the **PROC** Key. The 'Procedures' Window is displayed.
- 2) Highlight 'Select Arrival'.
- 3) Press the **ENT** Key. The 'PROC - Arrival Loading' Page is displayed.
- 4) Use the **FMS** Knob to select an airport, if necessary, and press the **ENT** Key.
- 5) Select an arrival from the list and press the **ENT** Key.
- 6) Select a transition (if required) and press the **ENT** Key.
- 7) Select a runway (if required) and press the **ENT** Key. 'Load?' is highlighted.
- 8) Press the **ENT** Key to load the arrival procedure.

REMOVING AN ARRIVAL FROM THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the active 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, and highlight 'Remove Arrival'.
- 3) Press the **ENT** Key. A confirmation window is displayed listing the arrival procedure.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the removal request, highlight 'CANCEL' and press the **ENT** Key.

HSI Annunciation	Description
LNAV	RNAV GPS approach using published LNAV minima
LNAV+V	RNAV GPS approach using published LNAV minima. Advisory vertical guidance is provided
L/VNAV (available only if SBAS available)	RNAV GPS approach using published LNAV/VNAV minima (downgrades to LNAV if SBAS unavailable)
LP (available only if SBAS available)	RNAV GPS approach using published LP minima (downgrades to LNAV if SBAS unavailable)
LP+V (available only if SBAS available)	RNAV GPS approach using published LP minima Advisory vertical guidance is provided (downgrades to LNAV if SBAS unavailable)
LPV (available only if SBAS available)	RNAV GPS approach using published LPV minima (downgrades to LNAV if SBAS unavailable)

Approach Service Levels

LOADING AN APPROACH INTO THE ACTIVE FLIGHT PLAN USING THE PROC KEY

- 1) Press the **PROC** Key. The 'Procedures' Window is displayed.
- 2) Highlight 'Select Approach', and press the **ENT** Key. The 'PROC - Approach Loading' Page is displayed.
- 3) Select the airport and approach:
 - a) Use the **FMS** Knob to select an airport, if necessary, and press the **ENT** Key.
 - b) Select an approach from the list and press the **ENT** Key.

Or:

 - a) If necessary, push the **FMS** Knob to exit the approach list, and use the large **FMS** Knob to move the cursor to the 'Approach Channel' Field.
 - b) Use the **FMS** Knob to enter the approach channel number, and press the **ENT** Key to accept the approach channel number. The airport and approach are selected.
- 4) Select a transition (if required) and press the **ENT** Key.
- 5) Minimums
 - a) To set 'Minimums', turn the small **FMS** Knob to select 'BARO' or 'TEMP COMP', and press the **ENT** Key. Turn the small **FMS** Knob to select the altitude, and press the **ENT** Key.
 - b) If 'TEMP COMP' was selected, the cursor moves to the 'TEMP At XXXX' Field. Turn the small **FMS** Knob to select the temperature, and press the **ENT** Key.

Or:

To skip setting minimums, press the **ENT** Key.
- 6) Press the **ENT** Key with 'Load?' highlighted to load the approach procedure; or turn the large **FMS** Knob to highlight 'Activate' and press the **ENT** Key to load and activate the approach procedure.

ACTIVATING A PREVIOUSLY LOADED APPROACH

- 1) Press the **PROC** Key. The 'Procedures' Window is displayed with 'Activate Approach' highlighted.
- 2) Press the **ENT** Key to activate the approach.

ACTIVATING A PREVIOUSLY LOADED APPROACH WITH VECTORS TO FINAL

- 1) Press the **PROC** Key to display the 'Procedures' Window.
- 2) Highlight 'Activate Vector-to-Final' and press the **ENT** Key.

REMOVING AN APPROACH FROM THE ACTIVE FLIGHT PLAN

- 1) Press the **FPL** Key to display the 'FPL - Active Flight Plan' Page (MFD) or the 'Flight Plan' Window (PFD).
- 2) Press the **MENU** Key, and highlight 'Remove Approach'.
- 3) Press the **ENT** Key. A confirmation window is displayed listing the approach procedure.
- 4) With 'OK' highlighted, press the **ENT** Key. To cancel the removal, highlight 'CANCEL' and press the **ENT** Key.

ACTIVATING A MISSED APPROACH IN THE ACTIVE FLIGHT PLAN

- 1) Press the **PROC** Key.
- 2) Turn the **FMS** Knob to highlight 'Activate Missed Approach'.
- 3) Press the **ENT** Key. The aircraft automatically sequences to the MAHP.

Or:

Press the Go-Around Button.

ACTIVATING TEMPERATURE COMPENSATED ALTITUDE

- 1) From the 'FPL - Active Flight Plan' Page, press the **MENU** Key. The 'Page Menu' is displayed.
- 2) Turn the **FMS** Knob to highlight 'Temperature Compensation'.
- 3) Press the **ENT** Key. The 'Temperature Compensation' Window is displayed.
- 4) Use the small **FMS** Knob to select the temperature at the <airport>. The compensated altitude is computed as the temperature is selected.
- 5) Press the **ENT** Key. 'Activate Compensation?' is highlighted.
- 6) Press the **ENT** Key. The compensated altitudes for the approach are shown in the flight plan.

CANCELLING TEMPERATURE COMPENSATED ALTITUDE

- 1) From the 'FPL - Active Flight Plan' Page, press the **MENU** Key. The 'Page Menu' is displayed.
- 2) Turn the **FMS** Knob to highlight 'Temperature Compensation'.
- 3) Press the **ENT** Key. The 'Temperature Compensation' Window is displayed.

- 4) Press the **ENT** Key. 'Cancel Compensation?' is highlighted.
- 5) Press the **ENT** Key. The temperature compensated altitude at the FAF is cancelled.

TRIP PLANNING

SELECTING AUTOMATIC OR MANUAL PAGE MODE

Press the **Auto** Softkey or the **Manual** Softkey; or press the **MENU** Key, highlight 'Auto Mode' or 'Manual Mode', and press the **ENT** Key.

SELECTING FLIGHT PLAN OR WAYPOINT MODE

Press the **FPL** Softkey or the **WPTs** Softkey; or press the **MENU** Key, highlight 'Flight Plan Mode' or 'Waypoints Mode', and press the **ENT** Key.

SELECTING A FLIGHT PLAN AND LEG FOR TRIP STATISTICS

- 1) Press the **FMS** Knob to activate the cursor in the 'FPL' (flight plan number) Field.
- 2) Turn the small **FMS** Knob to select the desired flight plan number.
- 3) Turn the large **FMS** Knob to highlight 'CUM' or 'REM'. The statistics for each leg can be viewed by turning the small **FMS** Knob to select the desired leg. The Inset Map also displays the selected data.

SELECTING WAYPOINTS FOR WAYPOINT MODE

- 1) Press the **WPTs** Softkey; or press the **MENU** Key, highlight 'Waypoints Mode', and press the **ENT** Key. The cursor is positioned in the waypoint Field directly below the 'FPL' Field.
- 2) Turn the **FMS** knobs to select the desired waypoint (or select from the 'Page Menu' 'Set WPT to Present Position' if that is what is desired), and press the **ENT** Key. The cursor moves to the second waypoint Field.
- 3) Turn the **FMS** knobs to select the desired waypoint, and press the **ENT** Key. The statistics for the selected leg are displayed.

ENTERING MANUAL DATA FOR TRIP STATISTICS CALCULATIONS

- 1) Press the **Manual** Softkey or select 'Manual Mode' from the Page Menu, and press the **ENT** Key. The cursor may now be positioned in any field in the top right two boxes.
- 2) Turn the **FMS** Knobs to move the cursor onto the Departure Time field and enter the desired value. Press the **ENT** Key. The statistics are calculated using the new value and the cursor moves to the next entry field. Repeat until all desired values have been entered.

RAIM PREDICTION

PREDICTING RAIM AVAILABILITY AT A SELECTED WAYPOINT

- 1) Select the 'Aux - GPS Status' Page.
- 2) If necessary, press the **RAIM** Softkey.
- 3) Press the **FMS** Knob. The 'Waypoint' Field is highlighted.
- 4) Turn the small **FMS** Knob to display the 'Waypoint Information' Window.
- 5) Enter the desired waypoint:

Use the **FMS** Knob to enter the desired waypoint by identifier, facility, or city name and press the **ENT** Key.

Or:

- a) Turn the small **FMS** Knob counter-clockwise to display a list of flight plan waypoints (the FPL list is populated only when navigating a flight plan).
 - b) Turn the small **FMS** Knob clockwise to display the Flight Plan, Nearest, Recent, or User waypoints, if required.
 - c) Turn the large **FMS** Knob clockwise to select the desired waypoint. The system automatically fills in the identifier, facility, and city fields with the information for the selected waypoint.
 - d) Press the **ENT** Key to accept the waypoint entry.
- 6) Use the **FMS** Knob to enter an arrival time and press the **ENT** Key.
 - 7) Use the **FMS** Knob to enter an arrival date and press the **ENT** Key.
 - 8) With the cursor highlighting 'Compute RAIM?', press the **ENT** Key. Once RAIM availability is computed, one of the following is displayed:
 - 'Compute RAIM?'—RAIM has not been computed for the current waypoint, time, and date combination
 - 'Computing Availability'—RAIM calculation in progress
 - 'RAIM Available'—RAIM is predicted to be available for the specified waypoint, time, and date
 - 'RAIM not Available'—RAIM is predicted to be unavailable for the specified waypoint, time, and date

PREDICTING RAIM AVAILABILITY AT THE AIRCRAFT PRESENT POSITION

- 1) Select the 'Aux - GPS Status' Page.
- 2) If necessary, press the **RAIM** Softkey.
- 3) Press the **FMS** Knob. The 'Waypoint' Field is highlighted.
- 4) Press the **MENU** Key.
- 5) With 'Set WPT to Present Position' highlighted, press the **ENT** Key.
- 6) Press the **ENT** Key to accept the waypoint entry.
- 7) Use the **FMS** Knob to enter an arrival time and press the **ENT** Key.
- 8) Use the **FMS** Knob to enter an arrival date and press the **ENT** Key.
- 9) With the cursor highlighting 'Compute RAIM?', press the **ENT** Key. Once RAIM availability is computed, one of the following is displayed:
 - 'Compute RAIM?'—RAIM has not been computed for the current waypoint, time, and date combination
 - 'Computing Availability'—RAIM calculation in progress
 - 'RAIM Available'—RAIM is predicted to be available for the specified waypoint, time, and date
 - 'RAIM not Available'—RAIM is predicted to be unavailable for the specified waypoint, time, and date

DISABLING SBAS

- 1) Select the 'Aux - GPS Status' Page.
- 2) If necessary, press the **SBAS** Softkey.
- 3) Press the **FMS** Knob, and turn the large FMS Knob to highlight 'EGNOS', 'MSAS' or 'WAAS'.
- 4) Press the **ENT** Key to uncheck the box.
- 5) Press the **FMS** Knob to remove the cursor.

HAZARD AVOIDANCE

Customizing the Hazard Displays on the 'Map - Navigation Map' Page

- 1) With the 'Map - Navigation Map' Page displayed, press the **MENU** Key to display the Navigation Map 'Page Menu'. The cursor flashes on the 'Map Settings' option.
- 2) Press the **ENT** Key. The Map Setup Menu is displayed. Turn the small **FMS** Knob to select 'Weather' to customize the display of weather features. Select 'Traffic' to customize the display of traffic.
- 3) Press the small **FMS** Knob to return to the Navigation Map Page.

Changing the display range of hazard displays

Turn the **Joystick** clockwise to display a larger area or rotate counter-clockwise to display a smaller area.

DATA LINK WEATHER



WARNING: Do not use data link weather information for maneuvering in, near, or around areas of hazardous weather. Information contained within data link weather products may not accurately depict current weather conditions.



WARNING: Do not use the indicated data link weather product age to determine the age of the weather information shown by the data link weather product. Due to time delays inherent in gathering and processing weather data for data link transmission, the weather information shown by the data link weather product may be older than the indicated weather product age.

Enabling/disabling the weather product age for PFD Maps

- 1) With the PFD Inset Map or HSI Map displayed, press the **Map/HSI** Softkey.
- 2) Press the **WX LGND** to show/remove the weather product age information for PFD maps.

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCs
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

SiriusXM Weather Product	Symbol	Expiration Time (Minutes)
NEXRAD		30
Cloud Top (CLD TOP)		60
Echo Top (ECHO TOP)		30
SiriusXM Lightning (LTNG)		30
Cell Movement		30
SIGMETs/AIRMETs		60
METARs		90
City Forecast		90
Surface Analysis		60
Freezing Levels		120
Winds Aloft		90
County Warnings		60
Cyclone Warnings		60
Icing Potential (CIP and SLD)		90

SiriusXM Weather Product Symbols and Data Timing

SiriusXM Weather Product	Symbol	Expiration Time (Minutes)
Pilot Weather Report (PIREP)		90
Air Report(AIREP)		90
Turbulence		180
Radar Coverage Not Available	No product image	30
Temporary Flight Restrictions (TFRs)	TFR	60
Terminal Aerodrome Forecast (TAF)	No product image	60

SiriusXM Weather Product Symbols and Data Timing

FIS-B Weather Product	Symbol	Expiration Time (Minutes)	Broadcast Rate (Minutes)
NEXRAD Composite (US)		30	15
NEXRAD Composite (Regional)		30	2.5
METARs		90	5
Pilot Weather Report (PIREP)		90	10
Winds Aloft		90	10
SIGMETs/AIRMETs	SIGM AIRM	60	5
No Radar Coverage	no product image	30	2.5
Terminal Aerodrome Forecast (TAF)	no product image	60	10
Temporary Flight Restrictions (TFRs)	TFR		10

FIS-B Weather Product Symbols and Data Timing

Flight Instruments	Garmin Connex Weather Product	Symbol	Expiration Time (Minutes)	Refresh Rate (Minutes)
EIS	Radar Precipitation		30	U.S.: 3* Canada: 3*† Europe: 15 Australia: 15^
Nav/Com/XPDR/Audio	Infrared Satellite		60	30
Flight Management	Datalink Lightning		30	Continuous
	SIGMETs/AIRMETs		60	Continuous
Hazard Avoidance	METARs		90	Continuous
	Winds Aloft		90	Continuous
AFCs	Pilot Weather Report (PIREPs)		90	Continuous
Additional Features	Temporary Flight Restrictions (TFRs)		60	Continuous
	Terminal Aerodrome Reports TAFs)	no product image	60	Continuous

* The composite precipitation image is updated every 3 minutes, but individual radar sites may take between 3 and 10 minutes to provide new data.

† Canadian radar precipitation data provided by Environment Canada.

^ Australian radar precipitation data provided by the Australian Bureau of Meteorology.

Garmin Connex Weather Product Symbols and Data Timing

Displaying Data Link Weather Products

Viewing the Weather Data Link page and changing the data link weather source, if applicable

- 1) Turn the large **FMS** Knob to select the Map Page Group.
- 2) Turn the small **FMS** Knob to select the Weather Data Link (XM or CNXT or FIS-B) Page. The currently selected data link weather source appears in the page title.

- 3) If the page title does not contain the desired weather source, press the **MENU** Key.
 - a) Turn the **FMS** Knob to highlight 'Display XM Weather', 'Display Connex Weather' or 'Display FIS-B Weather' (choices may vary depending on the installed equipment).
 - b) Press the **ENT** Key.

Viewing legends for displayed weather products on the Weather Data Link page

- 1) Select the Weather Data Link Page.
- 2) Press the **Legend** Softkey to display the legends for the displayed weather products.
- 3) Turn the **FMS** Knob to scroll through the legends if more are available than fit in the window.
- 4) To remove the Weather Legends Window, press the **Legend** Softkey, the **ENT** or the **CLR** Key, or press the **FMS** Knob.

Setting up and customizing the Weather Data Link page

- 1) Select the Weather Data Link page.
- 2) Press the **MENU** Key.
- 3) Turn the **FMS** Knob to highlight 'Weather Setup', then press the **ENT** Key.
- 4) Turn the small **FMS** Knob to select 'Product Group 1' or 'Product Group 2', and press the **ENT** Key.
- 5) Turn the large **FMS** Knob or press the **ENT** Key to scroll through product selections.
- 6) Turn the small **FMS** Knob to scroll through options for each product (ON/OFF, range settings, etc.).
- 7) Press the **ENT** Key to select an option.
- 8) Press the **FMS** Knob or **CLR** Key to return to the Weather Data Link page with the changed settings.



NOTE: 'Product Group 2' is only applicable to SiriusXM and Garmin Connex Weather.

Restoring default Weather Data Link page settings

- 1) Select the Weather Data Link Page.
- 2) Press the **MENU** Key.

Flight Instruments

EIS

Nav/Com/XPDR/Audio

Flight Management

Hazard Avoidance

AFCs

Additional Features

Abnormal Operation

Annun/Alerts

Appendix

Index

- 3) Turn the FMS Knob to highlight 'Weather Setup', then press the **ENT** Key.
- 4) Press the **MENU** Key.
- 5) Highlight the desired default(s) to restore (all or for selection) and press **ENT** Key.
- 6) When finished, press the **FMS** Knob or press the **CLR** Key.

Displaying Data Link Weather Products on the 'Map - Navigation Map' Page

- 1) Select the 'Map - Navigation Map' Page.
- 2) Press the **Map Opt** Softkey.
- 3) Press the softkey to enable/disable the desired weather product.

Showing/removing the weather legend on the 'Map - Navigation Map' Page

- 1) Select the 'Map - Navigation Map' Page.
- 2) Press the **Map Opt** Softkey.
- 3) Press the **Legend** Softkey to show the weather legends window.
- 4) When finished, press the **Legend** Softkey again, or press the **FMS** Knob or the **CLR** Key to remove the window.

Setting up and customizing weather data for the navigation maps

- 1) Select the 'Map - Navigation Map' Page.
- 2) Press the **MENU** Key.
- 3) With 'Map Settings' highlighted, press the **ENT** Key.
- 4) Turn the small **FMS** Knob to select the 'Weather' Group and press the **ENT** Key.
- 5) Turn the large **FMS** Knob or press the **ENT** Key to scroll through product selections.
- 6) Turn the small **FMS** Knob to scroll through options for each product (On/Off range settings).
- 7) Press the **ENT** Key to select an option.
- 8) Press the **FMS** Knob or **CLR** Key to return to the Navigation Map Page with the changed settings.

Displaying Data Link Weather products on the PFD

- 1) On the PFD, press the **Map/HSI** Softkey.
- 2) Press the desired weather product softkey(s) to enable/disable the display of data link products on the PFD map.

- 3) To enable/disable a data link lightning weather product on a PFD, press the **Lightning** Softkey.

- a) Press the **Datalink** Softkey to enable data link lightning or press the **LTNG Off** Softkey.

- b) Press the Back Softkey twice to return to the top-level PFD Softkeys.

Enabling/disabling the weather product icon and age display (PFD maps)

- 1) On the PFD, press the **Map/HSI** Softkey.
- 2) Press the **Layout** Softkey.
- 3) Press the **WX LGND** Softkey to enable/disable the weather product age, source, and icon box display on PFD Maps.

Connex Data Requests

The Connex Data Request Menu provides the pilot with the options to define the requested weather coverage area(s), choose automatic weather update intervals (if desired), and the ability to send or cancel weather data requests. A Request Status Window inside the menu shows the status of the Connex Data Request.

Manually Requesting Garmin Connex Weather information

- 1) Select the 'Map - Weather Data Link (CNXT)' Page.
- 2) Press the **MENU** Key.
- 3) With 'Connex Data Request' highlighted, press the **ENT** Key.
- 4) Turn the large **FMS** Knob to highlight the desired coverage option(s) and press the **ENT** Key to show or hide a green check mark to select one of more of the following coverage selections:
 - Present Position – Requests data based on current location.
 - Destination – Requests data based on the active flight plan destination (Direct-To destinations excluded). See the Flight Management section for more information about entering and activating flight plans.
 - FPL – Requests data along an active flight plan, if one currently exists. Turn the small **FMS** Knob to select the desired flight plan look-ahead distance option (or choose 'Remaining FPL' to request weather data for the remainder of the flight plan), then press the **ENT** Key.
 - Waypoint – Requests data based on a waypoint (which may be off-route). Turn the large and small **FMS** Knobs to enter a waypoint, then press the **ENT** Key.

- 5) Turn the large **FMS** Knob highlight to the 'Diameter / Route Width' distance field and turn the small **FMS** Knob to select the desired diameter and route width of the request, then press the **ENT** Key.
- 6) Turn the large **FMS** Knob until the 'Send Request' field is highlighted. Press the **ENT** Key to initiate the request immediately or press the **FMS** Knob to return to the Weather Data Link (CNXT) Page without requesting weather data.

Cancelling a Connex Data Request in Progress

- 1) Select the 'Map - Weather Data Link (CNXT)' Page.
- 2) Press the **MENU** Key.
- 3) With 'Connex Data Request' highlighted, press the **ENT** Key.
- 4) Turn the large **FMS** Knob to select the 'Cancel Request' field and press the **ENT** Key. The request status box indicates 'Request Cancelled'.
- 5) Press the **FMS** Knob to return to the 'Map - Weather Data Link (CNXT)' Page.

Enabling/disabling automatic Connex Data Requests

- 1) Select the 'Map - Weather Data Link (CNXT)' Page.
- 2) Press the **MENU** Key.
- 3) With 'Connex Data Request' highlighted, press the **ENT** Key.
- 4) Choose the desired weather coverage options.
- 5) Turn the large **FMS** Knob to select the 'Update Rate' field. Then turn the small **FMS** Knob to highlight the desired automatic update frequency (Off, 5 Min, 10 Min, 15 Min, 20 Min, 30 Min, 45 Min, or 60 Min), then press the **ENT** Key.
- 6) The 'Send Request' field is highlighted and a countdown timer is displayed in the 'Request Status' Window based on the currently selected update rate. Press the **ENT** Key to immediately send an immediate Connex Data Request.

Or:

Press the **FMS** Knob to return to the 'Map - Weather Data Link (CNXT)' Page

NEXRAD (SiriusXM)

Displaying the NEXRAD weather product on the 'Map - Weather Data Link (XM)' Page

- 1) Select the 'Map - Weather Data Link (XM)' Page.
- 2) Press the **NEXRAD** Softkey.

Displaying the NEXRAD weather product on the 'Map - Navigation Map' Page

- 1) Press the **Map Opt** Softkey.
- 2) Press the **NEXRAD** Softkey.

Displaying the NEXRAD weather product on PFD maps

- 1) Press the **Map/HSI** Softkey.
- 2) Press the **NEXRAD** Softkey to enable/disable the display of NEXRAD information.

Selecting the NEXRAD Coverage Region on the 'Map - Weather Data Link (XM)' Page

- 1) Press the **MENU** Key.
- 2) Turn the **FMS** Knob to select 'Weather Setup' and press the **ENT** Key.
- 3) With Product Group 1 selected, turn the large **FMS** Knob to highlight the NEXRAD Region field.
- 4) Turn the small FMS Knob to select 'US' or 'CNDA', then press the **ENT** Key.
- 5) To remove the menu, push the **FMS** Knob or the **CLR** Key.

NEXRAD (FIS-B)

Displaying the FIS-B NEXRAD weather product on the 'Map - Weather Data Link (FIS-B)' Page

- 1) Select the 'Map - Weather Data Link (FIS-B)' Page.
- 2) Press the **NEXRAD** Softkey. Each press cycles through a coverage option as the softkey name changes (**US**, **RGNL**, or **US/RGNL**).

Displaying the FIS-B NEXRAD weather product on the 'Map - Navigation Map' Page

- 1) Press the **Map Opt** Softkey.
- 2) Press the **NEXRAD** Softkey.
- 3) To change the type of NEXRAD displayed, press the **MENU** Key.

- 4) With 'Map Settings' highlighted, press the ENT Key.
- 5) Turn the small **FMS** Knob to select the 'Weather' Group, then press the **ENT** Key.
- 6) Turn the large **FMS** Knob to highlight the NEXRAD Data Region field.
- 7) Turn the small **FMS** Knob to highlight 'CONUS' (continental United States), 'RGNL' (regional), or 'Combined', then press the **ENT** Key. This selection also affects display of NEXRAD on the PFD Maps.
- 8) When finished, press the **FMS** Knob or press the **CLR** Key.

Displaying the FIS-B NEXRAD weather product on PFD maps

- 1) Press the **Map/HSI** Softkey.
- 2) Press the **NEXRAD** Softkey to enable/disable the display of NEXRAD information.

Precipitation (Garmin Connex)

Displaying Precipitation weather information

- 1) Press the **Map Opt** Softkey (for PFD maps, press the **Map/HSI** Softkey). This step is not necessary on the 'Map - Weather Data Link (CNXT)' Page.
- 2) Press the **PRECIP** Softkey.

Echo Tops (SiriusXM)

Displaying Echo Tops information

- 1) Select the 'Weather Data Link (XM)' Page.
- 2) Press the **Echo Top** Softkey.

Cloud Tops (SiriusXM)

Displaying Cloud Tops information

- 1) Select the 'Weather Data Link (XM)' Page with the **FMS** Knob.
- 2) Press the **CLD Top** Softkey.

Data Link Lightning (SiriusXM, Garmin Connex)

Displaying Data Link Lightning information on Weather Data Link Page

- 1) Turn the **FMS** Knob to select the Weather Data Link Page.
- 2) Press the **XM LTNG** or **DL LTNG** Softkey (softkey name depends on the selected weather source).

Displaying Data Link Lightning information on the 'Map - Navigation Map' Page

- 1) Turn the **FMS** Knob to select the Navigation Map Page.
- 2) Press the **Map Opt** Softkey.
- 3) Press the **XM LTNG** or **DL LTNG** Softkey.

Enabling/disabling Data Link Lightning information on PFD maps

- 1) On the PFD, press the **Map/HSI** Softkey.
- 2) Press the **Lightning** Softkey.
- 3) Press the **Datalink** Softkey to enable data link lightning from the selected data link weather source, or press the **LTNG Off** Softkey to disable data link lightning.
- 4) When finished, press the **Back** Softkey.

Cell Movement (SiriusXM)

Displaying Cell Movement information on the 'Map - Weather Data Link (XM)' Page

- 1) Select the Weather Data Link (XM) Page using the **FMS** Knob.
- 2) Press the **Cell MOV** Softkey.

Setting up the system to display Cell Movement with NEXRAD on navigation maps

- 1) Use the **FMS** Knob to select the Navigation Map Page.
- 2) Press the **MENU** Key.
- 3) With 'Map Settings' highlighted, press the **ENT** Key.
- 4) Turn the small **FMS** Knob to highlight 'Weather' and press the **ENT** Key.
- 5) Turn the large **FMS** Knob to 'On' or 'Off' for the Cell Movement menu option. When set to 'On', Cell Movement is enabled/disabled with the NEXRAD weather product on navigation maps. When set to 'Off', Cell Movement is not displayed on navigation maps.
- 6) When finished, push the **FMS** Knob or **CLR** Key to remove the menu.

Infrared Satellite (Garmin ConnexT)

Displaying Intrared Satellite information

- 1) Select the Weather Data Link (CNXT) Page.
- 2) Select the **IR SAT** Softkey.

SIGMETs and AIRMETs

Displaying SIGMETs and AIRMETs

- 1) Select the Weather Data Link Page.
- 2) Press the **SIG/AIR** Softkey.
- 3) To view the text of the SIGMET or AIRMET, press the **Joystick** and move the Map Pointer over the icon.
- 4) Press the **ENT** key. The following figure shows sample SIGMET text.

METARs and TAFs

Displaying METAR and TAF text on the MFD

- 1) On the Weather Data Link Page, press the **METAR** Softkey.
- 2) Press the **Joystick** and pan to the desired airport.
- 3) Press the **ENT** Key. The Weather Information Page is shown with METAR and TAF text.
- 4) Use the **FMS** Knob or the **ENT** Key to scroll through the METAR and TAF text. METAR text must be completely scrolled through before scrolling through the TAF text.
- 5) Press the **FMS** Knob or the **CLR** Key to return to the Weather Data Link Page.

Displaying original METAR text on the Active Flight Plan Page

- 1) Select the 'FPL - Active Flight Plan' Page on the MFD.
- 2) Press the **FMS** Knob to activate the cursor.
- 3) Turn the large **FMS** Knob to highlight a waypoint with an available METAR (indicated with a METAR flag next to it). The METAR text will appear in the 'Selected Waypoint Weather' Window below.
- 4) When finished, press the **FMS** Knob to remove the cursor or press the **FPL** Key to exit the Active Flight Plan Page.

Displaying original METAR text information on the PFD Inset Map

- 1) On the PFD, press the **Map/HSI** Softkey.
- 2) Press the **METAR** Softkey.
- 3) Press the **Joystick** and pan to the desired METAR flag. Original METAR text appears on the map.
- 4) When finished, press the **Joystick** to remove the Map Pointer.

Surface Analysis and City Forecast (SiriusXM)

Displaying Surface Analysis and City Forecast information

- 1) Select the 'Map - Weather Data Link (XM)' Page.
- 2) Press the **More WX** Softkey.
- 3) Press the **SFC** Softkey.
- 4) Press the softkey for the desired forecast time: **Current, 12 HR, 24 HR, 36 HR, or 48 HR**. The **SFC** Softkey label changes to show the forecast time selected.

Or:

Press the **Off** Softkey to disable the display of the weather product.

Freezing Level (SiriusXM)

Displaying Freezing Level information

- 1) Select the 'Map - Weather Data Link (XM)' Page.
- 2) Press the **More WX** Softkey.
- 3) Press the **FRZ LVL** Softkey.

Winds Aloft

Displaying the Winds Aloft weather product

- 1) Select the 'Map - Weather Data Link (XM)' Page with the **FMS** Knob.
- 2) Press the **More WX** Softkey.
- 3) Press the **Wind** Softkey.
- 4) Press a softkey for the desired altitude level: **SFC** (surface) up to 42,000 feet. Press the **Next** or **Prev** Softkey to cycle through the altitude softkeys. The **Wind** Softkey label changes to reflect the altitude selected.

Enabling/disabling the Vertical Situation Display (containing winds aloft data)

- 1) Select the 'Map - Navigation Map' Page.
- 2) Press the **Map Opt** Softkey.
- 3) Press the **Inset** Softkey.
- 4) Press the **VSD** Softkey to enable/disable the Vertical Situation Display.

Enabling/disabling winds aloft data display for the VSD

- 1) Select the 'Map - Navigation Map' Page.
- 2) Press the **MENU** Key.

- 3) With 'Map Settings' highlighted, press the **ENT** Key.
- 4) Turn the small **FMS** Knob to select 'VSD' and press the **ENT** Key.
- 5) Turn the large **FMS** Knob to highlight the Winds On/Off field.
- 6) Turn the small **FMS** Knob to select 'On' or 'Off'.
- 7) Press the **FMS** Knob or **CLR** Key to return to the 'Map - Navigation Map' Page with the changed settings.

County Warnings (SiriusXM)

Displaying County Warning information

- 1) Select the 'Map - Weather Data Link (XM)' Page with the **FMS** Knob.
- 2) Press the **More WX** Softkey.
- 3) Press the **County** Softkey.

Cyclone (SiriusXM)

Displaying cyclone (hurricane) track information

- 1) Select the 'Map - Weather Data Link (XM)' Page with the **FMS** Knob.
- 2) Press the **More WX** Softkey.
- 3) Press the **Cyclone** Softkey.

Icing (CIP & SLD) (SiriusXM)

Displaying Icing data

- 1) Select the 'Map - Weather Data Link (XM)' Page.
- 2) Press the **More WX** Softkey.
- 3) Press the **ICNG** Softkey.
- 4) Press a softkey for the desired altitude level: 1,000 feet up to 30,000 feet. Press the **Next** or **PREV** Softkey to cycle through the altitude softkeys. The **ICNG** Softkey label changes to indicate the altitude selected.

Turbulence (SiriusXM)

Displaying Turbulence data

- 1) Select the 'Map - Weather Data Link (XM)' Page.
- 2) Press the **More WX** Softkey.
- 3) Press the **TURB** Softkey.
- 4) Press a softkey for the desired altitude: 21,000 feet up to 45,000 feet. Press the **Next** or **PREV** Softkey to cycle through the altitude softkeys. The **TURB** Softkey label changes to indicate the altitude selection.

PIREPs and AIREPs

Displaying PIREP and AIREP text

- 1) Select the 'Map - Weather Data Link (XM)' Page.
- 2) Press the **More WX** Softkey.
- 3) Press the **PIREPS** or **AIREPS** Softkey. (Note the **AIREPS** Softkey is only available with the SiriusXM Weather service.)
- 4) Press the **Joystick** and pan to the desired weather report. A gray circle will appear around the weather report when it is selected.
- 5) Press the **ENT** Key. The Weather Information Page is shown with PIREP or AIREP text. The data is first displayed in a decoded fashion, followed by the original text. Note the original text may contain additional information not present in the decoded version.
- 6) Use the **FMS** Knob or the **ENT** Key to scroll through the PIREP or AIREP text.
- 7) Press the **FMS** Knob or the **CLR** Key to return to the 'Map - Weather Data Link (XM)' Page.

TFRS

Displaying TFR Data

- 1) Select the 'Map - Weather Data Link (XM)' Page or the 'Map - Navigation Map' Page.
- 2) Press the **Joystick** and pan the map pointer over a TFR to highlight it. The system displays TFR summary information above the map.
- 3) Press the **ENT** Key. The system displays a pop-up menu.
- 4) If necessary, turn the **FMS** Knob to select 'Review Airspaces' and press the **ENT** Key. The system displays the TFR Information window.
- 5) Press the **FMS** Knob or the **CLR** Key to remove the TFR Information window.

Setting up and customizing TFR data for maps on which TFR data can be displayed

- 1) Select the 'Map - Navigation Map' Page.
- 2) Press the **MENU** Key.
- 3) With 'Map Settings' highlighted, press the **ENT** Key.
- 4) Turn the small **FMS** Knob to select the 'Aviation' Group and press the **ENT** Key.

- 5) Turn the large **FMS** Knob to scroll to the TFR product range setting.
- 6) Turn the small **FMS** Knob to scroll through options (Off, range settings).
- 7) Press the **ENT** Key to select an option.
- 8) Press the **FMS** Knob or **CLR** Key to return to the Navigation Map Page with the changed settings.

FIS-B WEATHER STATUS

Viewing FIS-B status

- 1) Turn the large **FMS** Knob to select the Aux Page Group.
- 2) Turn the small **FMS** Knob to select the ‘Aux - ADS-B Status’ Page.

Enabling/disabling the FIS-B weather feature

- 1) Select the ‘Map - Weather Data Link (FIS-B)’ Page.
- 2) Press the **MENU** Key.
- 3) Turn the small **FMS** Knob to highlight ‘Enable FIS-B Weather’ or ‘Disable FIS-B Weather’, and press the **ENT** Key.

STORMSCOPE LIGHTNING DETECTION SYSTEM



WARNING: Do not rely on information from the lightning detection system display as the sole basis for hazardous weather avoidance. Range limitations and interference may cause the system to display inaccurate or incomplete information. Refer to documentation from the lightning detection system manufacturer for detailed information about the system.

Lightning Age	Symbol
Strike is less than 6 seconds old	
Strike is between 6 and 60 seconds old	
Strike is between 1 and 2 minutes old	
Strike is between 2 and 3 minutes old	

Lightning Age and Symbols

Adjusting the Stormscope Map Range

- 1) Turn the large **FMS** Knob to select the Map Page Group.
- 2) Turn the small **FMS** Knob to select the Stormscope Page.
- 3) Turn the **Joystick** clockwise to increase the map range or counter-clockwise to decrease the map range.

Selecting 'cell' or 'strike' mode

- 1) Select the 'Map - Stormscope®' Page.
- 2) Press the **Mode** Softkey. The **Cell** and **Strike** softkeys are displayed.
- 3) Press the **Cell** Softkey to display 'CELL' data or press the **Strike** Softkey to display 'STRIKE' data. 'CELL' or 'STRIKE' is displayed in the mode box in the upper right corner of the Stormscope Page.
- 4) Press the **Back** Softkey to return to the top level softkeys for the Stormscope Page.

Or:

- 1) Press the **MENU** Key on the 'Map - Navigation Map' Page.
- 2) With 'Map Settings' selected, press the **ENT** Key.
- 3) Turn the **FMS** Knob to select the 'Weather' group and press the **ENT** Key.
- 4) Turn the large **FMS** Knob to select the Stormscope Mode field.
- 5) Turn the small **FMS** Knob to change between 'Cell' and 'Strike' options. When the desired item is selected, press the **ENT** Key.
- 6) Press the **FMS** knob to return to the 'Map - Navigation Map' Page.

Manually clearing Stormscope cell or strike information

- 1) Select the Stormscope Page.
- 2) Press the **Clear** Softkey.

Or:

- 1) Select the Navigation Map Page.
- 2) Press the **MENU** Key.
- 3) Turn the **FMS** Knob to highlight 'Clear Stormscope® Lightning', then press the **ENT** Key.

Displaying Stormscope information on MFD navigation maps

- 1) Press the **Map Opt** Softkey.
- 2) Press the **STRMSCP** Softkey.

Displaying Stormscope information on PFD maps

- 1) On the PFD, press the **Map/HSI** Softkey.
- 2) Press the **Lightning** Softkey.
- 3) Press the **STRMSCP** Softkey.

TERRAIN DISPLAYS



WARNING: Do not use terrain avoidance displays as the sole source of information for maintaining separation from terrain and obstacles. Garmin obtains terrain and obstacle data from third party sources and cannot independently verify the accuracy of the information.



NOTE: Terrain data is not displayed when the aircraft is outside of the installed terrain database coverage area.



NOTE: Terrain and obstacle alerting is not available north of 89° North latitude and south of 89° South latitude. This is due to limitations present within the Terrain database and the system's ability to process the data representing the affected areas.

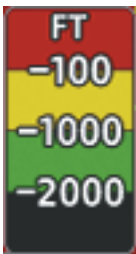


NOTE: Terrain and obstacle alerting requires the Terrain-SVT option included with Garmin SVT. No terrain or obstacle alerting occurs for the Terrain Proximity feature.

On-Ground Legend



In-Air Legend



Relative Terrain Legend

Enabling/disabling relative terrain information on MFD navigation maps

- 1) Press the **Map Opt** Softkey.
- 2) Press the **TER** Softkey to cycle through a terrain option with each press. The selected mode is displayed in cyan: Off, Topo, REL. Press the **TER** Softkey as needed until 'REL' is displayed on the softkey.

Displaying relative terrain information on PFD Inset Map or HSI Map

- 1) Press the **Map/HSI** Softkey.
- 2) Press the **Rel Ter** Softkey.

Customizing terrain and obstacle display on the Navigation Map Page

- 1) Select the Navigation Map Page.
- 2) Press the **MENU** Key.
- 3) With 'Map Settings' highlighted, press the **ENT** Key.
- 4) Turn the small **FMS** Knob to select the 'Map' Group and press the **ENT** Key.
- 5) Turn the large **FMS** Knob or press the **ENT** Key to scroll through product selections.
 - Terrain Display – Turns the display of relative ('REL') terrain data on or off and sets maximum range at which terrain is shown
 - Obstacle Data – Turns the display of obstacle data on or off and sets maximum range at which obstacles are shown
- 6) Turn the small **FMS** Knob to scroll through options for each product (On/Off, range settings, etc.).
- 7) Press the **ENT** Key to select an option.
- 8) Press the **FMS** Knob or **CLR** Key to return to the Navigation Map Page with the changed settings.

Displaying the terrain page

- 1) Turn the large **FMS** Knob to select the Map Page Group.
- 2) Turn the small **FMS** Knob to select the Terrain Proximity Page/Terrain-SVT Page.

Showing/hiding aviation information on the terrain page

- 1) Press the **MENU** Key.
- 2) Select 'Show Aviation Data' or 'Hide Aviation Data' (choice dependent on current state) and press the **ENT** Key.

TRAFFIC INFORMATION SERVICE (TIS)



WARNING: Do not rely solely upon the display of traffic information for collision avoidance maneuvering. The traffic display does not provide collision avoidance resolution advisories and does not under any circumstances or conditions relieve the pilot's responsibility to see and avoid other aircraft.



NOTE: TIS is available only when the aircraft is within the service volume of a TIS-capable terminal radar site. Aircraft without an operating transponder are invisible to both Traffic Advisory Systems (TAS), Traffic Alert and Collision Avoidance Systems (TCAS) and TIS. Aircraft without altitude reporting capability are shown without altitude separation data or climb descent indication.



NOTE: Mode S TIS is disabled if another traffic system such as TAS or Automatic Dependent Surveillance-Broadcast (ADS-B) is installed.

TIS Symbol	Description
	Non-Threat Traffic
	Traffic Advisory (TA)
	Traffic Advisory Off Scale

TIS Traffic Symbols

Displaying Traffic Data

Enabling/disabling TIS traffic overlays (MFD navigation maps)

- 1) Press the **Map Opt** Softkey.
- 2) Press the **Traffic** Softkey. Traffic is now displayed on the navigation map.

Enabling/disabling TIS traffic information on the PFD Inset Map or HSI Map

- 1) On the PFD, press the **Map/HSI** Softkey.
- 2) Press the **Traffic** Softkey to enable/disable the display traffic information on the Inset Map or HSI Map.

Customizing TIS traffic display on the 'Map - Navigation Map' Page

- 1) Select the Navigation Map Page.
- 2) Press the **MENU** Key.
- 3) With 'Map Setup' highlighted, press the **ENT** Key.
- 4) Turn the small **FMS** Knob to select the 'Traffic' Group and press the **ENT** Key.
- 5) Turn the large **FMS** Knob or press the **ENT** Key to scroll through product selections.
 - Traffic – Turns the display of traffic data on or off
 - Traffic Mode – Selects the traffic mode for display; select from:
 - All Traffic - Displays all traffic
 - TA Only - Displays Traffic Alerts only
 - Traffic Symbols – Selects the maximum range at which traffic symbols are shown
 - Traffic Labels – Selects the maximum range at which traffic labels are shown (with the option to turn off)
- 6) Turn the small **FMS** Knob to scroll through options for each product (On/Off, range settings, etc.).
- 7) Press the **ENT** Key to select an option.
- 8) Press the **FMS** Knob or **CLR** Key to return to the Navigation Map Page with the changed settings.

Displaying TIS traffic on the 'Map - Traffic Map' Page

- 1) Turn the large **FMS** Knob to select the Map Page Group.
- 2) Turn the small **FMS** Knob to select the 'Map - Traffic Map' Page.
- 3) Press the **Operate** Softkey to begin displaying traffic. 'OPERATING' is displayed in the Traffic Mode field.
- 4) Press the **Standby** Softkey to place the system in the Standby Mode. 'STANDBY' is displayed in the Traffic Mode field.
- 5) Rotate the **Joystick** clockwise to increase the map range or rotate counter-clockwise to decrease the map range.

Muting the "TIS Not Available" voice alert

- 1) Select the 'Map - Traffic Map' Page.
- 2) Press the **TNA Mute** Softkey. The status is displayed in the upper left corner of the 'Map - Traffic Map' Page..

AVIDYNE TAS600 TRAFFIC



WARNING: Do not rely solely upon the display of traffic information for collision avoidance maneuvering. The traffic display does not provide collision avoidance resolution advisories and does not under any circumstances or conditions relieve the pilot's responsibility to see and avoid other aircraft.



WARNING: Do not rely solely upon the display of traffic information to accurately depict all of the traffic information within range of the aircraft. Due to lack of equipment, poor signal reception, and/or inaccurate information from other aircraft, traffic may be present but not represented on the display.



NOTE: Pilots should be aware of traffic system limitations. Traffic systems require transponders of other aircraft to respond to system interrogations. If the transponders do not respond to interrogations due to phenomena such as antenna shading or marginal transponder performance, traffic may be displayed intermittently, or not at all. Aircraft without altitude reporting capability are shown without altitude separation data or climb descent indication. Pilots should remain vigilant for traffic at all times.



NOTE: Mode-S TIS is disabled when another traffic system is installed.



NOTE: Refer to the Avidyne TAS600 Series Pilot's Guide for a detailed discussion of the Avidyne TAS600 System.

TAS Symbology

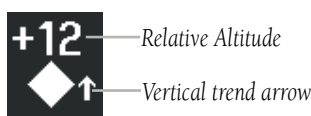
If a Avidyne TAS600 is installed with a Garmin GTX 335 transponder, the system uses the following symbols to represent traffic.

TAS Symbol	Description
	Non-Threat Traffic
	Proximity Advisory (PA)

TAS Symbol	Description
	Traffic Advisory (TA)
	Traffic Advisory Off Scale

TAS Symbol Description with GTX 335 Transponder

The system displays the altitude vertical trend as an up/down arrow (for speeds greater than 500 fpm in either direction) to the right of the intruder symbol.



Intruder Altitude and Vertical Trend Arrow

System Test

Testing the traffic system

- 1) Turn the large **FMS** Knob to select the Map Page Group.
- 2) Turn the small **FMS** Knob to select the 'Map - Traffic Map' Page.
- 3) Turn the **Joystick** to set the range to 2/6 nm to allow for full test pattern to be displayed during test.
- 4) Press the **Standby** or **TAS STBY** Softkey.
- 5) Press the **Test** Softkey.

Operation

Changing traffic system modes on the 'Map - Traffic Map' Page

- 1) Turn the large **FMS** Knob to select the Map Page Group.
- 2) Turn the small **FMS** Knob to select the 'Map - Traffic Map' Page.
- 3) Press the **Operate** or **TAS OPER** Softkey to begin displaying traffic. 'OPERATING' is displayed in the Traffic mode field.
- 4) Press the **Standby** or **TAS STBY** Softkey to place the system in the Standby mode. 'STANDBY' is displayed in the Traffic mode field.

Changing the altitude range

- 1) On the 'Map - Traffic Map' Page, select the **ALT Mode** Softkey.
- 2) Select one of the following Softkeys:
 - **Above:** Displays Other Non-Threat and Proximity Advisory traffic from 9900 feet above the aircraft to 2700 feet below the aircraft. Typically used during climb phase of flight.
 - **Normal:** Displays Other Non-Threat and Proximity Advisory traffic from 2700 feet above the aircraft to 2700 feet below the aircraft. Typically used during enroute phase of flight.
 - **Below:** Displays Other Non-Threat and Proximity Advisory traffic from 2700 feet above the aircraft to 9900 feet below the aircraft. Typically used during descent phase of flight.
 - **UNREST** (unrestricted): All traffic is displayed from 9900 feet above and 9900 feet below the aircraft.
- 3) To return to the 'Map - Traffic Map' Page, select the **Back** Softkey.

Enabling/disabling traffic information (MFD maps other than the Traffic Map Page)

- 1) Press the **Map Opt** Softkey.
- 2) Press the **Traffic** Softkey. Traffic is now displayed on the map.

Customizing the traffic display on the 'Map - Navigation Map' Page

- 1) Select the 'Map - Navigation Map' Page.
- 2) Press the **MENU** Key.
- 3) With 'Map Settings' highlighted, press the **ENT** Key.
- 4) Turn the small **FMS** Knob to select the Traffic Group and press the **ENT** Key.
- 5) Turn the large **FMS** Knob or press the **ENT** Key to scroll through the selections.
 - Traffic – Turns the display of traffic data on or off
 - Traffic Mode – Selects the traffic mode for display; select from:
 - All Traffic - Displays all traffic
 - TA/PA - Displays only Traffic Advisories and Proximity Advisories
 - TA Only - Displays Traffic Advisories only
 - Traffic Symbols – Selects the maximum range at which traffic symbols are shown

- Traffic Labels – Selects the maximum range at which traffic labels (relative altitude, vertical trend) are shown with the option to turn off
- 6) Turn the small **FMS** Knob to scroll through options (On/Off, range settings, etc.).
 - 7) Press the **ENT** Key to select an option.
 - 8) Press the **FMS** Knob or **CLR** Key to return to the Navigation Map Page.

Enabling/disabling traffic overlay on PFD navigation maps

- 1) With the Inset Map or HSI Map displayed, press the **Map/HSI** Softkey on the PFD.
- 2) Press the **Traffic** Softkey to enable/disable the display traffic information.

ADS-B TRAFFIC



WARNING: Do not rely solely upon the display of traffic information for collision avoidance maneuvering. The traffic display does not provide collision avoidance resolution advisories and does not under any circumstances or conditions relieve the pilot's responsibility to see and avoid other aircraft.



WARNING: Do not rely solely upon the display of traffic information to accurately depict all of the traffic information within range of the aircraft. Due to lack of equipment, poor signal reception, and/or inaccurate information from other aircraft, traffic may be present but not represented on the display.

The following table describes the aircraft that are observed by a GTX 345R-equipped aircraft according to the level of equipment installed in the other aircraft.

Other Aircraft Equipment	Viewable by GTX 345 UAT Equipped Aircraft
1090ES Out Equipped	Yes
UAT Receive Only Capable	No
UAT Transmit Equipped	Yes
No Transponder, No ADS-B	No
Non ADS-B Equipped, but with Mode C or S Transponder	Yes*

* Only when in ADS-B ground station coverage and when the other aircraft is in ATC radar coverage.

Aircraft Available for Viewing by an ADS-B Equipped Own Aircraft

	Symbol	Description
Flight Instruments		Traffic Advisory with directional information. Points in the direction of the intruder aircraft track.
EIS		Traffic Advisory without directional information.
Nav/Com/XPDR/Audio		Traffic Advisory out of the selected display range with directional information. Displayed at outer range ring at proper bearing.
Flight Management		Traffic Advisory out of the selected display range without directional information. Displayed at outer range ring at proper bearing.
		Proximity Advisory with directional information. Points in the direction of the aircraft track.
Hazard Avoidance		Proximity Advisory without directional information.
AFCS		Other Non-Threat traffic with directional information. Points in the direction of the intruder aircraft track.
		Other Non-Threat traffic without directional information.
Additional Features		Traffic located on the ground with directional information. Points in the direction of the aircraft track. Ground traffic is only displayed when ADS-B is in Surface (SURF) Mode or own aircraft is on the ground.
Abnormal Operation		Ground traffic without directional information. Ground traffic is only displayed when ADS-B is in Surface (SURF) Mode or own aircraft is on the ground.
Annun/Alerts		Non-aircraft ground traffic with ADS-B directional information. Pointed end indicates direction of travel. Ground traffic is only displayed when ADS-B is in Surface (SURF) Mode or own aircraft is on the ground.
		Non-aircraft ground traffic without ADS-B directional information. Ground traffic is only displayed when ADS-B is in Surface (SURF) Mode or own aircraft is on the ground.

ADS-B Traffic Symbology

Operation

Enabling/disabling the display of ADS-B traffic

- 1) Select the 'Map - Traffic Map' Page.
- 2) Press the **ADS-B** Softkey.

Testing the display of ADS-B traffic

- 1) Select the 'Map - Traffic Map' Page.
- 2) If necessary, turn the Joystick to select a map range of 2 and 6 nm to ensure full test pattern display.
- 3) Ensure the **ADS-B** Softkey is disabled.
- 4) If the optional TAS is installed, ensure the **TAS STBY** Softkey is enabled.
- 5) Press the **Test** Softkey.

Changing the altitude range

- 1) On the 'Map - Traffic Map' Page, press the **ALT Mode** Softkey.
- 2) Press one of the following softkeys:
 - **Above:** Displays Other Non-Threat and proximity traffic from 9000 feet above the aircraft to 2700 feet below the aircraft. Typically used during climb phase of flight.
 - **Normal:** Displays Other Non-Threat and proximity traffic from 2700 feet above the aircraft to 2700 feet below the aircraft. Typically used during enroute phase of flight.
 - **Below:** Displays Other Non-Threat and proximity traffic from 2700 feet above the aircraft to 9000 feet below the aircraft. Typically used during descent phase of flight.
 - **UNREST** (unrestricted): All traffic is displayed from 9900 feet above and 9900 feet below the aircraft.
- 3) To return to the 'Map - Traffic Map' Page, select the **Back** Softkey.

Enabling/disabling the Motion Vector display

- 1) Select the Traffic Map Page.
- 2) Press the **Motion** Softkey.
- 3) Press one of the following softkeys:
 - **Absolute:** Displays the motion vector pointing in the absolute direction.
 - **Relative:** Displays the motion vector relative to own aircraft
 - **Off:** Disables the display of the motion vector.

Adjusting the duration for the Motion Vector projected time

- 1) Select the 'Map - Traffic Map' Page.
- 2) Press the **Motion** Softkey.

- 3) Press the **Duration** Softkey.
- 4) Press a softkey for the desired duration (**30 SEC, 1 MIN, 2 MIN, 5 MIN**).
- 5) When finished, press the **Back** Softkey to return to the 'Map - Traffic Map' Page.

Showing additional traffic information

- 1) Select the 'Map - Traffic Map' Page.
- 2) Press the **FMS** Knob. A cyan border appears on the first selected traffic symbol. Additional information appears in a window in the lower-left corner of the Traffic Map Page.
- 3) To select a different aircraft symbol, turn the **FMS** Knob to move the cyan border until another symbol is selected.
- 4) When finished, press the **FMS** Knob again to disable the traffic selection.

Changing the display range

Turn the **Joystick** clockwise to display a larger area or rotate counter-clockwise to display a smaller area.

Viewing ADS-B Traffic Status

- 1) Turn the large **FMS** Knob to select the Aux Page Group.
- 2) Turn the small **FMS** Knob to select the 'Aux - ADS-B Status' Page.

AUTOMATIC FLIGHT CONTROL SYSTEM



NOTE: If the attitude information required for the default flight director modes becomes invalid or unavailable, the autopilot automatically disengages.

ACTIVATING THE FLIGHT DIRECTOR

An initial press of a key listed in following table (when the Flight Director is not active) activates the Flight Director in the listed modes.

Control Pressed	Modes Selected			
	Lateral		Vertical	
FD Key	Roll Hold (default)	ROL	Pitch Hold (default)	PIT
AP Key	Roll Hold (default)	ROL	Pitch Hold (default)	PIT
CWS Button	Roll Hold (default)	ROL	Pitch Hold (default)	PIT
GO-AROUND Button	Takeoff (on ground)	TO	Takeoff (on ground)	TO
	Go Around (in air)	GA	Go Around (in air)	GA
ALT Key	Roll Hold (default)	ROL	Altitude Hold	ALT
VS Key	Roll Hold (default)	ROL	Vertical Speed	VS
VNV Key	Roll Hold (default)	ROL	Vertical Path Tracking*	VPTH
NAV Key	Navigation**	GPS VOR LOC BC	Pitch Hold (default)	PIT
APR Key	Approach**	GPS VOR LOC	Pitch Hold (default)	PIT GP GS
HDG Key	Heading Select	HDG	Pitch Hold (default)	PIT

*Valid VNV flight plan must be entered before **VNV** Key press activates the Flight Director.

The selected navigation receiver must have a valid VOR or LOC signal or active GPS course before **NAV or **APR** Key press activates the Flight Director.

VERTICAL MODES

Vertical Mode	Description	Control	Annunciation	
Pitch Hold	Holds the current aircraft pitch attitude; may be used to climb/descend to the Selected Altitude	(default)	PIT	
Selected Altitude Capture	Captures the Selected Altitude	*	ALTS	
Altitude Hold	Holds the current Altitude Reference	ALT Key	ALT	nnnnn FT
Vertical Speed	Maintains the current aircraft vertical speed; may be used to climb/descend to the Selected Altitude	VS Key	VS	nnnn FPM
Flight Level Change	Maintains the current aircraft airspeed while the aircraft is climbing/descending to the Selected Altitude	FLC Key	FLC	nnn KT

* *ALTS armed automatically when PIT, VS, FLC, TO, or GA active, and under VPTH when Selected Altitude is to be captured instead of VNV Target Altitude*

LATERAL MODES

Lateral Mode	Description	Control	Annunciation	
Roll Hold	Holds the current aircraft roll attitude or rolls the wings level, depending on the commanded bank angle	(default)	ROL	
Heading Select	Captures and tracks the Selected Heading	HDG Key	HDG	
Navigation, GPS	Captures and tracks the selected navigation source (GPS, VOR, LOC)	NAV Key	GPS	
Navigation, VOR Enroute Arm/Capture/Track			VOR	
Navigation, LOC Capture/Track (No Glideslope)			LOC	

COMBINATION MODES (VNV, APR, NAV, BC, GA)

Mode	Description	Control	Annunciation
Vertical Path Tracking	Captures and tracks descent legs of an active vertical profile	VNV Key	VPTH
VNV Target Altitude Capture	Captures the Vertical Navigation (VNV) Target Altitude	*	ALTV
Glidepath	Captures and tracks the SBAS glidepath on approach	APR Key	GP
Glideslope	Captures and tracks the ILS glideslope on approach		GS
Backcourse Arm/Capture/Track	Captures and tracks a localizer signal for backcourse approaches	BC Key	BC
Approach, GPS Arm/Capture/Track	Captures and tracks the selected navigation source (GPS, VOR, LOC)	APR Key	GPS
Approach, VOR Arm/Capture/Track			VAPP
Approach, ILS Arm/Capture/Track (Glideslope Mode automatically armed)			LOC
Takeoff	Commands a constant pitch angle and wings level on the ground in preparation for takeoff	GO-AROUND Button	TO
Go Around**	Commands a constant pitch angle and wings level in the air	GO-AROUND Button	GA

* *ALTV is armed automatically under VPTH when VNV Target Altitude is to be captured instead of Selected Altitude.*

***Go Around mode disengages the autopilot unless a compatible lift computer is installed.*

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ADDITIONAL FEATURES

TERMINAL PROCEDURE CHARTS



NOTE: With the availability of SafeTaxi®, ChartView, or FliteCharts®, it may be necessary to carry another source of charts on-board the aircraft.

SafeTaxi®

SafeTaxi® is an enhanced feature that gives greater map detail as the map range is adjusted in on the airport. The airport display on the map reveals runways with numbers, taxiways identifiers, and airport landmarks including ramps, buildings, control towers, and other prominent features. Resolution is greater at lower map ranges. The aircraft symbol provides situational awareness while taxiing.

Pressing the **Detail** Softkey (declutter) once removes the taxiway markings and airport identification labels. Pressing the **Detail** Softkey twice removes VOR station ID, the VOR symbol, and intersection names if within the airport plan view. Pressing the **Detail** Softkey a third time removes the airport runway layout, unless the airport in view is part of an active route structure. Pressing the **Detail** Softkey again cycles back to the original map detail.

The SafeTaxi database contains detailed airport diagrams for selected airports. These diagrams provide the pilot with situational awareness by displaying the aircraft position in relation to taxiways, ramps, runways, terminals, and services. This information should not be used by the pilot as the basis for maneuvering the aircraft on the ground. This database is updated on a 56-day cycle.

ChartView

ChartView resembles the paper version of Jeppesen terminal procedures charts. The charts are displayed in full color with high-resolution. The MFD depiction shows the aircraft position on the moving map in the plan view of most approach charts and on airport diagrams.

The ChartView database is updated on a 14-day cycle. If the ChartView database is not updated within 70 days of the expiration date, ChartView will no longer function.

FliteCharts®

FliteCharts® resemble the paper version of AeroNav Services terminal procedures charts. The charts are displayed with high-resolution and in color for applicable charts. The MFD depiction shows the aircraft position on the moving map in the plan view of most approach charts and on airport diagrams.

The FliteCharts database contains procedure charts for the United States only. This database is updated on a 28-day cycle. If not updated within 180 days of the expiration date, FliteCharts will no longer function.

Selecting Terminal Procedures Charts

While viewing the 'Map-Navigation Map' Page, 'NRST-Nearest Airport' Page, or 'FPL-Active Flight Plan' Page, press the **Charts** Softkey.

Or:

- 1) Press the **MENU** Key to display the 'Page Menu.'
- 2) Turn the large **FMS** Knob to scroll through the 'Options' Menu to Charts.
- 3) Press the **ENT** Key to display the chart.

Selecting Additional Information

- 1) While viewing the Airport Taxi Diagram, press the **Full SCN** Softkey to display the information windows (Airport, Info).
- 2) Press the **FMS** Knob to activate the cursor.
- 3) Turn the large **FMS** Knob to highlight the Airport, Info, Runways, or Frequencies Box.
- 4) Turn the small **FMS** Knob to select the Info Box choices. If multiple choices are available, scroll to the desired choice with the large **FMS** Knob and press the **ENT** Key to complete the selection.
- 5) Press the **FMS** Knob again to deactivate the cursor.

Selecting full screen On or Off

- 1) While viewing a terminal chart press the **MENU** Key to display the 'Page Menu' Options.
- 2) Turn the large **FMS** Knob to highlight the 'Chart Setup' Menu Option and press the **ENT** Key.
- 3) Turn the large **FMS** Knob to move between the 'Full Screen' and 'Color Scheme' Options.
- 4) Turn the small **FMS** Knob to choose between the 'On' and 'Off' 'Full Screen' Options.

Selecting Day, Night, or Automatic View

- 1) While viewing a terminal chart press the **MENU** Key to display the 'Page Menu' Options.

- 2) Turn the large **FMS** Knob to highlight the 'Chart Setup' Menu Option and press the **ENT** Key.
- 3) Turn the large **FMS** Knob to move to the 'Color Scheme' Option.
- 4) Turn the small **FMS** Knob to choose between 'Day', 'Auto', and 'Night' Options.
- 5) If Auto Mode is selected, turn the large **FMS** Knob to select the percentage field. Use the small **FMS** Knob to change the percentage value. The percentage value is the day/night crossover point based on the percentage of backlighting intensity. For example, if the value is set to 15%, the day/night display changes when the display backlight reaches 15% of full brightness.

The display must be changed in order for the new setting to become active. This may be accomplished by selecting another page or changing the display range.

- 6) Press the **FMS** Knob when finished to remove the 'Chart Setup' Menu.

IFR/VFR Charts

IFR/VFR charts resemble the paper version and are displayed with high-resolution and in color for applicable charts.

Selecting IFR Low, IFR High, VFR Charts

- 1) Select the 'Map - IFR/VFR Charts' Page.
- 2) Press the **VFR**, **IFR Low**, or **IFR High** Softkey to display the desired chart.

AIRPORT DIRECTORY

The Aircraft Owners and Pilots Association (AOPA) or AC-U-KWIK Airport Directory database offers detailed information regarding services, hours of operation, lodging options, and more for various airports. This information is viewed on the 'WPT-Airport Information' Page.

The Airport Directory databases are revised every 56 days. The Airport Directory is always available for use after the expiration date. Check fly.garmin.com for the current database.

Selecting the Airport Directory Page

- 1) Turn the large **FMS** Knob to select the 'WPT' page group.
- 2) Turn the small **FMS** Knob to select the Airport Information Page. Initially, information for the airport closest to the aircraft's present position is displayed.
- 3) If necessary, press the Info softkey until **Info 2** is displayed.

SIRIUSXM™ RADIO ENTERTAINMENT

The 'Aux - XM Radio' Page provides information and control of the audio entertainment features of the SiriusXM™ Satellite Radio.

Selecting the XM Radio Page

- 1) Turn the large **FMS** Knob to select the Auxiliary Page Group.
- 2) Turn the small **FMS** Knob to select the displayed 'Aux - XM Radio' Page.

Active Channel and Channel List

The 'Active Channel' Box on the 'Aux - XM Radio' Page displays the currently selected channel. The 'Channels List' Box of the 'Aux - XM Radio' Page shows a list of the available channels for the selected category.

Selecting a channel from the channel list

- 1) While on the 'Aux - XM Radio' Page, press the **Channel** Softkey.
- 2) Press the **CH +** Softkey to go up through the list in the Channel Box, or move down the list with the **CH –** Softkey.

Or:

- 1) Press the **FMS** Knob to highlight the channel list and turn the large **FMS** Knob to scroll through the channels.
- 2) Press the **ENT** Key to activate the selected channel.

Selecting a channel directly

- 1) While on the 'Aux - XM Radio' Page, press the **Channel** Softkey.
- 2) Press the **Direct CH** Softkey. The channel number in the 'Active Channel' Box is highlighted.
- 3) Press the numbered softkeys located on the bottom of the display to directly select the desired channel number.
- 4) Press the **ENT** Key to activate the selected channel.

Selecting a Category

The 'Category' Box of the 'Aux - XM Radio' Page displays the currently selected category of audio.

Selecting a Category

- 1) Press the **Category** Softkey on the 'Aux - XM Radio' Page.
- 2) Press the **CAT +** and **CAT -** Softkeys to cycle through the categories.

Or:

Turn the small **FMS** Knob to display the Categories list. Highlight the desired category with the small **FMS** Knob and press the **ENT** Key. Selecting All Categories places all channels in the list.

Setting a preset channel number

- 1) On the 'Aux - XM Radio' Page, while listening to an Active Channel that is wanted for a preset, press the **Presets** Softkey to access the first five preset channels (**Preset 1 - Preset 5**).
- 2) Press the **More** Softkey to access the next five channels (**Preset 6 – Preset 10**), and again to access the last five channels (**Preset 11 – Preset 15**). Pressing the **More** Softkey repeatedly cycles through the preset channels.
- 3) Press any one of the (**Preset 1 - Preset 15**) softkeys to assign a number to the active channel.
- 4) Press the **Set** Softkey on the desired channel number to save the channel as a preset.

Adjusting Volume

- 1) With the 'Aux - XM Radio' Page displayed, press the **Volume** Softkey.
- 2) Press the **VOL –** Softkey to reduce volume or press the **VOL +** Softkey to increase volume. (Once the **VOL** Softkey is pressed, the volume can also be adjusted using the small **FMS** Knob.)

SiriusXM Radio volume may also be adjusted on each passenger headset.

Muting SiriusXM audio

- 1) Select the 'Aux - XM Radio' Page or 'Aux - XM Information' Page.
- 2) Press the **Mute** Softkey to mute the audio. Press the **Mute** Softkey again to unmute the audio.

SATELLITE TELEPHONE

Disabling/enabling telephone and low speed data services

- 1) Turn the large **FMS** Knob on the MFD to select the Aux page group.
- 2) Turn the small **FMS** Knob to select the 'Aux - Telephone' Page.
- 3) Press the **MENU** Key. The Page Menu window is now displayed.
- 4) Turn the **FMS** Knob to select 'Disable Iridium Transmission' in the menu list.
- 5) Press the **ENT** Key. The Iridium transceiver is now disabled.

- 6) To enable the Iridium transceiver, repeat steps 1 through 3, then select 'Enable Iridium Transceiver'.

Viewing the Telephone Page

- 1) Turn the large **FMS** Knob on the MFD to select the Aux page group.
- 2) Turn the small **FMS** Knob to select the 'Aux - Telephone' or 'Aux - Text Messaging' Page.
- 3) If necessary, press the **Phone** Softkey to display the 'Aux - Telephone' Page.

Internal Phone	External Phone	Description
 Idle	 Idle	Phone is Idle
 Ringing	 Ringing	Phone is ringing
 Connected	 Connected	Phone has a dial tone (off hook) or connected to another phone
 	 	Phone dialed is busy
 Dialing	 Dialing	Phone is dialing another phone
 	 	Phone has failed
 	 	Phone status not known
	 Disabled	Phone is disabled

Internal Phone	External Phone	Description
		Phone is reserved for data transmission
		Calling other phone or incoming call from other phone
		Other phone is on hold
		Phones are connected

Entering a new contact

- 1) With the 'Aux - Telephone' Page displayed, press the **FMS** Knob to display the cursor.
- 2) If necessary, turn either **FMS** Knob to place the cursor on 'New Entry'.
- 3) Press the **ENT** Key. The cursor moves the 'Name' field of the 'Contact Details' window.
- 4) Enter the desired name of the new contact. Entry can be accomplished through the alphanumeric keys on the MFD Controller, or the **FMS** Knobs on the controller or the MFD.
- 5) Press the **ENT** Key. The cursor moves to the 'Phone Number' field.
- 6) Enter the desired telephone number. Entry can be accomplished through the alphanumeric keys on the MFD Controller, or the **FMS** Knobs on the controller or the MFD.
- 7) Press the **ENT** Key. The cursor moves to the 'Email' field.
- 8) Enter the desired email address. Entry can be accomplished through the alphanumeric keys on the MFD Controller, or the **FMS** Knobs on the controller or the MFD.
- 9) Press the **ENT** Key. The **Save** button is highlighted.
- 10) Press the **ENT** Key. The new contact entry is added to the list of saved contacts.

Delete a contact

- 1) With the 'Aux - Telephone' Page displayed, press the **FMS** Knob to display the cursor.
- 2) Turn either **FMS** Knob to place the cursor on the desired contact name.
- 3) Press the **Delete** Softkey. A confirmation window is displayed.
- 4) With 'OK' highlighted, press the **ENT** Key to delete the selected contact.

Editing a contact

- 1) With the 'Aux - Telephone' Page displayed, press the **FMS** Knob to display the cursor.
- 2) Turn either **FMS** Knob to place the cursor on the desired contact name.
- 3) Press the **Edit** Softkey. The cursor is placed in the 'Name' field. Enter the desired changes. Entry can be accomplished through the alphanumeric keys on the MFD Controller, or the **FMS** Knobs on the controller or the MFD.
- 4) Press the **ENT** Key when each field is complete. The **Save** Button is now highlighted.
- 5) Press the **ENT** Key to save the changes.

Incoming Calls



NOTE: The Push-to-Talk switch is not utilized for telephone communication. The microphone is active after pressing the **Answer** Softkey, and stays active until the call is terminated.

Answering an incoming call in the cockpit

- 1) Press the **TEL** Key on the appropriate audio panel.
- 2) Press the **Answer** Softkey on the MFD.

Pressing the **Ignore** Softkey will extinguish the pop-up window and leave the call unanswered. Pressing the **Phone** Softkey will display the 'Aux - Telephone' Page allowing additional call information to be viewed before answering.

Muting incoming call alerts

- 1) With the 'Aux - Telephone' Page displayed, press the **MENU** Key on the MFD to display the Page Menu.
- 2) Turn either **FMS** Knob to place the cursor on 'Disable Incoming Call Alerts'.
- 3) Press the **ENT** Key. The voice and pop-up alert will not be displayed now when an incoming call is received.

Outgoing Calls

Voice calls can be made from the cockpit to the handset in the cabin, or to an external phone through the Iridium Satellite Network.

Making a call to the cabin

- 1) Press the **TEL** Key on the appropriate audio panel.
- 2) Press the **Dial** Softkey on the MFD.
- 3) Press the **ENT** Key. The cursor will move from 'Cabin' to 'OK'.
- 4) Press the **ENT** Key again. The cabin phone will now ring.

To exit the call, press the **Hangup** Softkey.

Making an external call from the cockpit using the Iridium satellite network

- 1) Press the **TEL** Key on the appropriate audio panel.
 - 2) Press the **Dial** Softkey on the MFD.
 - 3) Turn the small **FMS** Knob to select 'Iridium'.
 - 4) Press the **ENT** Key. The cursor has now moved to the phone number entry field.
 - 5) Enter the desired telephone number (country code first) by pressing the number softkeys on the MFD or by pressing the numeric keys on the MFD Control Unit.
 - 6) Press the **ENT** Key. 'OK' is highlighted.
 - 7) Press the **ENT** Key. The system will begin calling the number.
- To exit the call, press the **Hangup** Softkey.

Making an external call from the cockpit by using the Contact List

- 1) Press the **TEL** Key on the appropriate audio panel.
- 2) Press the **FMS** Knob to activate the cursor.
- 3) Turn the small **FMS** Knob to select the desired contact name in the list of contacts.
- 4) Press the **Call** Softkey. The external call is initiated and the number associated with the contact name is dialed.

Placing a call on hold

Press the **Hold** Softkey on the MFD.

Transferring a call

- 1) Press the **Trans** Softkey on the MFD.
- 2) Press the **ENT** Key. The cursor now highlights the phone number entry field.
- 3) Enter the phone number to which the call is being transferred, as discussed earlier for making an external call on the Iridium satellite network.
- 4) Press the **ENT** Key. 'OK' is now highlighted.
- 5) Press the **ENT** Key again to place the call.

Adding another phone to an active call

- 1) Press the **Conf** Softkey on the MFD.
- 2) Press the **ENT** Key. The cursor now highlights the phone number entry field.
- 3) Enter the phone number to be added to the call, as discussed earlier for making an external call on the Iridium satellite network.
- 4) Press the **ENT** Key. 'OK' is now highlighted.
- 5) Press the **ENT** Key again to place the call.

Viewing the Text Messaging Page

- 1) Turn the large **FMS** Knob on the MFD to select the Aux page group.
- 2) Turn the small **FMS** Knob to select the 'Aux - Telephone' or 'Aux - Text Messaging' Page.
- 3) If necessary, press the **SMS** Softkey to display the 'Aux - Text Messaging' Page.

Message Symbol	Description
	Received text message that has not been opened
	Received text message that has been opened
	Saved text message, draft not sent
	System is sending text message

Message Symbol	Description
	Text message has been sent
	System failed to send text message
	Predefined text message

Text Message Symbols

Viewing a Text Message When Received

Press the **View** Softkey to view the message. Pressing the **Ignore** Softkey will extinguish the pop-up window and leave the text message unopened. Pressing the **Ignore All** Softkey will extinguish the pop-window and ignore all future incoming text messages. Pressing the **SMS** Softkey will display the 'Aux - Text Messaging' Page.

Enabling/disabling incoming text message pop-up alerts

- 1) With the 'Aux - Text Messaging' Page displayed, press the **MENU** Key on the MFD to display the Page Menu.
- 2) Turn either **FMS** Knob to place the cursor on 'Disable New Message Popups' or 'Enable New Message Popups'.
- 3) Press the **ENT** Key. The pop-up alert will not be displayed when an incoming text message is received.

Replying to a text message

While viewing the text message, press the **Reply** Softkey.

Sending a new text message

- 1) While viewing the 'Aux - Text Messaging' Page, press the **New** Softkey.
- 2) The SMS Text Message Draft Window is now displayed with the cursor in the 'To' field. Enter the desired telephone number or email address. Entry can be accomplished through the alphanumeric keys on the MFD Controller, or combination of the **FMS** Knob on the controller and softkeys on the MFD. The **FMS** Knob is used to enter letters and numbers, or numbers can be entered from the MFD by pressing the **Numbers** Softkey. Press the **CapsLock** Softkey to create upper and lower case alpha characters. Special characters can be accessed by pressing the **Symbols** Softkey.
- 3) Press the **ENT** Key. The cursor is now displayed in the 'Message' field.

- 4) Enter the desired message using any combination of entry methods as described in step 2.
- 5) Press the **ENT** Key.
- 6) Confirm you wish to send the message by pressing the **ENT** Key with 'Yes' highlighted. Select 'No' to return to the message entry field and either press the **Cancel** Softkey to delete the message or press the **Save** Softkey to save the message as a draft.

Sending a new text message/email to a saved contact

- 1) With the 'Aux - Telephone' Page displayed, press the **FMS** Knob to display the cursor.
- 2) Turn either **FMS** Knob to place the cursor on the desired contact name.
- 3) Press either the **Text Email** Softkey to send to the email address saved for the contact or **Text Phone** Softkey to send to the phone number saved for the contact. The cursor is placed in the 'To' field. Confirm the contact you wish to send a message to by pressing the **ENT** Key.
- 4) Enter the desired message, then press the **ENT** Key.
- 5) Confirm you wish to send the message by pressing the **ENT** Key with 'Yes' highlighted. Select 'No' to return to the message entry field and either press the **Cancel** Softkey to delete the message or press the **Save** Softkey to save the message as a draft.

Creating a predefined text message

- 1) While viewing the 'Aux - Text Messaging' Page, press the **MENU** Key to display the Page Menu.
- 2) Turn either **FMS** Knob to select 'Edit Predefined Messages'.
- 3) Press the **ENT** Key. The Predefined Messages view is now displayed.
- 4) Press the **New** Softkey.
- 5) The cursor is displayed in the 'Title' field. Enter the desired message title. Entry can be accomplished through the alphanumeric keys on the MFD Controller, or combination of the **FMS** Knob on the controller and softkeys on the MFD. The **FMS** Knob is used to enter letters and numbers, or numbers can be entered from the MFD by pressing the **Numbers** Softkey. Press the **CapsLock** Softkey to create upper and lower case alpha characters. Special characters can be accessed by pressing the **Symbols** Softkey.

- 6) Press the **ENT** Key. The cursor is now displayed in the 'Message' field.
- 7) Enter the desired message using any combination of entry methods as described in step 5.
- 8) Press the **ENT** Key.
- 9) Press the **Save** Softkey. The new predefined message is now shown in the displayed list. Pressing the **Cancel** Softkey will delete the message without saving.
- 10) Press the **MENU** Key to display the Page Menu.
- 11) Turn either **FMS** Knob to place the cursor on 'Cancel Drafting Message'.
- 12) Press the **ENT** Key.

Sending a predefined text message

- 1) While viewing the 'Aux - Text Messaging' Page, press the New Softkey.
- 2) The Predefined SMS Text Message Window is now displayed with the cursor in the 'TO' field. Enter the desired telephone number or email address. Entry can be accomplished through the alphanumeric keys on the MFD Controller, or combination of the **FMS** Knob on the controller and softkeys on the MFD. The **FMS** Knob is used to enter letters and numbers, or numbers can be entered from the MFD by pressing the **Numbers** Softkey. Press the **CapsLock** Softkey to create upper and lower case alpha characters. Special characters can be accessed by pressing the **Symbols** Softkey.
- 3) Press the **ENT** Key. The cursor is now displayed in the 'Message' field.
- 4) Press the **PREDEFD** Softkey. The Predefined Message Menu Window is displayed.
- 5) Turn either **FMS** Knob to select the desired predefined message.
- 6) Press the **ENT** Key. The predefined message text is inserted into the message field. If desired, the message can be edited by using the FMS Knobs.
- 7) Press the **ENT** Key.
- 8) Press the **Send** Softkey to transmit the message.

Showing Inbox messages

While viewing the 'Aux - Text Messaging' Page, press the **Arrange** Softkey, then press the **Inbox** Softkey.

Showing Outbox messages

While viewing the 'Aux - Text Messaging' Page, press the **Arrange** Softkey, then press the **Outbox** Softkey.

Showing Draft messages

While viewing the 'Aux - Text Messaging' Page, press the **Arrange** Softkey, then press the **Drafts** Softkey.

Viewing messages sorted by message date/time

While viewing the 'Aux - Text Messaging' Page, press the **Arrange** Softkey, then press the **Time** Softkey.

Viewing messages sorted by message type

While viewing the 'Aux - Text Messaging' Page, press the **Arrange** Softkey, then press the **Type** Softkey.

Viewing messages sorted by address

While viewing the 'Aux - Text Messaging' Page, press the **Arrange** Softkey, then press the **Address** Softkey.

Viewing the content of a text message

- 1) While viewing the 'Aux - Text Messaging' Page, select the desired message box.
- 2) Press the **FMS** Knob to activate the cursor.
- 3) Turn either **FMS** Knob to select the desired message.
- 4) Press the **VIEW** Softkey.

Or:

Press the **ENT** Key. Message content is displayed.

- 5) To close the text message, press the **Close** Softkey.

Marking selected message as read

- 1) While viewing the Inbox on the 'Aux - Text Messaging' Page, press the **FMS** Knob to activate the cursor.
- 2) Turn either **FMS** Knob to select the desired message.
- 3) Press the **MRK Read** Softkey.

The message symbol now indicates the message has been opened.

Marking all messages as read

- 1) While viewing the Inbox on the 'Aux - Text Messaging' Page, press the **MENU** Key to display the Page Menu.
- 2) Turn either **FMS** Knob to place the cursor on 'Mark All New Messages As Read'.
- 3) Press the **ENT** Key. A confirmation window is displayed.
- 4) With cursor highlighting 'YES', press the **ENT** Key. The message symbols now indicate all the message have been opened.

Deleting a message

- 1) While viewing the Inbox on the 'Aux - Text Messaging' Page, press the **FMS** Knob to activate the cursor.
- 2) Turn either **FMS** Knob to select the desired message.
- 3) Press the **Delete** Softkey.

SCHEDULER

The Scheduler feature can be used to enter and display reminder messages (e.g., Change oil, Switch fuel tanks, or Altimeter-Transponder Check) in the Alerts Window on the PFD. Messages can be set to display based on a specific date and time (event), once the message timer reaches zero (one-time; default setting), or recurrently whenever the message timer reaches zero (periodic). Message timers set to periodic alerting automatically reset to the original timer value once the message is displayed. When power is cycled, all messages are retained until deleted, and message timer countdown is resumed

Entering a scheduler message

- 1) Select the 'Aux - Utility' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight the first empty scheduler message naming field.
- 4) Use the **FMS** Knob to enter the message text to be displayed in the Messages Window and press the **ENT** Key.
- 5) Press the **ENT** Key again or use the large **FMS** Knob to move the cursor to the field next to Type.

- 6) Turn the small **FMS** Knob to select the message type:
 - Event—Message issued at the specified date/time
 - One time—Message issued when the message timer reaches zero (default setting)
 - Periodic—Message issued each time the message timer reaches zero
- 7) Press the **ENT** Key again or use the large **FMS** Knob to move the cursor to the next field.
- 8) For periodic and one-time messages, use the **FMS** Knob to enter the timer value (HHH:MM:SS) from which to countdown and press the **ENT** Key.
- 9) For event-based messages:
 - a) Use the **FMS** Knob to enter the desired date (DD-MM-YY) and press the **ENT** Key.
 - b) Press the **ENT** Key again or use the large **FMS** Knob to move the cursor to the next field.
 - c) Use the **FMS** Knob to enter the desired time (HH:MM) and press the **ENT** Key.
- 10) Press the **ENT** Key again or use the large **FMS** Knob to move the cursor to enter the next message.

Deleting a scheduler message

- 1) Select the 'Aux - Utility' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight the name field of the scheduler message to be deleted.
- 4) Press the **CLR** Key to clear the message text. If the **CLR** Key is pressed again, the message is restored.
- 5) Press the **ENT** Key while the message line is cleared to clear the message time.

CREW PROFILES

Creating a profile

- 1) Select the 'Aux - System Setup (X)' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight 'Create' in the Crew Profile Box.

- 4) Press the **ENT** Key. A 'Create Profile' Window is displayed.
 - 5) Use the **FMS** Knob to enter a profile name up to 16 characters long and press the **ENT** Key. Crew profile names cannot begin with a blank as the first letter.
 - 6) In the next field, use the small **FMS** Knob to select the desired settings upon which to base the new profile. Profiles can be created based on Garmin factory defaults, default profile settings (initially based on Garmin factory defaults unless edited by the pilot), or other previously created profile settings.
 - 7) Press the **ENT** Key.
 - 8) With 'Create' highlighted, press the **ENT** Key to create the profile
- Or:**
- Use the large **FMS** Knob to select 'Create & Activate' and press the **ENT** Key to activate the new profile.
- 9) To cancel the process, select 'Cancel' with the large FMS Knob and press the **ENT** Key.

Selecting an active profile

- 1) Select the 'Aux - System Setup (X)' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight the active profile field in the 'Crew Profile' Box.
- 4) Turn the small **FMS** Knob to display the crew profile list and highlight the desired profile.
- 5) Press the **ENT** Key. The system loads and displays the system settings for the selected profile.

Renaming a profile

- 1) Select the 'Aux - System Setup (X)' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight 'Rename' in the 'Crew Profile' Box.
- 4) Press the **ENT** Key.
- 5) In the 'Rename Profile' Window, turn the **FMS** Knob to select the profile to rename.

- 6) Press the **ENT** Key.
- 7) Use the **FMS** Knob to enter a new profile name up to 16 characters long and press the **ENT** Key.
- 8) With 'Rename' highlighted, press the **ENT** Key.
- 9) To cancel the process, use the large **FMS** Knob to select 'Cancel' and press the **ENT** Key.

Deleting a profile

- 1) Select the 'Aux - System Setup (X)' Page.
- 2) Press the **FMS** Knob momentarily to activate the flashing cursor.
- 3) Turn the large **FMS** Knob to highlight 'Delete' in the 'Crew Profile' Box.
- 4) Press the **ENT** Key.
- 5) In the 'Delete Profile' Window, turn the **FMS** Knob to select the profile to delete.
- 6) Press the **ENT** Key.
- 7) With 'Delete' highlighted, press the **ENT** Key.
- 8) To cancel the process, use the large **FMS** Knob to select 'Cancel' and press the **ENT** Key.

Importing a profile from an SD card

- 1) Insert an SD card containing the crew profile(s) into the top card slot on the MFD.
- 2) Turn the **FMS** Knob to select the 'Aux - System Setup (X)' Page.
- 3) Press the **Import** Softkey.
- 4) The system displays the 'Crew Profile Importing' Window with 'Import' highlighted. Turn the large **FMS** Knob to highlight the 'Profile Name' Field, then scroll to the desired profile name with the large and small **FMS** Knobs, then press the **ENT** Key. Then press the **ENT** Key with 'Import' highlighted.
- 5) If the imported profile name is the same as an existing profile on the system, the system displays an 'Overwrite existing profile? OK or Cancel' prompt. Press the **ENT** Key to replace profile on the system with the profile imported from the SD card, or turn the **FMS** Knob to highlight 'Cancel' and press the **ENT** Key to return to the 'Crew Profile Importing' Window.

- 6) If successful, the system displays 'Crew profile import succeeded.' in the window below. With 'OK' highlighted, press the **ENT** or **CLR** Keys or press the **FMS** Knob to return to the 'Aux - System Setup (X)' Page. The imported profile becomes the active profile.

Exporting a profile to an SD card

- 1) Insert the SD card for storing the Crew Profile into the top card slot on the MFD.
- 2) Turn the **FMS** Knob to select the 'Aux - System Setup (X)' Page.
- 3) Press the **Export** Softkey. The system displays the 'Crew Profile Exporting' Window.
- 4) To export the crew profile using the current selected profile, press the **ENT** Key with 'Export' highlighted. To change the selected profile, turn the large **FMS** Knob to highlight the 'Profile Name' field, then scroll to the desired profile name with the large and small FMS Knobs, then press the **ENT** Key. Then press the **ENT** Key with 'Export' highlighted.
- 5) If the selected profile to be exported is the same as an existing profile file name on the SD card, the system displays an 'Overwrite existing profile? OK or Cancel' prompt. Press the **ENT** Key to replace the profile on the SD card with the profile to be exported, or turn the FMS Knob to highlight 'Cancel' and press the **ENT** Key to return to the 'Pilot Profile Exporting' Window without exporting the profile.
- 6) If successful, the window displays 'Crew profile export succeeded.' With 'OK' highlighted, press the **ENT** or **CLR** Keys, or press the **FMS** Knob to return to the 'Aux - System Setup (X)' Page.

CONNEXT® BLUETOOTH® MANAGEMENT

The Connex Setup Page allows for setting up the installed optional Flight Stream device for a Bluetooth connection between the system and a mobile device running the Garmin Pilot™ application.

The mobile device must be 'paired' with the system in order to use the various functions. Pairing is accomplished by first placing the system in pairing mode by displaying the Connex Setup Page. The system is 'discoverable' whenever this page is displayed. The pairing operation is completed from the mobile device and the Garmin Pilot application. See the device Bluetooth pairing instructions and the connection instructions in the Garmin Pilot application.

Viewing the Connex Setup Page

- 1) Turn the large **FMS** Knob on the MFD to select the Aux page group.
- 2) Turn the small **FMS** Knob to select the Connex Setup page.

Changing the Bluetooth Name

- 1) While viewing the 'Aux - Connex Setup' Page, press the **FMS** Knob to activate the cursor.
- 2) Turn the large **FMS** Knob to place the cursor in the 'Bluetooth Name' field.
- 3) Enter the desired name by using the large **FMS** Knob to select the character field, and the small **FMS** Knob select the desired alphanumeric character for that field.
- 4) Press the **ENT** Key. The cursor is removed and the new name is displayed.

Enabling/Disabling Flight Plan Importing from Garmin Pilot

- 1) While viewing the 'Aux - Connex Setup' Page, press the **FMS** Knob to activate the cursor.
- 2) Turn the large **FMS** Knob to place the cursor in the 'Flight Plan Import' field.
- 3) Turn the small **FMS** Knob to select 'Enabled' or 'Disabled'.
- 4) Press the **FMS** Knob to remove the cursor.

Enabling/Disabling WiFi Database Importing from Garmin Pilot

- 1) While viewing the Connex Setup Page, press the **FMS** Knob to activate the cursor.
- 2) Turn the large **FMS** Knob to place the cursor in the 'WiFi Database Import' field.
- 3) Turn the small **FMS** Knob to select 'Enabled' or 'Disabled'.
- 4) Press the **FMS** Knob to remove the cursor.

Enabling/Disabling Automatic Reconnection of a Specific Paired Device

- 1) While viewing the 'Aux - Connex Setup' Page, press the **FMS** Knob to activate the cursor.
- 2) Turn the large **FMS** Knob to highlight the desired paired device.
- 3) Turn the small **FMS** Knob to select 'Enabled' or 'Disabled'. Selecting 'Enabled' allows the system to automatically connect to a previously paired device when detected.
- 4) Press the **FMS** Knob to remove the cursor.

Remove a Specific Paired Device from the List of Paired Devices

- 1) While viewing the 'Aux - Connex Setup' Page, press the **FMS** Knob to activate the cursor.
- 2) Turn the large **FMS** Knob to highlight the desired paired device.
- 3) Press the **Remove** Softkey. A confirmation screen is displayed.
- 4) If necessary, turn the large **FMS** Knob to select 'YES'.
- 5) Press the **ENT** Key to remove the device from the list of paired devices.

ELECTRONIC STABILITY AND PROTECTION (ESP™)

Enabling/disabling ESP

- 1) Turn the large **FMS** Knob to select the Aux Page Group.
- 2) Turn the small **FMS** Knob to select the System Setup Page.
- 3) If necessary, press the **Setup 2** Softkey to display the 'Aux - System Setup 2' Page.
- 4) Press the **FMS** Knob to activate the cursor.
- 5) Turn the large **FMS** Knob to place the cursor in the 'Stability & Protection' field.
- 6) Turn the small **FMS** Knob to select 'Enable' or 'Disable'.
- 7) Press the **FMS** Knob to remove the cursor.

ELECTRONIC CHECKLISTS



NOTE: The optional checklists presented here are for example only and may differ from checklists available for the DA42NG. The information described in this section is not intended to replace the checklist information described in the AFM or the Pilot Safety and Warning Supplements document.



NOTE: Garmin is not responsible for the content of checklists. Checklists are created by the aircraft manufacturer. Modifications or updates to the checklists are coordinated through the aircraft manufacturer. The user cannot edit these checklists.

The MFD is able to display optional electronic checklists which allow a pilot to quickly find the proper procedure on the ground and during each phase of flight. The system accesses the checklists from an SD card inserted into the bezel slot. If the SD card contains an invalid checklist file or no checklist, the Power-up Page messages display 'Checklist File: Invalid' or 'Checklist File: N/A (not available)' and the **Checklist** Softkey is not available.

Accessing and navigating checklists

- 1) From any page on the MFD (except the EIS Pages), press the **Checklist** Softkey or turn the large **FMS** Knob to select the 'Checklist' Page.
- 2) Turn the large **FMS** Knob to select the 'Group' field.
- 3) Turn the small **FMS** Knob to select the desired procedure and press the **ENT** Key.
- 4) Turn the large **FMS** Knob to select the 'Checklist' field.
- 5) Turn the **FMS** Knob to select the desired checklist and press the **ENT** Key. The selected checklist item is indicated with white text surrounded by a white box.
- 6) Press the **ENT** Key or **Check** Softkey to check the selected checklist item. The line item turns green and a checkmark is placed in the associated box. The next line item is automatically selected for checking.

Either **FMS** Knob can be used to scroll through the checklist and select the desired checklist item.

Press the **CLR** Key or **Uncheck** Softkey to remove a check mark from an item.

- 7) When all checklist items have been checked, '*Checklist Finished*' is displayed in green text at the bottom left of the checklist window. If all items in the checklist have not be checked, '*Checklist Not Finished*' will be displayed in yellow text.
- 8) Press the **ENT** Key. 'Go To Next Checklist?' will be highlighted by the cursor.
- 9) Press the **ENT** Key to advance to the next checklist.
- 10) Press the **Exit** Softkey to exit the Checklist Page and return to the page last viewed.

Accessing emergency procedures

- 1) From any page on the MFD (except the EIS Pages), press the **Checklist** Softkey or turn the large **FMS** Knob to select the Checklist Page.
- 2) Press the **EMER** Softkey.
- 3) Turn the **FMS** Knob to select the desired emergency checklist and press the **ENT** Key.
- 4) Press the **ENT** Key or **Check** Softkey to check the selected emergency checklist item. The line item turns green and a checkmark is placed in the box next to it. The next line item is automatically highlighted for checking. Either **FMS** Knob can be used to scroll through the checklist and select the desired checklist item.

Press the **CLR** Key or **Uncheck** Softkey to remove a check mark from an item.

- 5) When all checklist items have been checked, '*Checklist Finished*' is displayed in green text at the bottom left of the checklist window. If all items in the checklist have not be checked, '*Checklist Not Finished*' will be displayed in yellow text.
- 6) Press the **ENT** Key. 'Go To Next Checklist?' will be highlighted by the cursor.
- 7) Press the **ENT** Key to advance to the next checklist.
- 8) Press the **Return** Softkey to return to the previous checklist.
- 9) Press the **Exit** Softkey to exit the Checklist Page and return to the page last viewed.

Flight
Instruments

EIS

Nav/Com/
XPDR/Audio

Flight
Management

Hazard
Avoidance

AFCS

Additional
Features

Abnormal
Operation

Annun/Alerts

Appendix

Index

Blank Page

ABNORMAL OPERATION

REVERSIONARY MODE

Should a system detected failure occur in either display, the system automatically enters reversionary mode. In reversionary mode, critical flight instrumentation is combined with engine instrumentation on the remaining display.

Reversionary display mode can be manually activated by pressing the **DISPLAY BACKUP** Button on the audio panel.



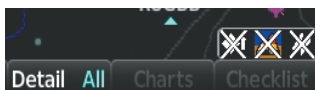
NOTE: *The Diamond DA42NG Pilot's Operating Handbook (POH) always takes precedence over the information found in this section.*

ABNORMAL COM OPERATION

When a COM tuning failure is detected by the system, the emergency frequency (121.500 MHz) is automatically loaded into the active frequency field of the COM radio for which the tuning failure was detected. In the event of a failure of both PFD and MFD, the emergency frequency (121.500 MHz) automatically becomes the active frequency on both COM radios.

HAZARD DISPLAYS WITH LOSS OF GPS POSITION

If GPS position is lost, or becomes invalid, selected hazards being displayed on the Navigation Map Page are removed until GPS position is again established.



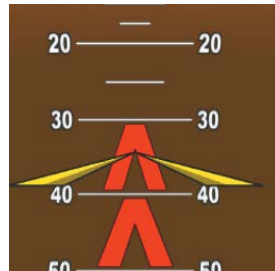
Loss of Hazard Functions with Loss of GPS Position

UNUSUAL ATTITUDES

The PFD ‘declutters’ when the aircraft enters an unusual attitude. Only the primary functions are displayed in these situations.

The following information is removed from the PFD (and corresponding softkeys are disabled) when the aircraft experiences unusual attitudes:

- Traffic Annunciations
- AFCS Annunciations
- Flight director Command Bars
- Inset Map
- Temperatures
- DME Information Window
- Wind Data
- Selected Heading Box
- Selected Course Box
- Transponder Status Box
- System Time
- PFD Setup Menu
- Windows displayed in the lower right corner of the PFD:
 - Timer/References
 - Nearest Airports
 - Flight Plan
 - Messages
 - Procedures
 - DME Tuning
- Barometric Minimum Descent Altitude Box
- Glideslope, Glide-path, and Vertical Deviation Indicators
- Altimeter Barometric Setting
- Selected Altitude
- VNV Target Altitude



Extreme Pitch Indication

DEAD RECKONING

While in Enroute or Oceanic phase of flight, if the system detects an invalid GPS solution or is unable to calculate a GPS position, the system automatically reverts to Dead Reckoning (DR) Mode. In DR Mode, the system uses its last-known position combined with continuously updated airspeed and heading data (when available) to calculate and display the aircraft's current estimated position.



NOTE: *Dead Reckoning Mode only functions in Enroute (ENR) or Oceanic (OCN) phase of flight. In all other phases, an invalid GPS solution produces a "NO GPS POSITION" annunciation on the map and the system stops navigating in GPS Mode.*

DR Mode is indicated on the system by the appearance of the letters 'DR' superimposed in yellow over the 'own aircraft' symbol as shown in the following figure. In addition, 'DR' is prominently displayed, also in yellow, on the HSI slightly above and to the right of the aircraft symbol on the CDI as shown in the following figure. The CDI deviation bar is displayed in yellow, but will be removed from the display after 20 minutes. Lastly, but at the same time, a 'GPS NAV LOST' alert message appears on the PFD.

Normal navigation using GPS/SBAS source data resumes automatically once a valid GPS solution is restored.

It is important to note that estimated navigation data supplied by the system in DR Mode may become increasingly unreliable and must not be used as a sole means of navigation. If, while in DR Mode, airspeed and/or heading data is also lost or not available, the DR function may not be capable of estimating your position and, consequently, the system may display a path that is different than the actual movement of the aircraft. Estimated position information displayed by the system through DR while there is no heading and/or airspeed data available should not be used for navigation.

DR Mode is inherently less accurate than the standard GPS/SBAS Mode due to the lack of satellite measurements needed to determine a position. Changes in wind speed and/or wind direction compounds the relative inaccuracy of DR Mode. Because of this degraded accuracy, the crew must maintain position awareness using other navigation equipment until GPS-derived position data is restored.



CDI 'DR' Indication on PFD



Symbolic Aircraft
(Map pages and Inset Map)

Dead Reckoning Indications

As a result of operating in DR Mode, all GPS-derived data is computed based upon an estimated position and is displayed as amber text on the display to denote degraded navigation source information. This data includes the following:

- Navigation Status Box fields except Active Leg, TAS, and DTK
- GPS Bearing Pointer
- Wind data and pointers in the Wind Data Box on the PFD
- Current Track Indicator
- All Bearing Pointer Distances
- Active Flight Plan distances, bearings, and ETE values

Also, while the system is in DR Mode, the autopilot will couple to GPS for up to 20 minutes. Terrain functions are also disabled. Additionally, the accuracy of all nearest information (airports, airspaces, and waypoints) is questionable. Finally, airspace alerts continue to function, but with degraded accuracy.

GARMIN CONNEXT WEATHER REQUEST STATUS

Weather Request Status Message	Description
Auto requests inhibited Send manual request to reset.	The system has disabled automatic weather data requests due to excessive errors. Automatic weather data requests have stopped. Send a manual weather data request to resume automatic updates.
Auto update retry: ## Seconds	The system will attempt another automatic weather data request after an error occurred during the previous request. Timer counts down until the next automatic request occurs.

Weather Request Status Message	Description
Connex Comm Error [1]	A general error has occurred. If the error persists, the system should be serviced.
Connex Comm Error [2]	A communications error has occurred with the GIA. The system should be serviced.
Connex Comm Error [3]	A general error has occurred. If the error persists, the system should be serviced.
Connex Comm Error [4]	This occurs if multiple automatic weather data requests have recently failed, or the GIA is off-line.
Connex Comm Error [5]	This can occur if the Iridium or Garmin Connex services are not accessible. Check Iridium signal strength. If this error persists, the system should be serviced.
Connex Comm Error [6]	A communications error has occurred. If this error persists, the system should be serviced.
Connex Comm Error [7]	A weather data transfer has timed out. Check Iridium signal strength and re-send the data request.
Connex Comm Error [8]	A server error has occurred or invalid data received.
Connex Comm Error [9]	An error occurred while reading or writing data. If the error persists, the system should be serviced.
Connex Login Invalid	There is a problem with the Garmin Connex registration. Contact Garmin at 1-866-739-5687 in the United States or 913-397-8200, wait for the operator, and request Extension 1135 for assistance.
Connex Server Temp Inop	The Garmin Connex Weather data server is temporarily out of service, but is expected to return to service in less than 30 minutes.
Connex Server Inop	The Garmin Connex Weather data server will be out of service for at least 30 minutes.
Invalid Coverage Area	The weather data request coverage area does not contain at least one of the following: a waypoint, a flight plan, or a flight plan destination. Verify at least one of the coverage options is enabled (checked) and contains required criteria, then re-send the data request.
No Connex Subscription	The system is not currently subscribed to Garmin Connex services or the access code is incorrect. Verify the access code. Contact Garmin at 1-866-739-5687 in the United States or 913-397-8200, wait for the operator, and request Extension 1135 for assistance.

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

Weather Request Status Message	Description
Reduce Request Area	The size of the received weather data has exceeded system memory limits. Reduce the size of the coverage area and issue another Connex Data Request to ensure all available weather data has been received.
Request Canceled	The user has cancelled a Connex Data Request.
Request Failed - Try Again	The weather data request timed-out. Re-send data request.

ANNUNCIATIONS & ALERTS

CREW ALERTING SYSTEM (CAS) MESSAGES

WARNING Annunciations

Annunciation Window Text	Alerts Window Message	Audio Alert
ALTN AMPS	Engine alternator output is > 70 amps.	Continuous Aural Tone
ALTN FAIL	Engine alternator has failed.	
DOOR OPEN	Front or rear door is not closed.	
ENG TEMP	Engine coolant temp is >105 deg. C.	
FUEL PRES	Engine fuel pressure is low.	
GBOX TEMP	Engine gearbox temp is > 120 deg. C.	
L FUEL TEMP	Left fuel temp is greater than 60 deg. C.	
OIL PRES	Engine oil pressure is less than 0.9 bar.	
OIL TEMP	Engine oil temp is greater than 140 deg. C.	
R FUEL TEMP	Right fuel temp is greater than 60 deg. C.	
STARTER	Engine starter is engaged.	

CAUTION Annunciations

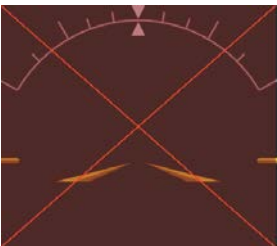
Annunciation Window Text	Alerts Window Message	Audio Alert
COOL LVL	Engine coolant level is low.	Single Aural Tone
ECU A FAIL	Engine ECU A has failed.	
ECU B FAIL	Engine ECU B has failed.	
FUEL LOW	Left fuel quantity is low	
VOLTS LOW	Bus voltage is less than 25 volts.	
PITOT FAIL	Pitot heat has failed.	
PITOT HT OFF	Pitot heat is off.	

ADVISORY Annunciations

Annunciation Window Text	Alerts Window Message	Audio Alert
GLOW ON	Engine glow plug active.	None
FUEL XFER	Fuel transfer in progress	

SYSTEM ANNUNCIATIONS

When an LRU or an LRU function fails, a large red or amber ‘X’ is typically displayed on windows associated with the failed data. Refer to the Pilot’s Operating Handbook (POH) for additional information regarding pilot responses to these annunciations.

System Annunciation	Comment
	Air Data, Attitude and Heading Reference System is aligning.
	Display system is not receiving attitude information from the AHRS.
	AHRS calibration incomplete or configuration module failure.
	GPS information is either not present or is invalid for navigation use. Note that AHRS utilizes GPS inputs during normal operation. AHRS operation may be degraded if GPS signals are not present (see POH).
	Display system is not receiving airspeed input from the air data computer.

System Annunciation	Comment
	Display system is not receiving vertical speed input from the air data computer.
	Display system is not receiving valid heading input from the AHRS or magnetometer.
	Display system is not receiving altitude input from the air data computer.
	Display system is not receiving valid OAT information from the air data computer.
	Display system is not receiving valid transponder information.
Other Various Red or Amber X Indications	An red or amber 'X' through any other display field (such as Engine instrumentation fields) indicates that the field is not receiving valid data.

AFCS ALERTS

Alert Condition	Annunciation	Description
Aileron Mistrim Right	AIL→	Roll servo providing sustained force in the indicated direction
Aileron Mistrim Left	←AIL	
Elevator Mistrim Down	↓ELE	Pitch servo providing sustained force in the indicated direction. May indicate a failure of the pitch trim servo or trim system.
Elevator Mistrim Up	↑ELE	

Alert Condition	Annunciation	Description
Pitch Trim Failure (or Stuck AP TRIM Switch)	PTRM	If AP engaged, take control of the aircraft and disengage AP If AP disengaged, move AP TRIM switches separately to unstick
Roll Failure	ROLL	Roll axis control failure; AP inoperative
Pitch Failure	PTCH	Pitch axis control failure; AP inoperative
System Failure	AFCS	AP and MEPT are unavailable; FD may still be available
Preflight Test	PFT	Performing preflight system test; aural alert sounds at completion Do not press the AP DISC Switch during servo power-up and preflight system tests. Doing so may cause the preflight system test to fail or never to start. Cycle power to the servos if this occurs.
	PFT	Preflight system test failed; aural alert sounds at failure

RELATIVE OBSTACLE SYMBOLS AND COLORS

Unlighted Obstacle		Lighted Obstacle		Obstacle Location
< 1000' AGL	> 1000' AGL	< 1000' AGL	> 1000' AGL	
				Red obstacle is above or within 100 ft below the aircraft altitude
				Yellow obstacle is between 100 ft and 1000 ft below the aircraft altitude
				White obstacle is more than 1000 ft below aircraft altitude

RELATIVE WIND TURBINE OBSTACLES AND COLORS

Unlighted Wind Turbine Obstacle	Lighted Wind Turbine Obstacle	Wind Turbine Obstacle Location
		Red obstacle is above or within 100 ft below the aircraft altitude
		Yellow obstacle is between 100 ft and 1000 ft below the aircraft altitude
		White obstacle is more than 1000 ft below aircraft altitude

TERRAIN-SVT AND TAWS-B POTENTIAL IMPACT AREA WITH ANNUNCIATIONS

Potential Impact Area Examples	Alert Type	Example Annunciation
 or 	Warning	TAWS-B Warning PULL UP Terrain-SVT Warning TERRAIN
 or 	Caution	TERRAIN

TERRAIN-SVT ALERTS

Alert Type	PFD/Terrain-SVT Annunciation	MFD Pop-Up Alert (except Terrain-SVT Page)	Voice Alert
Reduced Required Terrain Clearance Warning (RTC)	TERRAIN	WARNING – TERRAIN	“Warning; Terrain, Terrain”
Imminent Terrain Impact Warning (ITI)	TERRAIN	WARNING – TERRAIN	“Warning; Terrain, Terrain”

Alert Type	PFD/Terrain-SVT Annunciation	MFD Pop-Up Alert (except Terrain-SVT Page)	Voice Alert
Reduced Required Obstacle Clearance Warning (ROC)	TERRAIN	WARNING – OBSTACLE	"Warning; Obstacle, Obstacle"
Imminent Obstacle Impact Warning (IOI)	TERRAIN	WARNING – OBSTACLE	"Warning; Obstacle, Obstacle"
Reduced Required Terrain Clearance Caution (RTC)	TERRAIN	CAUTION – TERRAIN	"Caution; Terrain, Terrain"
Imminent Terrain Impact Caution (ITI)	TERRAIN	CAUTION – TERRAIN	"Caution; Terrain, Terrain"
Reduced Required Obstacle Clearance Caution (ROC)	TERRAIN	CAUTION – OBSTACLE	"Caution; Obstacle, Obstacle"
Imminent Obstacle Impact Caution (IOI)	TERRAIN	CAUTION – OBSTACLE	"Caution; Obstacle, Obstacle"

* Annunciation is displayed on the MFD when terrain display is enabled.

TERRAIN-SVT SYSTEM STATUS ANNUNCIATIONS

Alert Type	PFD/MFD [†] Alert Annunciation	Terrain-SVT Page Center Banner Annunciation	Voice Alert
System Test in Progress	TER TEST	TERRAIN TEST	None
System Test Pass	None	None	"Terrain System Test OK"
Terrain Alerting Inhibited	TER INH	None	None
No GPS position	TER N/A	NO GPS POSITION	"Terrain System Not Available" *
Excessively degraded GPS signal; or Out of database coverage area	TER N/A	None	"Terrain System Not Available" *
Terrain System Test Fail; Terrain or Obstacle database unavailable or invalid; Invalid software configuration; or System audio fault	TER FAIL	TERRAIN FAIL	"Terrain System Failure"

Alert Type	PFD/MFD [†] Alert Annunciation	Terrain-SVT Page Center Banner Annunciation	Voice Alert
MFD Terrain or Obstacle database unavailable or invalid, and Terrain-SVT operating with PFD Terrain or Obstacle databases	None	TERRAIN DATABASE FAILURE	None

[†] Annunciation is shown on Terrain-SVT Page and the Navigation Map Page when Terrain is enabled.
* “Terrain System Available” will be heard when sufficient GPS signal is received, or Terrain database coverage area re-entered.

TIS SYSTEM STATUS

Traffic Map Page Center Banner Annunciation	Description
NO DATA	Data is not being received from the transponder*
DATA FAILED	Data is being received from the transponder, but a failure is detected in the data stream*
FAILED	The transponder has failed*
UNAVAILABLE	TIS is unavailable or out of range

* Contact a service center or Garmin dealer for corrective action

Traffic Map Mode Annunciation	Traffic Map Page Center Banner Annunciation	Traffic Overlay Status Icon (Navigation Maps)	Description
DATA FAILED	TRFC FAIL		Data is being received from the transponder, but a failure is detected in the data stream*
NO DATA	TRFC FAIL		Data is not being received from the transponder*
OPERATING	None		TIS is operating and is receiving traffic data from a data link.
OPERATING	UNAVAILABLE		TIS is operating, but the traffic service is currently unavailable or is out of reception range.

Traffic Map Mode Annunciation	Traffic Map Center Banner Annunciation	Traffic Overlay Status Icon (Navigation Maps)	Description
STANDBY	STANDBY		TIS is in Standby Mode.
UNIT FAILED	TRFC FAIL		The transponder has failed*

* Contact a service center or Garmin dealer for corrective action

TIS TRAFFIC STATUS ANNUNCIATIONS

Traffic Status Banner Annunciation	Description
TA OFF SCALE	A Traffic Advisory is outside the selected display range* Annunciation is removed when traffic comes within the selected display range
TA X.X ± XX ↓	System cannot determine bearing of Traffic Advisory** Annunciation indicates distance in nm, altitude separation in hundreds of feet, and altitude trend arrow (climbing/descending)
AGE MM:SS	Appears if traffic data is not refreshed within 6 seconds If after another 6 seconds data is not received, traffic is removed from the display The quality of displayed traffic information is reduced as the age increases
TRFC COAST	The displayed data is not current (6 to 12 seconds since last message) The quality of displayed traffic information is reduced when this message is displayed
TRFC RMVD	Traffic is removed because it is too old for coasting (12 to 60 seconds since last message). Traffic may exist within the selected display range, but it is not displayed
TRFC FAIL	Traffic data has failed
NO TRFC DATA	Traffic has not been detected
TRFC UNAVAIL	The traffic service is unavailable or out of range

*Shown as symbol on Traffic Map Page

**Shown in center of Traffic Map Page

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

TAS TRAFFIC MODES

Mode	Traffic Mode Annunciation (Traffic Map Page)	Traffic Display Status Icon (Other Maps)
Traffic System Test Initiated	TEST (‘TEST MODE’ shown in center of page)	
Operating	OPERATING	
Standby	STANDBY (also shown in white in center of page)	
Traffic System Failed*	FAIL	

TAS FAILURE ANNUNCIATIONS

Traffic Map Page Center Annunciation	Description
NO DATA	Data is not being received from the TAS unit
DATA FAILED	Data is being received from the TAS unit, but the unit is self-reporting a failure
FAILED	Incorrect data format received from the TAS unit

TAS TRAFFIC STATUS ANNUNCIATIONS

Traffic Status Banner Annunciation	Description
TA OFF SCALE	A Traffic Advisory is outside the selected display range*. Annunciation is removed when traffic comes within the selected display range.
TA X.X ± XX ↓	System cannot determine bearing of Traffic Advisory**. Annunciation indicates distance in nm, altitude separation in hundreds of feet, and altitude trend arrow (climbing/descending).
TRFC FAIL	TAS unit has failed (unit is self-reporting a failure or sending incorrectly formatted data)
NO TRFC DATA	Data is not being received from the TAS unit

*Shown as symbol on Traffic Map Page

**Shown in center of Traffic Map Page

ADS-B TRAFFIC MODES

ADS-B Mode	Traffic Mode Annunciation (Traffic Map Page)	Traffic Map Page Center Banner Annunciation	Traffic Display Status Icon (Other Maps)
ADS-B System Test Initiated	ADS-B: TEST	TEST MODE	
ADS-B Operating in Airborne Mode	ADS-B: AIRB	None	
ADS-B Operating in Surface Mode	ADS-B: SURF	None	
ADS-B Traffic Off	ADS-B: OFF	ADS-B TRFC OFF	
ADS-B Traffic Not Available	ADS-B: N/A	ADS-B TRFC N/A	
ADS-B Failed*	ADS-B: FAIL	ADS-B TRFC FAIL	

ADS-B TRAFFIC FAILURE ANNUNCIATIONS

Traffic Map Page Center Annunciation	Description
NO DATA	Data is not being received from the traffic unit
DATA FAILED	Data is being received from the traffic unit, but the unit is self-reporting a failure
FAILED	Incorrect data format received from the traffic unit

ADS-B TRAFFIC STATUS ANNUNCIATIONS

Traffic Status Banner Annunciation	Description
TA OFF SCALE	A Traffic Advisory is outside the selected display range*. Annunciation is removed when traffic comes within the selected display range.
TA X.X ± XX ↓	System cannot determine bearing of Traffic Advisory**. Annunciation indicates distance in nm, altitude separation in hundreds of feet, and altitude trend arrow (climbing/descending).

Traffic Status Banner Annunciation	Description
TRFC FAIL	Traffic unit has failed (unit is self-reporting a failure or sending incorrectly formatted data)
NO TRFC DATA	Data is not being received from the traffic unit

*Shown as symbol on Traffic Map Page

**Shown in center of Traffic Map Page

AUX-ADS-B STATUS PAGE MESSAGES

ADS-B Status Page Item	Status Message	Description
Traffic Application Status: Airborne (AIRB), Surface (SURF), Airborne Alerts (CSA)	On	Traffic application is currently on. Required input data is available, and it meets performance requirements.
	Available to Run	Traffic application is not currently active, but application is ready to run when condition(s) determine the application should be active. Required input data is available, and it meets performance requirements.
	Not Available	Traffic application is not available. Required input data is available, but it does not meet performance requirements.
	Fault	Traffic application is not available. Required input data is not available or the application has failed.
	Not Configured	Traffic application is not available, because it has not been configured. If this annunciation persists, the system should be serviced.
	----- ----	Traffic application status is invalid or unknown.
TIS-B/ADS-R Coverage	Available	The system is receiving the ADS-R coverage from an FAA ground station.
	Not Available	The system is not receiving the ADS-R coverage from an FAA ground station.
	----- ----	ADS-R coverage is invalid or unknown.

ADS-B Status Page Item	Status Message	Description
GPS Status: GPS Source	External #1	The system is using the #1 GPS receiver for the GPS position source.
	External #2	The system is using the #2 GPS receiver for the GPS position source.
	-----	The GPS source is invalid or unknown.
Ground Uplink Status: Last Uplink	Number of minutes, or '-----'	Displays the number of minutes since the last uplink from a ground station occurred. If no uplink has been received, or the status is invalid, dashes appear instead of the number of minutes.

ADS-B GROUND STATION STATUS

ADS-B Ground Station Reception Status	'Traffic Map' Pane Icon
Receiving ADS-B traffic services from a ground station	
Not receiving ADS-B traffic services from a ground station	

GDL 69A SXM DATA LINK RECEIVER MESSAGES

Message	Message Location	Description
CHECK ANTENNA	XM Information Page (MFD)	Data Link Receiver antenna error; service required
UPDATING	XM Information Page (MFD))	Data Link Receiver updating encryption code
NO SIGNAL	XM Information Page Weather Datalink Page (MFD)	Loss of signal; signal strength too low for receiver
LOADING	XM Radio Page (MFD)	Acquiring channel audio or information

Message	Message Location	Description
OFF AIR	XM Radio Page (MFD)	Channel not in service
-----	XM Radio Page (MFD)	Missing channel information
WEATHER DATA LINK FAILED	Weather Datalink Page (MFD)	No communication from Data Link Receiver within last 5 minutes
ACTIVATION REQUIRED	XM Information Page (MFD)	SiriusXM subscription is not activated
DETECTING ACTIVATION	Weather Datalink Page (MFD)	SiriusXM subscription is activating.
WAITING FOR DATA...	Weather Datalink Page (MFD)	SiriusXM subscription confirmed downloading weather data.

VOICE ALERTS

Voice Alert	Alert Trigger
"Minimums, Minimums"	Issued when the aircraft transitions through the minimum descent altitude/decision height (MDA/DH)
"Vertical Track"	Aircraft is one minute from Top of Descent - issued only when vertical navigation is enabled
"Traffic"	TIS voice alert - issued when a Traffic Advisory (TA) is issued
"Traffic Not Available"	TIS voice alert - issued when the TIS traffic system fails or cannot communicate
"Traffic, Traffic"	TAS voice alert - issued when a Traffic Advisory (TA) is issued
"Traffic Advisory System Test Passed"	TAS voice alert - issued when the TAS passes a pilot-initiated self test
"Traffic Advisory System Test Failed"	TAS voice alert - issued when the TAS fails a pilot-initiated self test

SYSTEM MESSAGE ADVISORIES

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
ADC1 ALT EC – ADC1 altitude error correction is unavailable	GDC1 is reporting that the altitude error correction is unavailable.
ADC1 AS EC – ADC1 airspeed error correction is unavailable.	GDC1 is reporting that the airspeed error correction is unavailable.
ADC1 SERVICE – ADC1 needs service. Return unit for repair.	The GDC1 should be serviced.
AHRS1 CAL – AHRS1 calibration version error. Svc req'd.	The #1 AHRS calibration version error. The system should be serviced.
AHRS1 CONFIG – AHRS1 config error. Config service req'd.	AHRS configuration settings do not match those of backup configuration memory. The system should be serviced.
AHRS1 GPS – AHRS1 using backup GPS source.	The #1 AHRS is using the backup GPS path. Primary GPS path has failed. The system should be serviced when possible.
AHRS1 GPS – AHRS1 not receiving any GPS information.	The #1 AHRS is not receiving any or any useful GPS information. Check the current version of the pertinent flight manual limitations. The system should be serviced.
AHRS1 GPS – AHRS1 not receiving backup GPS information.	The #1 AHRS is not receiving backup GPS information. The system should be serviced.
AHRS1 GPS – AHRS1 operating exclusively in no-GPS mode.	The #1 AHRS is operating exclusively in no-GPS mode. The system should be serviced.

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
AHRS1 SERVICE – AHRS1 Magnetic-field model needs update.	The #1 AHRS earth magnetic field model is out of date. Update magnetic field model when practical.
AHRS1 TAS – AHRS1 not receiving airspeed.	The #1 AHRS is not receiving true airspeed from the air data computer. The AHRS relies on GPS information to augment the lack of airspeed. The system should be serviced.
APR DWNGRADE – Approach downgraded.	Use LNAV minima when approach is downgraded.
ARSPC AHEAD – Airspace ahead less than 10 minutes.	Special use airspace is ahead of aircraft. The aircraft will penetrate the airspace within 10 minutes.
ARSPC NEAR – Airspace near – less than 2 nm.	Special use airspace is within 2 nm of the aircraft position.
ARSPC NEAR – Airspace near and ahead.	Special use airspace is near and ahead of the aircraft position.
APR INACTV – Approach is not active.	The system notifies the pilot that the loaded approach is not active. Activate approach when required.
AUDIO MANIFEST - Audio software mismatch, communication halted.	The GMA has incorrect software installed. The system should be serviced.
CHECK CRS – Database course for LOC2 / [LOC ID] is [CRS]°.	Selected course for LOC2 differs from published localizer course by more than 10 degrees.
CNFG MODULE – PFD1 configuration module is inoperative.	The PFD1 configuration module backup memory has failed. The system should be serviced.
COM1 CONFIG – COM1 config error. Config service req'd.	The COM1 configuration settings do not match backup configuration memory. The system should be serviced
COM #[1, 2] INOP - CAL - Check COM calibration.	COM 1 and/or COM 2 calibration version error. Check COM calibration.

	Message	Comments
Flight Instruments	ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
EIS	COM #[1, 2] INOP - CRNT - Check COM current.	COM 1 and/or COM 2 current is low. Check COM current.
Nav/Com/XPDR/Audio	COM #[1, 2] INOP - NTRL - Com internal fault.	COM 1 and/or COM 2 has an internal fault.
Flight Management	COM #[1, 2] REDUCED TX POWER - COM synthesizer lock fault.	COM 1 and/or COM 2 has a reduced transmission power.
	COM #[1, 2] INOP - SYNTH - COM synthesizer lock fault.	The COM 1 and/or COM 2 has a synthesizer lock fault.
Hazard Avoidance	COM1 PTT – COM1 push-to-talk key is stuck.	The COM1 external push-to-talk switch is stuck in the enable (or “pressed”) position. Press the PTT switch again to cycle its operation. If the problem persists, the system should be serviced.
AFCs	COM1 RMT XFR – COM1 remote transfer key is stuck.	The COM1 transfer switch is stuck in the enabled (or “pressed”) position. Press the transfer switch again to cycle its operation. If the problem persists, the system should be serviced.
Additional Features	COM1 SERVICE – COM1 needs service. Return unit for repair.	The system has detected a failure in COM1. COM1 may still be usable. The system should be serviced when possible.
Abnormal Operation	COM1 TEMP – COM1 over temp. Reducing transmitter power.	The system has detected an over temperature condition in COM1. The transmitter operates at reduced power. If the problem persists, the system should be serviced.
Annun/Alerts	COM2 CONFIG – COM2 config error. Config service req'd.	The COM2 configuration settings do not match backup configuration memory. The system should be serviced
Appendix	COM2 PTT – COM2 push-to-talk key is stuck.	The COM2 external push-to-talk switch is stuck in the enable (or “pressed”) position. Press the PTT switch again to cycle its operation. If the problem persists, the system should be serviced.
Index		

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
COM2 RMT XFR – COM2 remote transfer key is stuck.	The COM2 transfer switch is stuck in the enabled (or “pressed”) position. Press the transfer switch again to cycle its operation. If the problem persists, the system should be serviced.
COM2 SERVICE – COM2 needs service. Return unit for repair.	The system has detected a failure in COM2. COM2 may still be usable. The system should be serviced when possible.
COM2 TEMP – COM2 over temp. Reducing transmitter power.	The system has detected an over temperature condition in COM2. The transmitter operates at reduced power. If the problem persists, the system should be serviced.
COPILOT RADIOS MUTED – Copilot radios are muted.	The copilot radios are set on mute.
DATA LOST – Pilot stored data was lost. Recheck settings.	The pilot profile data was lost. System reverts to default pilot profile and settings. The pilot may reconfigure the MFD & PFD with preferred settings, if desired.
DB CHANGE – Database changed. Verify user modified procedures.	This occurs when a stored flight plan contains procedures that have been manually edited. This alert is issued only after an navigation database update. Verify that the user-modified procedures in stored flight plans are correct and up to date.
DB CHANGE – Database changed. Verify stored airways.	This occurs when a stored flight plan contains an airway that is no longer consistent with the navigation database. This alert is issued only after an navigation database update. Verify use of airways in stored flight plans and reload airways as needed.

	Message	Comments
Flight Instruments	ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
EIS	DB MISMATCH – Navigation database mismatch. Xtalk is off.	The PFD and MFD have different navigation database versions or types installed. Crossfill is off. Check the Aux-System Status Page to determine versions or regions. Also, check the Aux-System Status Page for a database synchronization function not completed. After synchronization is complete, power must be turned off, then on.
Nav/Com/XPDR/Audio		
Flight Management	DB MISMATCH – Obstacle database mismatch.	The PFD and MFD have different obstacle database versions or types installed. Check the Aux-System Status Page to determine versions or regions. Also, check the Aux-System Status Page for a database synchronization function not completed. After synchronization is complete, power must be turned off, then on.
Hazard Avoidance		
AFCS	DB MISMATCH – Terrain database mismatch.	The PFD and MFD have different terrain database versions or types installed. Check the Aux-System Status Page to determine versions or regions. Also, check the Aux-System Status Page for a database synchronization function not completed. After synchronization is complete, power must be turned off, then on.
Additional Features		
Abnormal Operation	DIG GMA1 MANIFEST – DIG GMA 1 software mismatch, communication halted.	The digital audio controller has incorrect software installed. The system should be serviced.
Annun/Alerts	DISABLE GP TO ARM CLIMB – Press APPR button to enable VNAV track change.	Disable AFCS APPR mode to enable VNAV
Appendix	DISABLE GS TO ARM CLIMB – Press APPR button to enable VNAV track change.	Disable AFCS APPR mode to enable VNAV
Index	DME CHECK RANGE – DME range disagreement. Check position sensors.	A failure or disagreement has been detected in a DME receiver. Check position sensors.

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
FAILED PATH – A data path has failed.	A data path connected to the GDU or the GIA 64W has failed.
FPL TRUNC – Flight plan has been truncated.	This occurs when a newly installed navigation database eliminates an obsolete approach or arrival used by a stored flight plan. The obsolete procedure is removed from the flight plan. Update flight plan with current arrival or approach.
FPL WPT LOCK – Flight plan waypoint is locked.	Upon power-up, the system detects that a stored flight plan waypoint is locked. This occurs when an navigation database update eliminates an obsolete waypoint. The flight plan cannot find the specified waypoint and flags this message. This can also occur with user waypoints in a flight plan that is deleted. Remove the waypoint from the flight plan if it no longer exists in any database, OR update the waypoint name/identifier to reflect the new information.
G/S1 FAIL – G/S1 is inoperative.	A failure has been detected in glideslope receiver 1. The system should be serviced.
G/S1 SERVICE – G/S1 needs service. Return unit for repair.	A failure has been detected in glideslope receiver 1. The receiver may still be available. The system should be serviced when possible.
G/S2 FAIL – G/S2 is inoperative.	A failure has been detected in glideslope receiver 2. The system should be serviced.
G/S2 SERVICE – G/S2 needs service. Return unit for repair.	A failure has been detected in glideslope receiver 2. The receiver may still be available. The system should be serviced when possible.
GDC1 MANIFEST – GDC1 software mismatch, communication halted.	The AHRS has incorrect software installed. The system should be serviced.

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

	Message	Comments
Flight Instruments	ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
EIS	GDL69 CONFIG – GDL 69 config error. Config service req'd.	GDL 69/69A SXM configuration settings do not match those of backup configuration memory. The system should be serviced.
Nav/Com/XPDR/Audio	GDL69 FAIL – GDL 69 has failed.	A failure has been detected in the GDL 69/69A or GDL 69/69A SXM. The receiver is unavailable. The system should be serviced.
Flight Management	GDL69 MANIFEST – GDL software mismatch, communication halted.	The GDL 69/69A SXM has incorrect software installed. The system should be serviced.
Hazard Avoidance	GEA1 CONFIG – GEA1 config error. Config service req'd.	The GEA1 configuration settings do not match those of backup configuration memory. The system should be serviced.
AFCs	GEA1 MANIFEST – GEA1 software mismatch, communication halted.	The #1 GEA has incorrect software installed. The system should be serviced.
Additional Features	GEO LIMITS – AHRS1 too far North/South, no magnetic compass.	The aircraft is outside geographical limits for approved AHRS operation. Heading is flagged as invalid.
Abnormal Operation	GFC MANIFEST – GFC software mismatch, communication halted.	Incorrect servo software is installed, or gain settings are incorrect.
Annun/Alerts	GIA1 CONFIG – GIA1 audio config error. Config service req'd.	The GIA1 have an error in the audio configuration. The system should be serviced.
	GIA1 CONFIG – GIA1 config error. Config service req'd.	The GIA1 configuration settings do not match backup configuration memory. The system should be serviced.
Appendix	GIA1 COOLING – GIA1 temperature too low.	The GIA1 and/or GIA2 temperature is too low to operate correctly. Allow units to warm up to operating temperature.
Index	GIA1 COOLING – GIA1 over temperature.	The GIA1 temperature is too high. If problem persists, the system should be serviced.

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
GIA1 MANIFEST – GIA1 software mismatch, communication halted.	The GIA1 1 has incorrect software installed. The system should be serviced.
GIA1 SERVICE – GIA1 needs service. Return the unit for repair.	The GIA1 self-test has detected a problem in the unit. The system should be serviced.
GIA2 CONFIG – GIA2 audio config error. Config service req'd.	The GIA2 have an error in the audio configuration. The system should be serviced.
GIA2 CONFIG – GIA2 config error. Config service req'd.	The GIA2 configuration settings do not match backup configuration memory. The system should be serviced.
GIA2 COOLING – GIA2 over temperature.	The GIA2 temperature is too high. If problem persists, the system should be serviced.
GIA2 COOLING – GIA2 temperature too low.	The GIA2 temperature is too low to operate correctly. Allow units to warm up to operating temperature.
GIA2 MANIFEST – GIA2 software mismatch, communication halted.	The GIA 2 has incorrect software installed. The system should be serviced.
GIA2 SERVICE – GIA2 needs service. Return the unit for repair.	The GIA2 self-test has detected a problem in the unit. The system should be serviced.
GMA1 AUX MANIFEST – GMA 1 AUX software mismatch, communication halted.	The digital audio controller has incorrect software installed. The system should be serviced.
GMA1 CONFIG – GMA1 config error. Config service req'd.	The audio panel configuration settings do not match backup configuration memory. The system should be serviced.
GMA1 FAIL – GMA1 is inoperative.	The audio panel self-test has detected a failure. The audio panel is unavailable. The system should be serviced.

	Message	Comments
Flight Instruments	ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
EIS	GMA1 MANIFEST – GMA1 software mismatch, communication halted.	The audio panel has incorrect software installed. The system should be serviced.
Nav/Com/XPDR/Audio	GMA1 SERVICE – GMA1 needs service. Return unit for repair.	The audio panel self-test has detected a problem in the unit. Certain audio functions may still be available, and the audio panel may still be usable. The system should be serviced when possible.
Flight Management	GMU1 MANIFEST – GMU1 software mismatch, communication halted.	The GMU has incorrect software installed. The system should be serviced.
Hazard Avoidance	GPS NAV LOST – Loss of GPS navigation. Insufficient satellites.	Loss of GPS navigation due to insufficient satellites.
AFCs	GPS NAV LOST – Loss of GPS navigation. Position error.	Loss of GPS navigation due to position error.
Additional Features	GPS NAV LOST – Loss of GPS navigation. GPS fail.	Loss of GPS navigation due to GPS failure.
Abnormal Operation	GPS1 SERVICE – GPS1 needs service. Return unit for repair.	A failure has been detected in the GPS1 and/or GPS2 receiver. The receiver may still be available. The system should be serviced.
	GPS2 SERVICE – GPS2 needs service. Return unit for repair.	
Annun/Alerts	GRS1 MANIFEST – GRS1 software mismatch, communication halted.	The AHRS has incorrect software installed. The system should be serviced.
Appendix	GTS CONFIG – GTS config error. Config service req'd.	GTS 800 configuration settings do not match those of the GDU configuration. The system should be serviced.
Index	GTS MANIFEST – GTS software mismatch, communication halted.	The GTS has incorrect software installed. The system should be serviced.

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
GTX1 MANIFEST – GTX1 software mismatch, communication halted.	The transponder has incorrect software installed. The system should be serviced.
HDG FAULT – AHRS1 magnetometer fault has occurred.	A fault has occurred in the #1 GMU. Heading is flagged as invalid. The AHRS uses GPS for backup mode operation. The system should be serviced.
HW MISMATCH – GIA hardware mismatch. GIA1 communication halted.	A GIA mismatch has been detected, where only one is SBAS capable.
HW MISMATCH – GIA hardware mismatch. GIA2 communication halted.	
INSIDE ARSPC – Inside airspace.	The aircraft is inside the airspace.
INVALID ADM – Invalid ADM: ATN communication halted.	Data link avionics were not configured correctly and therefore will not be able to communicate with the ground network.
LOCKED FPL – Cannot navigate locked flight plan.	This occurs when the pilot attempts to activate a stored flight plan that contains locked waypoint. Remove locked waypoint from flight plan. Update flight plan with current waypoint.
LOI – GPS integrity lost. Crosscheck with other NAVS.	GPS integrity is insufficient for the current phase of flight.
LRG MAG VAR – Verify all course angles.	The GDU's internal model cannot determine the exact magnetic variance for geographic locations near the magnetic poles. Displayed magnetic course angles may differ from the actual magnetic heading by more than 2°.

	Message	Comments
Flight Instruments	ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
EIS	MANIFEST – MFD1 software mismatch, communication halted.	The PFD and/or MFD has incorrect software installed. The system should be serviced.
Nav/Com/XPDR/Audio	MANIFEST – PFD1 software mismatch, communication halted.	
Flight Management	MFD SOFTWARE – MFD mismatch, communication halted.	The specified GDU has different software versions installed. The system should be serviced.
Hazard Avoidance	MFD TERRAIN DSP – MFD Terrain awareness display unavailable.	One of the terrain or obstacle databases required for TAWS in the specified GDU is missing or invalid.
AFCs	MFD1 BACKLIGHT CALIBRATION – MFD1 calibration. Return for repair.	The specified GDU's backlight calibration cannot be found or is invalid. The system should be serviced.
Additional Features	MFD1 CONFIG – MFD1 config error. Config service req'd.	The MFD configuration settings do not match backup configuration memory. The system should be serviced.
Abnormal Operation	MFD1 COOLING – MFD1 has poor cooling. Reducing power usage.	The MFD is overheating and is reducing power consumption by dimming the display. If problem persists, the system should be serviced.
Annun/Alerts	MFD1 DB ERR – MFD1 obstacle database error exists.	The MFD detected a failure in the obstacle database. Ensure that the data card is properly inserted. Replace data card. If problem persists, the system should be serviced.
Appendix	MFD1 DB ERR – MFD1 obstacle database missing.	The obstacle database is present on another LRU, but is missing on the specified LRU.
Index	MFD1 DB ERR – MFD1 terrain database error exists.	The MFD detected a failure in the terrain database. Ensure that the terrain card is properly inserted in display. Replace terrain card. If problem persists, the system should be serviced.

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
MFD1 DB ERR – MFD1 terrain database missing.	The terrain database is present on another LRU, but is missing on the specified LRU.
MFD1 SERVICE – MFD1 needs service. Return unit for repair.	The MFD self-test has detected a problem. The system should be serviced.
MFD1 KEYSTK – MFD1 [key name] is stuck.	A key is stuck on the MFD bezel. Attempt to free the stuck key by pressing it several times. The system should be serviced if the problem persists.
MFD1 VOLTAGE – MFD1 has low voltage. Reducing power usage	The MFD voltage is low. The system should be serviced.
NAV1 MANIFEST – NAV1 software mismatch, communication halted.	NAV1 software mismatch. The system should be serviced.
NAV1 RMT XFR – NAV1 remote transfer key is stuck.	The remote NAV1 transfer switch is stuck in the enabled (or “pressed”) state. Press the transfer switch again to cycle its operation. If the problem persists, the system should be serviced.
NAV1 SERVICE – NAV1 needs service. Return unit for repair.	A failure has been detected in the NAV1 receiver. The receiver may still be available. The system should be serviced.
NAV2 MANIFEST – NAV2 software mismatch, communication halted.	NAV2 software mismatch. The system should be serviced.
NAV2 RMT XFR – NAV2 remote transfer key is stuck.	The remote NAV2 transfer switch is stuck in the enabled (or “pressed”) state. Press the transfer switch again to cycle its operation. If the problem persists, the system should be serviced.
NAV2 SERVICE – NAV2 needs service. Return unit for repair.	A failure has been detected in the NAV2 receiver. The receiver may still be available. The system should be serviced.

Flight Instruments	Message	Comments
	ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
EIS	NON-MAGNETIC UNITS – Non-magnetic NAV ANGLE display units are active.	Navigation angle is not set to MAGNETIC at power-up.
Nav/Com/ XPD/R/Audio	NON WGS84 WPT – Do not use GPS for navigation to [xxxx]	The position of the selected waypoint [xxxx] is not calculated based on the WGS84 map reference datum and may be positioned in error as displayed. Do not use GPS to navigate to the selected non-WGS84 waypoint..
Flight Management	PFD1 BACKLIGHT CALIBRATION – PFD1 calibration lost. Return for repair.	The PFD1 backlight calibration cannot be found or is invalid. The system should be serviced.
Hazard Avoidance	PFD1 CONFIG – PFD1 config error. Config service req'd.	The PFD configuration settings do not match backup configuration memory. The system should be serviced.
AFCs	PFD1 COOLING – PFD1 has poor cooling. Reducing power usage.	The PFD is overheating and is reducing power consumption by dimming the display. If problem persists, the system should be serviced.
Additional Features	PFD1 DB ERR – PFD1 obstacle database missing.	The obstacle database is present on another LRU, but is missing on the specified LRU.
Abnormal Operation	PFD1 DB ERR – PFD1 terrain database error exists.	The PFD detected a failure in the terrain database. Ensure that the terrain card is properly inserted in display. Replace terrain card. If problem persists, the system should be serviced.

Annun/Alerts

Appendix

Index

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
PFD1 DB ERR – PFD1 terrain database missing.	The terrain database is present on another LRU, but is missing on the specified LRU.
PFD1 KEYSTK – PFD1 [key name] is stuck.	A key is stuck on the PFD bezel. Attempt to free the stuck key by pressing it several times. The system should be serviced if the problem persists.
[PFD1 or MFD1] CARD 1 REM – Card 1 was removed. Reinsert card.	The SD card was removed from the top card slot of the specified PFD or MFD. The SD card needs to be reinserted.
[PFD1 or MFD1] CARD 2 REM – Card 2 was removed. Reinsert card.	The SD card was removed from the bottom card slot of the specified PFD or MFD. The SD card needs to be reinserted.
[PFD1 or MFD1] CARD 1 ERR – Card 1 is invalid.	The SD card in the top card slot of the specified PFD or MFD contains invalid data.
[PFD1 or MFD1] CARD 2 ERR – Card 2 is invalid.	The SD card in the bottom card slot of the specified PFD or MFD contains invalid data.
PFD1 VOLTAGE – PFD1 has low voltage. Reducing power usage	The PFD1 voltage is low. The system should be serviced.
PFD1 KEYSTK – PFD2 [key name] is stuck.	A key is stuck on the PFD bezel. Attempt to free the stuck key by pressing it several times. The system should be serviced if the problem persists.
PFD1 SERVICE – PFD1 needs service. Return unit for repair.	The PFD self-test has detected a problem. The system should be serviced.
PFD1 TERRAIN DSP – PFD1 Terrain awareness display unavailable.	One of the terrain or obstacle databases required for TAWS in PFD1 is missing or invalid.
PFD1 VOLTAGE – PFD2 has low voltage. Reducing power usage	The PFD2 voltage is low. The system should be serviced.
PILOT RADIOS MUTED – Pilot radios are muted.	The pilot radios are set on mute.

	Message	Comments
Flight Instruments	ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
EIS	PTK FAIL – Parallel track unavailable: invalid leg type.	Invalid leg type for parallel offset.
Nav/Com/XPDR/Audio	PTK FAIL – Parallel track unavailable: past IAF.	IAF waypoint for parallel offset has been passed.
	PTK FAIL – Parallel track unavailable: bad geometry.	Bad parallel track geometry.
Flight Management	SCHEDULER [#] – <message>.	Message criteria entered by the user.
Hazard Avoidance	SLCT FREQ – Select appropriate frequency for approach.	The system notifies the pilot to load the approach frequency for the appropriate NAV receiver. Select the correct frequency for the approach.
AFCs	SLCT MAG – Select MAGNETIC NAV ANGLE display units.	The system notifies the pilot to set the Nav Angle units on the Avionics Settings Screen to Magnetic.
Additional Features	SLCT NAV – Select NAV on CDI for approach.	The system notifies the pilot to set the CDI to the correct NAV receiver. Set the CDI to the correct NAV receiver.
Abnormal Operation	SLCT NON-MAG – Select alternate NAV ANGLE display units.	The system notifies the pilot to set the Nav Angle units on the Avionics Settings Screen to True.
	STEEP TURN – Steep turn ahead.	A steep turn is 15 seconds ahead. Prepare to turn.
Annun/Alerts	STRMSCP FAIL – Stormscope has failed.	Stormscope has failed. The system should be serviced.
Appendix	SVT DISABLED – Out of available terrain region.	Synthetic Vision is disabled because the aircraft is not within the boundaries of the installed terrain database.
Index	SVT DISABLED – Terrain DB resolution too low.	Synthetic Vision is disabled because a terrain database of sufficient resolution is not currently installed.

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
SW MISMATCH – GDU software version mismatch. Xtalk is off.	The MFD and PFD have different software versions installed. The system should be serviced.
TERRAIN AUD CFG – Trn Awareness audio config error. Service req'd.	TAWS is disabled because the audio configuration is invalid. The system should be serviced.
TERRAIN DISABLED – Terrain Awareness DB resolution too low.	TAWS is disabled because a terrain database of sufficient resolution is not currently installed.
TIMER EXPIRD – Timer has expired.	The system notifies the pilot that the timer has expired.
TRAFFIC FAIL – Traffic device has failed.	The system is no longer receiving data from the traffic system. The traffic device should be serviced.
TRN AUD FAIL – Trn Awareness audio source unavailable	Terrain Awareness audio is unavailable. The system should be serviced.
UNABLE V WPT – Can't reach current vertical waypoint.	The current vertical waypoint can not be reached within the maximum flight path angle and vertical speed constraints. The system automatically transitions to the next vertical waypoint.
VNV – Unavailable. Excessive crosstrack error.	The current crosstrack exceeds the limit, causing vertical deviation to go invalid.
VNV – Unavailable. Excessive track angle error.	The current track angle error exceeds the limit, causing the vertical deviation to go invalid.
VNV – Unavailable. Parallel course selected.	A parallel course has been selected, causing the vertical deviation to go invalid.
VNV – Unavailable. Unsupported leg type in flight plan.	The lateral flight plan contains a procedure turn, vector, or other unsupported leg type prior to the active vertical waypoint. This prevents vertical guidance to the active vertical waypoint.

	Message	Comments
Flight Instruments	ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
EIS	WPT ARRIVAL – Arriving at waypoint -[xxxx]	Arriving at waypoint [xxxx], where [xxxx] is the waypoint name.
Nav/Com/XPDR/Audio	XPDR1 ADS-B 1090 – Datalink: ADS-B 1090 receiver has failed.	A failure has been detected in the 1090 receiver.
Flight Management	XPDR1 ADS-B FAIL – Transponder: XPDR1 is unable to transmit ADS-B messages.	ADS-B is inoperative. The transponder may not be receiving a valid GPS position. Other transponder functions may be available. Service when possible.
Hazard Avoidance	XPDR1 ADS-B NO POS – Transponder: ADS-B is not transmitting position.	The transponder is not able to receive position information.
AFCS	XPDR1 ADS-B TRFC – Transponder: ADS-B traffic has failed	The Transponder is incapable of processing traffic information.
Additional Features	XPDR1 ADS-B UAT – Datalink: ADS-B in UAT receiver has failed.	A failure has been detected in the UAT receiver.
Abnormal Operation	XPDR1 CONFIG – XPDR1 config error. Config service req'd.	The transponder configuration settings do not match those of backup configuration memory. The system should be serviced.
Annun/Alerts	XPDR1 CSA FAIL - Traffic: ADS-B In traffic alerting has failed.	ADS-B Conflict Situational Awareness (CSA) is unavailable.
Appendix	XPDR1 FAIL – XPDR1 is inoperative.	There is no communication with the #1 or #2 transponder.
Index	XPDR1 FAULT – Datalink: ADS-B in has failed.	The transponder is unable to receive ADS-B information.
	XPDR1 FIS-B WX – Datalink: FIS-B Weather has failed.	The transponder is unable to receive FIS-B weather information.

Message	Comments
ABORT APR – Loss of GPS navigation. Abort approach.	Abort approach due to loss of GPS navigation.
XPDR1 OVER TEMP - Transponder: Transponder over temp.	The system has detected an over temperature condition in XPDR1. The transmitter operates at reduced power. If the problem persists, the system should be serviced.
XPDR1 PRES ALT – Transponder: ADS-B no pressure altitude.	Unable to provide pressure altitude information.
XPDR1 SRVC – XPDR1 needs service. Return unit for repair.	The #1 transponder should be serviced when possible.
XPDR1 UNDER TEMP - Transponder: Transponder under temp.	The system has detected an under temperature condition in XPDR1. The transmitter operates at reduced power. If the problem persists, the system should be serviced.
XTALK ERROR – A flight display crosstalk error has occurred.	The MFD and PFD are not communicating with each other. The system should be serviced.
XTALK ERROR – A flight display crosstalk error has occurred.	The MFD and PFD are not communicating with each other. The system should be serviced.

FLIGHT PLAN IMPORT/EXPORT MESSAGES

In some circumstances, some messages may appear in conjunction with others.

Flight Plan Import/Export Results	Description
'Flight plan successfully imported.'	A flight plan file stored on the SD card was successfully imported as a stored flight plan.
'File contained user waypoints only. User waypoints imported successfully. No stored flight plan data was modified.'	The file stored on the SD card did not contain a flight plan, only user waypoints. These waypoints have been saved to the system user waypoints. No flight plans stored in the system have been modified.
'No flight plan files found to import.'	The SD card contains no flight plan data.

Flight Instruments	Flight Plan Import/Export Results	Description
	'Flight plan import failed.'	Flight plan data was not successfully imported from the SD card.
EIS	'File contained user waypoints only.'	The file stored on the SD card did not contain a flight plan, only user waypoints. One or more of these waypoints did not import successfully.
Nav/Com/XPDR/Audio	'Flight plan partially imported.'	Some flight plan waypoints were successfully imported from the SD card, however others had errors and were not imported. A partial stored flight plan now exists in the system.
Flight Management	'Too many points. Flight plan truncated.'	The flight plan on the SD card contains more waypoints than the system can support. The flight plan was imported with as many waypoints as possible.
Hazard Avoidance	'Some waypoints not loaded. Waypoints locked.'	The flight plan on the SD card contains one or more waypoints that the system cannot find in the navigation database. The flight plan has been imported, but must be edited within the system before it can be activated for use.
AFCs	'User waypoint database full. Not all loaded.'	The flight plan file on the SD card contains user waypoints. The quantity of stored user waypoints has exceeded system capacity, therefore not all the user waypoints on the SD card have been imported. Any flight plan user waypoints that were not imported are locked in the flight plan. The flight plan must be edited within the system before it can be activated for use.
Additional Features		
Abnormal Operation	'One or more user waypoints renamed.'	One or more imported user waypoints were renamed when imported due to naming conflicts with waypoints already existing in the system.
Annun/Alerts	'Flight plan successfully exported.'	The stored flight plan was successfully exported to the SD card.
	'Flight plan export failed.'	The stored flight plan was not successfully exported to the SD card. The SD card may not have sufficient available memory or the card may have been removed prematurely.
Appendix		
Index		

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

CREW PROFILE IMPORT/EXPORT MESSAGES

Crew Profile Import/Export Results	Description
'No pilot profile plan files found to import.'	Displayed if the SD card does not have one or more valid pilot profile filenames.
'Overwrite existing profile?'	Displayed if the profile name matches the name of existing profile.
'Profile name invalid. Enter a different profile name.'	Displayed if the profile name is invalid.
'All available pilot profiles in use. Delete a profile before importing another.'	Displayed if the maximum number for pilot profiles has been reached.
'Pilot profile import failed.'	Displayed if the importing operation fails for any other reason.
'Pilot profile import succeeded.'	Displayed if the importing operation succeeds.
'Overwrite existing file?'	Displayed if the filename matches the name of an existing file on the SD card.
'Pilot profile export failed.'	Displayed if the export operation fails.
'Pilot profile export succeeded.'	Displayed if the export operation succeeds.

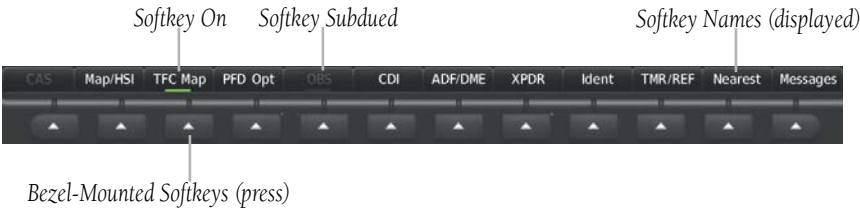
Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

Blank Page

APPENDIX

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

SOFTKEY FUNCTION



Softkeys (First-Level PFD Configuration)

PFD SOFTKEY MAPS

Level 1	Level 2	Level 3	Description
CAS			Displays the scroll key. Only selectable when the number of CAS messages exceeds the capacity of the display window.
	CAS Up		Scroll up list of CAS messages.
	CAS Down		Scroll down list of CAS messages.
Map/HSI			Displays the PFD Map display settings softkeys.
	Layout		Displays the PFD Map selection softkeys.
		Map Off	Removes the PFD map from display (Inset, HSI, or Traffic).
		Inset Map	Displays the Inset Map.
		HSI Map	Displays the HSI Map.
		Inset Trfc	Replaces the Inset Map with a dedicated traffic display.
		HSI Trfc	Replaces the HSI Map with a dedicated traffic display.

Flight Instruments	Level 1	Level 2	Level 3	Description
EIS		Detail		Selects desired amount of map detail: All (No Declutter): All map features visible Detail 3 : Declutters land data Detail 2 : Declutters land and SUA data Detail 1 : Removes everything except for the active flight plan
Nav/Com/XPDR/Audio		Traffic		Displays traffic information on PFD Map.
Flight Management		TER		Topo : Displays topographical data (e.g., coastlines, terrain, rivers, lakes) and elevation scale on PFD Map. REL : Displays relative terrain information on the PFD Map. Off : Removes terrain
Hazard Avoidance		WX LGND		Displays/removes the name of the selected weather data provider (SiriusXM) and the weather product icon and age box (for enabled weather products).
AFCS		NEXRAD		Displays XM NEXRAD weather and coverage on PFD Map (subscription optional)
		METAR		Displays METAR information on Inset Map (subscription optional)
Additional Features		Lightning		Adds/removes the display of SiriusXM or Connexxt lightning information (based on weather data source selection) on the PFD Map.
Abnormal Operation			LTNG Off	Disables lightning function on PFD Map. The softkey annunciator is green when the lightning function is off.
			Datalink	Selects the datalink weather source for the PFD Map
Annun/Alerts			STRMSCP	Adds or removes the display of Stormscope information on the PFD Map. The softkey annunciator is green when the function is on. When the function is off, the annunciator is gray.
Appendix	TFC Map			Replaces the PFD Map with a dedicated traffic display.
	PFD Opt			Displays second-level softkeys for additional PFD options.
		SVT		Displays additional SVT overlay softkeys.
			Pathways	Displays Pathway Boxes on the Synthetic Vision Display.
			Terrain	Enables synthetic terrain depiction.
Index			HDG LBL	Displays compass heading along the Zero-Pitch line.

Level 1	Level 2	Level 3	Description
		APT Sign	Displays position markers for airports within approximately 15 nm of the current aircraft position. Airport identifiers are displayed when the airport is within approximately 9 nm.
	Wind		Displays the wind option softkeys.
		Off	Wind information not displayed.
		Option 1	Headwind/Tailwind and crosswind arrows with numeric speed components.
		Option 2	Total wind direction arrow and numeric speed.
		Option 3	Total wind direction arrow with digital numeric direction and speed.
	DME		Displays 'DME Information' Window (optional).
	Bearing 1		Cycles the Bearing 1 Information Window through NAV1, NAV2, GPS/waypoint identifier and GPS-derived distance information, ADF/frequency, and Off.
	Bearing 2		Cycles the Bearing 2 Information Window through NAV1, NAV2, GPS/waypoint identifier and GPS-derived distance information, ADF/frequency, and Off.
	ALT Units		Displays softkeys to select altitude unit parameters.
		Meters	When enabled, displays altimeter in meters.
		IN	Press to display the BARO setting as inches of mercury.
		HPA	Press to display the BARO setting as hectopascals.
	STD Baro		Sets barometric pressure to 29.92 in Hg (1013 hPa if metric units are selected).
OBS			Selects OBS mode on the CDI when navigating by GPS (only available with active leg). When OBS is on, the softkey annunciator is green.
CDI			Cycles through FMS, NAV1, and NAV2 navigation modes on the CDI.
ADF/DME			Displays the ADF/DME Tuning Window, allowing tuning and selection of the ADF and DME (optional).
XPDR			Displays the transponder selection softkeys.

Level 1	Level 2	Level 3	Description
	Standby		Selects transponder Standby Mode (transponder does not reply to any interrogations).
	On		Activates transponder (transponder replies to identification interrogations).
	ALT		Altitude Reporting Mode (transponder replies to identification and altitude interrogations).
	VFR		Automatically enters the VFR code (1200 in the U.S.A. only).
	Code		Displays transponder code selection softkeys 0-7.
		0 - 7	Use numbers to enter code.
		Ident	Activates the Special Position Identification (SPI) pulse for 18 seconds, identifying the transponder return on the ATC screen.
		BKSP	Removes numbers entered, one at a time.
	Ident		Activates the Special Position Identification (SPI) pulse for 18 seconds, identifying the transponder return on the ATC screen.
Ident			Activates the Special Position Identification (SPI) pulse for 18 seconds, identifying the transponder return on the ATC screen.
TMR/REF			Displays 'References' Window.
Nearest			Displays 'Nearest Airports' Window.
Alerts or Message			Displays the 'Alerts' Window when pressed. System generated messages cause the Alerts Softkey label to change to a flashing 'Message' label. Pressing the Message Softkey opens the 'Alerts Window, acknowledges the message, and the softkey reverts to the 'Alerts' label.

MFD SOFTKEY MAPS

Level 1	Level 2	Level 3	Description
Engine			Displays EIS - Engine Page and second-level engine softkeys; press again to exit page (see the EIS Section for more information).
	Engine		Returns to previously shown softkeys.

Level 1	Level 2	Level 3	Description
	DEC Fuel		Decreases calculated fuel remaining by 1 gallon for each softkey press.
	INC Fuel		Increases calculated fuel remaining by 1 gallon for each softkey press.
	RST Fuel		Resets calculated fuel remaining to default and resets fuel used to zero.
Map Opt			Displays system oil pressure and temperature, fuel calculations, and electrical system information.
	Traffic		Displays traffic information on Navigation Map Page.
	Inset		Displays inset window second level softkeys.
		Off	Removes VSD inset from Navigation Map Page.
		VSD	Displays VSD profile information of terrain/obstacles along the current track, vertical track vector, and selected altitude. Auto: Automatically displays either VSD profile information for active flight plan information or along current track with no active flight plan. FPL: Displays VSD profile information for active flight plan. TRK: Displays VSD profile information along current track.
	TER		Displays terrain on the map; cycles through the following: Off: No terrain information shown on MFD Map. Topo: Displays topographical data (e.g., coastlines, terrain, rivers, lakes) and elevation scale on MFD Map. REL: Displays relative terrain information on the MFD Map.
	AWY		Displays airways on the map; cycles through the following: Off: No airways are displayed. On: All airways are displayed. LO: Only low altitude airways are displayed. HI: Only high altitude airways are displayed.
	STRMSCP		Displays Stormscope information on Navigation Map (optional).

Flight Instruments
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

	Level 1	Level 2	Level 3	Description
Flight Instruments		NEXRAD		Displays XM NEXRAD weather and coverage on Navigation Map Page (optional).
EIS		XM LTNG		Displays XM lightning information on Navigation Map Page (optional).
Nav/Com/XPDR/Audio		METAR		Displays METAR information on Inset Map (subscription optional).
Flight Management		Legend		Displays legends for the displayed XM Weather products (optional).
Hazard Avoidance	Detail			Selects desired amount of map detail; cycles through the following levels: Detail All: All map features visible. Detail-3: Declutters land data. Detail-2: Declutters land and SUA data. Detail-1: Removes everything except for the active flight plan.
AFCS	Charts			When available, displays optional airport and terminal procedure charts.
Additional Features		CHRT Opt		Displays chart display settings softkeys (if available).
Abnormal Operation			ROT CCW	Rotates chart 90 degrees counter clockwise.
Annun/Alerts			ROT CW	Rotates chart 90 degrees clockwise.
Appendix			Fit WDTN	Chart zoom adjusted to fill width of display.
Index			Full SCN	When the annunciator bar is green, full screen mode is enabled. The annunciator bar is grey when in split screen mode.
		SYNC		Displays the chart associated with the current phase of flight.
		Info		Returns to the selected airport information chart (Airport Diagram, Alternate Minimums, Climb/Descent Table, Diverse Vector Area, Hot Spot, INOP Components, LAHSO, and Takeoff Minimums)
		DP		Displays departure procedure chart.
		STAR		Displays standard terminal arrival procedure chart.

Level 1	Level 2	Level 3	Description
	APR		Displays approach procedure chart.
Checklist			When available, displays optional checklists.

LOADING UPDATED DATABASES



CAUTION: *Never disconnect power to the system when loading a database. Power interruption during the database loading process could result in maintenance being required to reboot the system.*



NOTE: *When loading database updates, the 'DB Mismatch' system messages will be displayed until database synchronization is complete, followed by turning system power off, then on. Synchronization can be monitored on the 'Aux-Database' Page.*

Databases may be loaded through Garmin Pilot and Flight Stream 510. When loading databases through Garmin Pilot and the Flight Stream 510, the Flight Stream 510 must be enabled on the system and the multimedia card inserted in the bottom SD slot of the MFD.

The cycles and dates for both standby and active databases are displayed on the “Aux – Databases” page on the MFD. Any active databases with expiration dates in the past will be highlighted with amber text. When an expired active database has a standby database that is ready to become effective, a cyan double-sided arrow will be displayed between the database cycles. When this arrow is visible, it indicates that the standby and active databases in that row will be switched on the next power cycle, activating the current standby database. Databases can also be manually selected (or deselected) by highlighting a list item and pressing the **ENT** Key, provided a valid, verified standby database is present.

In some cases it may be necessary to obtain an unlock code from Garmin in order to make the database product functional. It may also be necessary to have the system configured by a Garmin authorized service facility in order to use some database features.

The navigation database contains the aeronautical data used by the system for the flight management and flight planning functions. Included is detailed data for way-points, procedures (arrivals, departures, approaches), and airways. The navigation database is updated every 28 days.

Flight Instruments

The basemap database contains data for the topography and land features, such as rivers, lakes, and towns. It is updated only periodically, with no set schedule. There is no expiration date.

EIS

The terrain database contains the terrain mapping data. These databases are updated periodically and have no expiration date.

Nav/Com/XPDR/Audio

The obstacle database contains data for obstacles, such as towers, that pose a potential hazard to aircraft. Obstacles 200 feet and higher are included in the obstacle database. It is very important to note that not all obstacles are necessarily charted and therefore may not be contained in the obstacle database. This database is updated on a 56-day cycle.

Flight Management



NOTE: *The data contained in the terrain and obstacle databases comes from government agencies. Garmin accurately processes and cross-validates the data, but cannot guarantee the accuracy and completeness of the data.*

Hazard Avoidance

The AOPA or AC-U-KWIK Airport Directory provides data on airports and heliports throughout the U.S. or worldwide, respectively. The AOPA Directory offers detailed information for over 5,300 U. S. airports, along with the names and phone numbers of thousands of FBOs. These databases are updated every 56 days. The AC-U-KWIK Directory offers detailed information for more than 8,000 airports with runways longer than 3,000 feet worldwide.

Additional Features

The SafeTaxi database contains detailed airport diagrams for selected airports. These diagrams aid in following ground control instructions by accurately displaying the aircraft position on the map in relation to taxiways, ramps, runways, terminals, and services. This database is updated on a 56-day cycle.

Abnormal Operation

The FliteCharts database contains procedure charts. This database is updated on a 28-day cycle. If not updated within 180 days of the expiration date, FliteCharts will no longer function.

Annun/Alerts

The ChartView database is updated on a 14-day cycle. If the ChartView database is not updated within 70 days of the expiration date, ChartView will no longer function. The ChartView database must be purchased directly from Jeppesen, but can be updated at jeppesen.com or flygarmin.com.

Appendix

The IFR/VFR charts database contains VFR and IFR raster charts. The VFR Charts are digital representations of the Sectional Aeronautical Charts and Terminal Area Charts. The IFR Charts include both IFR High (designed for navigation at or above 18,000 ft) and IFR Low (designed for navigation below 18,000 ft). IFR/VFR Charts are updated every 28 days except for Canadian IFR/VFR Charts which are updated every 56 days.

Index

Updated Databases Using a Supplemental Data Card

All databases are updated through a single SD card in the bottom slot of the MFD. When the card is inserted, the databases on the card will be copied to standby and synchronized across all powered, configured units. After update, the card is removed and the databases are stored on the system. When in standby, databases are not immediately available for use, but stored to be activated at a later time.

Database updates can be obtained by following the instructions detailed in the 'Aviation Databases' section of the Garmin website (flygarmin.com). Once the updated files have been downloaded from the website, a PC equipped with an appropriate SD card reader is used to unpack and program the new databases onto an existing Supplemental Data Card. Equipment required to perform the update is as follows:

- Windows-compatible PC computer
- SD Card Reader: SanDisk SDDR-93, SanDisk SDDR-99, Verbatim #96504, or equivalent
- Updated database obtained from the Garmin website
- Supplemental SD Cards

Updating Databases

- 1) With the system OFF, remove an SD Card from the top SD card slot of the MFD.
- 2) Download and install the databases on an SD card.
- 3) Put the SD Card in the top SD card slot of the MFD.
- 4) Turn the system ON.
- 5) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
- 6) Turn the large **FMS** Knob and select Aux.
- 7) Turn the small **FMS** Knob and select Databases.
- 8) Monitor the Sync Status on the 'Aux-Databases' Page. Wait for all databases to complete syncing, indicated by 'Sync Complete' being displayed. A cyan double-arrow will appear between the Standby and Active columns to show which Standby databases will be transferred to Active at the next power cycle.
- 9) Verify the correct database cycle information is shown in the Standby column.

- 10) Remove the SD card from the bottom slot of the MFD if desired.
- 11) Remove and reapply power to the system.
- 12) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
- 13) Turn the large **FMS** Knob and select Aux.
- 14) Turn the small **FMS** Knob and select the Databases.
- 15) Verify that the standby databases transferred and are now in the Active column.
- 16) To manually activate any databases that did not transfer to the active column:
 - a) Press the **FMS** Knob. The first database title on the screen will be selected.
 - b) Turn the small **FMS** Knob as necessary to select the database title.
 - c) Press the **ENT** Key. A cyan double-sided arrow will appear indicating that the standby database will become active.

Or:

Press the **Menu** Key and select 'Swap Stby/Actv' using the small **FMS** Knob. Press the **ENT** Key. A cyan double-sided arrow will appear indicating that the standby database will become active.

 - d) Remove and reapply power to the system.
 - e) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
 - f) Turn the large **FMS** Knob and select Aux.
 - g) Turn the small **FMS** Knob and select Databases.
 - h) Verify that the standby databases transferred and are now in the Active column.
- 17) For additional information on each database, press and then turn the **FMS** Knob to select the database, and then press the **Details** Softkey. Press the **ENT** Key or the **FMS** Knob to exit.

Or:

Press the **Menu** Key and select 'Details' using the small **FMS** Knob and press the **ENT** Key. Press the **ENT** Key or the **FMS** Knob to exit.

- 18) To view database information for an individual display:
 - a) Turn the large **FMS** Knob and select Aux.
 - b) Turn the small **FMS** Knob and select System Status.
 - c) Press the Display Database Selection Softkey (MFD1 DB, PFD1 DB, PFD2 DB) to show database information for each display. Use the small **FMS** Knob to scroll through the database information. Press the **ENT** Key or the **FMS** Knob to exit.

Updating Databases Using the Flight Stream 510

In order to load databases through Garmin Pilot and the Flight Stream 510, the Flight Stream 510 must be enabled on the system and inserted in the bottom SD slot of the MFD. A mobile device with Garmin Pilot must be paired with the Flight Stream 510 over Bluetooth (refer to the Additional Features section). When there is at least one paired device available to connect, the Flight Stream 510 will automatically connect to the system's preferred mobile device. The preferred device can be selected on the 'Aux - Databases' Page from a menu list of paired devices.

Once a connection to the paired mobile device is made, Garmin Pilot makes available databases that can be transferred to the Flight Stream 510. If any of these databases is more recent than the respective standby database on the system, (or if there is no standby database on the system) those databases will be automatically selected to load. The database updates may be initiated from the 'Aux - Databases' Page, or from other pages on the MFD.



NOTE: The system will only provide a WiFi connection if new databases have been detected for download on Garmin Pilot via a valid Bluetooth connection. If there are no database updates required the system will not provide a WiFi signal.



NOTE: If the mobile device has previously connected to the FS510, and is not connected to another WiFi source, the mobile device should connect automatically to the FS510. If the mobile device is connected to another WiFi source (i.e. hangar wifi), then the FS510 will not connect automatically.

Updating Databases from any MFD page (except the 'Aux- Databases' Page)

- 1) Insert the Flight Stream 510 SD Card in the bottom slot of the MFD if not already inserted.
- 2) Turn the system ON.
- 3) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
- 4) On the mobile device, start Garmin Pilot and touch **Home > Connex > Database Concierge**.
- 5) Turn the large **FMS** Knob and select Aux.
- 6) Turn the small **FMS** Knob and select the Connex Setup.
- 7) Ensure that Wifi Database Import is enabled in the 'Device' Window (refer to Additional Features section for instructions to enable Wifi Database Import).
- 8) Verify that the mobile device is enabled via Bluetooth in the Bluetooth settings on the mobile device.
- 9) In the 'Paired Devices' Window on the 'Connex Setup' Page, ensure the system is paired with the mobile device in use. (refer to Additional Features for instructions on connection to a preferred device).



NOTE: The database updates may now be continued from any MFD page, however, the update windows shown in these instructions will not be shown on the 'Aux - Databases' Page. Use the proceeding instructions in this section to update databases from the 'Aux - Databases' Page.

- 10) Press the **Update** Softkey when the following window appears. (Pressing the **View** Softkey will allow database updates to be viewed from the 'Aux Databases' Page, however, the windows shown below will not appear on the 'Aux Databases' Page. Pressing the **Ignore** Softkey will postpone the updates until further action is taken.)
- 11) If using a device that has not been previously paired with the system, a password prompt will appear on the mobile device. Enter the password shown in the 'Password' Field of the 'Aux - Connex Setup' Page.
- 12) The following window will appear. Database update progress may be monitored on the mobile device.
- 13) When the transfer is complete, the following screen will appear.
- 14) Press the **Close** Softkey.

- 15) Remove and reapply power to the system.
 - 16) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
 - 17) Turn the large **FMS** Knob and select Aux.
 - 18) Turn the small **FMS** Knob and select Databases.
 - 19) Verify that the standby databases transferred and are now in the Active column.
 - 20) To manually activate any databases that did not transfer to the active column:
 - a) Press the **FMS** Knob. The first database title on the screen will be selected.
 - b) Turn the small **FMS** Knob as necessary to select the database title.
 - c) Press the **ENT** Key. A cyan double-sided arrow will appear indicating that the standby database will become active.

Or:

Press the **Menu** Key and select 'Swap Stby/Actv' using the small **FMS** Knob. Press the **ENT** Key. A cyan double-sided arrow will appear indicating that the standby database will become active.

 - d) Remove and reapply power to the system.
 - e) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
 - f) Turn the large **FMS** Knob and select Aux.
 - g) Turn the small **FMS** Knob and select Databases.
 - h) Verify that the standby databases transferred and are now in the Active column.
 - 21) For additional information on each database, press and then turn the **FMS** Knob to select the database, and then press the **Details** Softkey. Press the **ENT** Key or the **FMS** Knob to exit.
- Or:**
- Press the **Menu** Key and select 'Details' using the small **FMS** Knob and press the **ENT** Key. Press the **ENT** Key or the **FMS** Knob to exit.

Flight
Instruments

EIS

Nav/Com/
XPDR/Audio

Flight
Management

Hazard
Avoidance

AFCS

Additional
Features

Abnormal
Operation

Annun/Alerts

Appendix

Index

- 22) To view database information for an individual display:
 - a) Turn the large **FMS** Knob and select Aux.
 - b) Turn the small **FMS** Knob and select System Status.
 - c) Press the Display Database Selection Softkey (MFD1 DB, PFD1 DB, PFD2 DB) to show database information for each display. Use the small **FMS** Knob to scroll through the database information. Press the **ENT** Key or the **FMS** Knob to exit.

Updating Databases from the 'Aux - Databases' page



NOTE: The system will only provide a WiFi connection if new databases have been detected for download on Garmin Pilot via a valid Bluetooth connection. If there are no database updates required the system will not provide a WiFi signal.



NOTE: If the mobile device has previously connected to the FS510, and is not connected to another WiFi source, the mobile device should connect automatically to the FS510. If the mobile device is connected to another WiFi source (i.e. hangar wifi), then the FS510 will not connect automatically.

- 1) Insert the Flight Stream 510 SD Card in the bottom slot of the MFD if not already inserted.
- 2) Turn the system ON.
- 3) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
- 4) On the mobile device, start Garmin Pilot and tap **Home > Connex > Database Concierge**.
- 5) Turn the large **FMS** Knob and select Aux.
- 6) Turn the small **FMS** Knob and select the Connex Setup.
- 7) Ensure that Wifi Database Import is enabled in the 'Device' Window (refer to Additional Features section for instructions to enable Wifi Database Import).
- 8) Verify that the mobile device is enabled via Bluetooth in the Bluetooth settings on the mobile device.
- 9) In the 'Paired Devices' Window on the 'Connex Setup' Page, ensure the system is paired with the mobile device in use. (refer to Additional Features for instructions on connection to a preferred device).

- 10) Press the **Device** Softkey to view databases that are ready to be loaded from the mobile device (pressing the **Stby/Actv** Softkey will again display the current Standby and Active databases).
- 11) The 'Aux - Databases' Page will show the databases connected to the mobile device in place of the active databases on the system. Databases selected to load to the system will be indicated by a single cyan arrow.
- 12) Press the **Update** Softkey.
- 13) If using a device that has not been previously paired with the system, a password prompt will appear on the mobile device. Enter the password shown in the 'Password' Field of the 'Aux - Connex Setup' Page.
- 14) Database Update status will appear in the 'Status' Window at the top of the page. Monitor update progress in the 'Status' Window, or on the mobile device.
- 15) When all databases have been successfully transferred from the mobile device and appear in the Standby column, remove and reapply power to the system.
- 16) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
- 17) Turn the large **FMS** Knob and select Aux.
- 18) Turn the small **FMS** Knob and select Databases.
- 19) Verify that the standby databases transferred and are now in the Active column.
- 20) To manually activate any databases that did not transfer to the active column:
 - a) Press the **FMS** Knob. The first database title on the screen will be selected.
 - b) Turn the small **FMS** Knob as necessary to select the database title.
 - c) Verify the correct database cycle information is shown for each database for each display.

Or:

Press the **Menu** Key and select 'Swap Stby/Actv' using the small **FMS** Knob. Press the **ENT** Key. A cyan double-sided arrow will appear indicating that the standby database will become active.

 - d) Remove and reapply power to the system.

- e) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.
- f) Turn the large **FMS** Knob and select Aux.
- g) Turn the small **FMS** Knob and select Databases.
- h) Verify that the standby databases transferred and are now in the Active column.

- 21) For additional information on each database, press and then turn the **FMS** Knob to select the database, and then press the **Details** Softkey. Press the **ENT** Key or the **FMS** Knob to exit.

Or:

Press the **Menu** Key and select 'Details' using the small **FMS** Knob and press the **ENT** Key. Press the **ENT** Key or the **FMS** Knob to exit.

- 22) To view database information for an individual display:
- a) Turn the large **FMS** Knob and select Aux.
 - b) Turn the small **FMS** Knob and select System Status.
 - c) Press the **ENT** Key. A cyan double-sided arrow will appear indicating that the standby database will become active.
 - d) Press the Display Database Selection Softkey (MFD1 DB, PFD1 DB, PFD2 DB) to show database information for each display. Use the small **FMS** Knob to scroll through the database information. Press the **ENT** Key or the **FMS** Knob to exit.

DATABASE DELETION FEATURE

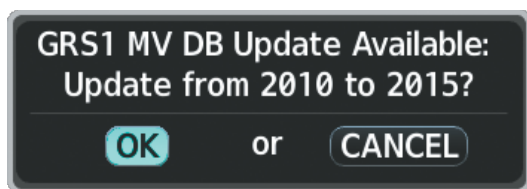
Deleting databases

- 1) Touch the **Menu** Key.
- 2) Turn the small **FMS** Knob to select 'Delete Databases.'
- 3) Push the **ENT** Key.
- 4) A prompt will appear to confirm deletion of all internal databases. Push the **ENT** Key.
- 5) Another prompt will appear to confirm deletion of all internal databases. Push the **ENT** Key.
- 6) Remove and reapply power to the system.
- 7) Press the **ENT** Key or the right-most softkey on MFD display to acknowledge the startup screen.

- 8) Turn the large **FMS** Knob and select Aux.
- 9) Turn the small **FMS** Knob and select Databases.
- 10) Confirm that all databases have been deleted from the system.

MAGNETIC FIELD VARIATION DATABASE UPDATE

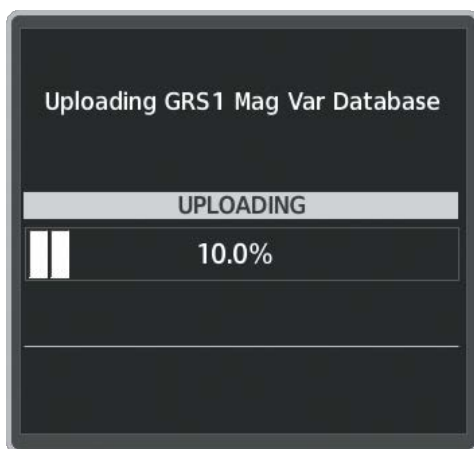
A copy of the current magnetic field variation database (MV DB) is included with the navigation database. At startup, the system compares this version of the MV DB with that presently being used by the ADAHRS. If the system determines the MV DB needs to be updated, a prompt is displayed on the Navigation Map Page.



Magnetic Field Variation Database Update Prompt

Loading the magnetic field variation database update

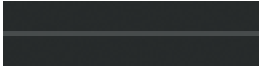
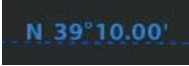
With 'OK' highlighted, press the **ENT** Key on the MFD. A progress monitor is displayed as shown. When the upload is complete, the system is ready for use.



Uploading Database

MAP SYMBOLS

Land Symbols

Land Symbols	Symbol	Default Range (nm)	Maximum Range (nm)
User Waypoint		25	40
Highways and Roads			
Interstate Highway (Freeway)		50	400
International Highway (Freeway)		50	400
US Highway (National Highway)		15	150
State Highway (Local Highway)		2.5	100
Local Road (Local Road)	N/A	4	25
Railroads (RAILROAD)		7.5	25
Large City (> 200,000)		100	1000
Medium City (> 50,000)		50	400
Small City (> 5,000)		25	100
State/Province		400	1000
River/Lake		75	100
Latitude/Longitude (LAT/LON)		1	1000

Aviation Symbols

Aviation Symbols	Symbol	Default Range (nm)	Maximum Range (nm)
Large Airport (Longest Runway ≥ 8100 ft)		100	1000
Medium Airport (8100 ft > Longest Runway ≥ 5000 ft)		50	400
Small Airport (Longest Runway < 5000 ft) (Medium Airport if it has a tower frequency)		25	150
Taxiways (SafeTaxi)	See Additional Features	1.5	5
Runway Extension		7.5	150
Missed Approach Preview On/Off (Missed APR)	N/A	N/A	N/A
Intersection (INT)		25	40
Non-directional Beacon (NDB)		25	50
VOR		50	250
VOR Compass Rose On/Off	N/A	N/A	N/A
Visual Reporting Point (VRP)		25	1000
Temporary Flight Restriction (TFR)		250	1000
VNAV Constraints		1000	1000
(VNAV Constraints) Show All (show all constraints within flight plan if VNAV Constraints are 'On' above)	N/A	N/A	N/A

Flight Instruments

EIS

Nav/Com/XPDR/Audio

Flight Management

Hazard Avoidance

AFCS

Additional Features

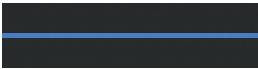
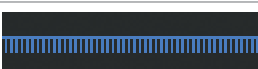
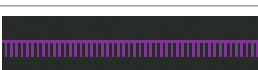
Abnormal Operation

Annun/Alerts

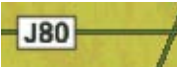
Appendix

Index

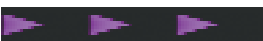
Airspace Symbols

Airspace Symbols	Symbol	Default Range (nm)	Maximum Range (nm)
Smart Airspace On/Off	N/A	N/A	N/A
Airspace Altitude Labels (Airspace ALT LBL) On/Off			
Class B Airspace Altitude Label (ceiling/floor)		N/A	N/A
Class C Airspace Altitude Label (ceiling/floor)		N/A	N/A
Class D Airspace Altitude Label (ceiling)		N/A	N/A
Class B Airspace/TMA (CLASS B/ TMA)		50	150
Class C Airspace/TCA (CLASS C/ TCA)		50	100
Class D Airspace (CLASS D)		10	100
Alert/Prohibited/Restricted/Warning Areas (RESTRICTED)		50	100
Military Operations Area [(MOA) (MILITARY)]		50	250
Other/Air Defense Interdiction Zone (OTHER/ADIZ)		50	250

Airways Symbols

Airways Symbols	Symbol	Default Range (nm)	Maximum Range (nm)
Low Altitude Airways (V Routes and T Routes)		50	100
High Altitude Airways (J Routes and Q Routes)		50	100

Active Flight Plan Leg Type Symbols

Flight Plan Leg Type	Symbol
Active Course Leg*	
Active Heading Leg*	
Active Roll Steering Path*†	
Course Leg in the current flight segment	
Course Leg not in the current flight segment	
Heading Leg	
Roll Steering Path †	
Future Roll Steering Path ‡	
Turn Anticipation Arc	

* The active leg or path is the one currently being flown, and is shown in magenta.
† A roll steering path is a computed transition between two disconnected legs.
‡ A roll steering path in the flight plan that is beyond the next leg appears as a future roll steering path. When a future roll steering path becomes the next leg in the flight plan, it appears as a roll steering path.

MISCELLANEOUS SYMBOLS

Item	Symbol
ARTCC Frequency or FSS Frequency	
North Arrow	
Map Pointer (when panning)	
Measuring Pointer	
User Waypoint	
Vertical Navigation Along Track Waypoint	
Parallel Track Waypoint	
Unanchored Flight Path Waypoint	
Displayed when aircraft GPS location is valid, but heading is invalid.	
Top of Descent (TOD)	
Bottom of Descent (BOD)	
Navigating using Dead Reckoning	

A

Active Channel 92
 AC-U-KWIK 91, 160
 ADAHRS 138, 141
 ADF 13, 14, 114
 AHRS 132, 133
 Airport
 Information 25
 Airport Directory 91, 160
 Airway
 Collapse/Expand Airways 33
 Along Track Offset 38
 Altitude
 Constraints 44
 Annunciations 120
 AOPA 91, 160
 Approach
 Activating 52, 53
 Loading 52
 Removing 53
 Approach activation 14
 Approach channel 52
 Arrival procedure 51
 Audio panel controls
 NAV1, NAV2 14
 Aviation Symbols 171, 172, 173

B

Bluetooth 107, 108, 109
 Bus voltage 11

C

CDI 4, 14
 ChartView 89
 Checklists 110–111
 Clearance Player/Recorder 15
 Closest Point 41

Cloud Tops 67
 Code selection softkeys 13
 COM 14, 15, 113
 Coolant temperature 10, 11
 Crew Alerting System (CAS) 12–18

D

Data Link Receiver 130, 131
 Day/Night views 90
 DB Mismatch 159
 Dead Reckoning (DR) Mode 115, 116
 DEC FUEL softkey 12
 Declutter 114
 Deleting
 An entire airway 38
 An entire procedure 38
 An individual waypoint 37, 38
 Departure
 Procedure 50
 Select 50, 51, 52
 Detail Softkey 89
 Direct-to 19, 29, 30
 DME 13, 14

E

Electronic Checklists 110, 111
 Electronic Stability and Protection (ESP)
 107, 109
 Emergency Checklist 111
 Engine load indicator 10, 11
 Export a Flight Plan 35

F

Flight Director 85
 Flight Plan, Active
 Creating 31
 Leg Type 173
 Flight plan import/export messages 149

Flight Stream 107

FliteCharts 89

FPA 44

Frequency Transfer 14

Fuel

Calculations 12

Flow indicator 10

Quantity 10, 11

Remaining 12

Temperature indicator 10

Used 12

G

Garmin Pilot 107, 108

H

Horizontal Situation Indicator (HSI) 51

HW Mismatch 141

I

Import a Flight Plan 34

Inverting a flight plan 34

IOI 124

Iridium satellite network 97, 98

J

Jeppesen 89

L

Land Symbols 170

Low speed data 93

M

MFD 14

Missed Approach 53

MKR/MUTE 15

Mode selection softkeys 13

N

NAV1/NAV2 4, 14

Navigation

Database 44

Nearest Airports 24, 25

NEXRAD 66

O

Oil

Pressure indicator 10, 11

Temperature Indicator 10, 11

Overspeed Protection 86

P

Pairing 107

Parallel Track 39

Pending flight plan 36, 37

PFD 14

Power-up Page 110

Predefined message 101

R

RAIM 55, 56

Reversionary mode 113

ROC 124

S

SafeTaxi 89

SBAS 56, 141

Scheduler 103

Secure Digital (SD) card 110

SiriusXM Weather

Icing 70

Softkeys 153–159, 154

EIS 156

Standby Navigation Database 161

Surface Analysis 69

Symbols, map 170, 174
 Synthetic Vision Technology (SVT) 146

T

Tachometer 10, 11
 TAS Traffic
 Non-Threat Traffic 78
 PA 78
 Proximity Advisory 78
 Telephone 93, 94
 Terminal procedures charts 90, 91
 Terrain 136
 Terrain-SVT 109
 Text message 99
 TOPO Data 21
 Total flight hours 11
 Traffic 57, 77
 Transfer from Wireless Connection: 35,
 36
 Transponder 13, 156
 True Airspeed (TAS) 133

U

User-Defined Holding Pattern 43

V

Vertical Navigation (VNAV) 30, 44
 Direct-to 44
 Guidance 45
 VNV 114
 VNV guidance
 Disabling 44
 Enabling 43
 VRP 26, 171
 Vspeed 1
 VS TGT 44

W

Waypoint Selection Submenu 29, 31,
 32

Flight Instrument
EIS
Nav/Com/XPDR/Audio
Flight Management
Hazard Avoidance
AFCS
Additional Features
Abnormal Operation
Annun/Alerts
Appendix
Index

Blank Page



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